PDE4ML

PDEs for Machine Learning

Gabriel Peyré
CNRS and ENS, PSL Université
gabriel.peyre@ens.fr

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Abstract

Machine learning increasingly relies on evolutions of probability measures. Training dynamics of wide neural networks, interacting particles, diffusion models, flow matching, attention layers and generative maps can all be described through transport, continuity equations, variational flows or stochastic PDE limits. This review develops an optimal-transport-oriented account of these PDE tools. The emphasis is not on PDE theory in isolation, but on the mechanisms by which PDEs organize learning algorithms: mass conservation, least-action principles, Wasserstein gradients, Fokker–Planck equations, McKean–Vlasov limits and transport-based generative dynamics.

The document is designed as a standalone PDE/ML review extracted and reorganized from the larger OT4ML project. It first gives a compact primer on Monge and Kantorovich optimal transport, Brenier maps and Wasserstein metric geometry. It then develops the material most directly relevant to PDE methods in machine learning: dynamic optimal transport, Wasserstein gradient flows, mean-field training dynamics, diffusion and flow-matching models, transportation views of transformers, and Gaussian closures. The goal is to expose a coherent mathematical language in which ML algorithms become evolutions on spaces of measures.

The source code for the full OT4ML project, including figure-generation notebooks, is available at `gpeyre/ot4ml`. Most computational figures were produced with the Python Optimal Transport (POT) library [35].

Guide to the Literature and Scope

This review uses optimal transport as the entry point to PDE methods for machine learning. The mathematical foundations of OT are covered in Villani's monographs [98, 99], Santambrogio's applied text [84], the probabilistic treatment of Rachev and Rüschendorf [78, 79], and the computational account of Peyré and Cuturi [75]. Gradient flows in metric spaces are developed systematically by Ambrosio, Gigli and Savaré [2], while the dynamic Benamou–Brenier viewpoint originates in [4].

Reference map. The review repeatedly returns to a small set of pillars. Static OT and Wasserstein geometry follow the classical line from Monge and Kantorovich to Brenier, McCann and Villani [65, 50, 8, 62, 99]. Dynamic OT and gradient flows use the Benamou–Brenier formula, the JKO scheme and Otto's formal Riemannian calculus [4, 5, 68, 2]. Machine-learning dynamics enter through mean-field neural-network limits and particle descriptions [19, 63, 81]. Generative modelling is organized around score-based diffusion, probability-flow ODEs, flow matching, rectified flows and stochastic interpolants [89, 56, 58, 1].

For machine learning, the relevant PDE themes include Fokker–Planck and McKean–Vlasov limits of particle systems, JKO schemes, entropy and interaction-energy flows, mean-field neural-network training, diffusion models, score-based generative modeling, flow matching and stochastic interpolants. The corresponding references are recalled locally in the text, so that each citation is tied to the mathematical mechanism being discussed.

The scope is deliberately selective. The primer only includes the OT material required later: push-forwards, Monge maps, Kantorovich couplings, Brenier's theorem, Wasserstein distances and geodesics. Algorithmic OT, duality, Sinkhorn convergence and generalized OT are treated in the full OT4ML manuscript; here they are invoked only when they support PDE or ML dynamics.

Reading map. The four sections move from static OT notation to dynamic least-action principles, then to Wasserstein gradient flows and particle limits, and finally to transportation dynamics for generative models and continuous-depth architectures.

Contents

SECTION 1

A Primer on Optimal Transport

This primer fixes the optimal-transport notation used throughout the review. The goal is not to reproduce the full OT4ML development, but to keep the minimal objects needed for PDE methods: push-forward maps, Monge and Kantorovich formulations, Brenier maps, Wasserstein distances and displacement geodesics. The figures are used as a visual glossary: each one is placed next to the definition or theorem whose geometry it makes concrete.

1.1 Measures and Push-Forwards

Let $\mathcal{X}$ and $\mathcal{Y}$ be Polish metric spaces, meaning complete separable metric spaces. This assumption is the natural background for probability theory because it gives enough measurable structure for disintegration, weak convergence and compactness arguments while still covering Euclidean spaces, manifolds, discrete spaces and many path spaces.

Definition 1.1 (Discrete measure)**.** A discrete probability measure on $\mathcal{X}$ is

$$\alpha = \sum_{i=1}^{n} a_i \delta_{x_i}, \qquad a_i \geqslant 0, \qquad \sum_{i=1}^{n} a_i = 1. \tag{1.1}$$

If $a_i = 1/n$, this is an empirical measure.

Definition 1.2 (Push-forward)**.** Let $T : \mathcal{X} \to \mathcal{Y}$ be measurable and let $\alpha \in \mathcal{P}(\mathcal{X})$. The push-forward $T_\sharp\alpha \in \mathcal{P}(\mathcal{Y})$ is defined by

$$(T_\sharp\alpha)(B) = \alpha(T^{-1}(B)) \tag{1.2}$$

for all Borel sets $B \subset \mathcal{Y}$. Equivalently, for all bounded measurable $g : \mathcal{Y} \to \mathbb{R}$,

$$\int_{\mathcal{Y}} g(y)\,\mathrm{d}(T_\sharp\alpha)(y) = \int_{\mathcal{X}} g(T(x))\,\mathrm{d}\alpha(x). \tag{1.3}$$

If X is a random variable with law α, then $T_\sharp\alpha$ is the law of $T(X)$. This simple identity is the bridge between deterministic maps, particle systems and measure-valued PDEs.

Proposition 1.3 (Push-forward formula for densities)**.** *Let $\mathcal{X} = \mathcal{Y} = \mathbb{R}^d$, let T be a diffeomorphism, and let $\alpha = \rho_\alpha(x)\mathrm{d}x$. Then $\beta = T_\sharp\alpha$ has density*

$$\rho_\beta(y) = \rho_\alpha(T^{-1}(y))\left|\det \nabla T(T^{-1}(y))\right|^{-1}. \tag{1.4}$$

Proof. Apply the change-of-variables formula to (1.3). □

Figure 1.1 makes the determinant factor visible. This is the first geometric warning behind transport PDEs: moving particles changes both their positions and the local volume element.

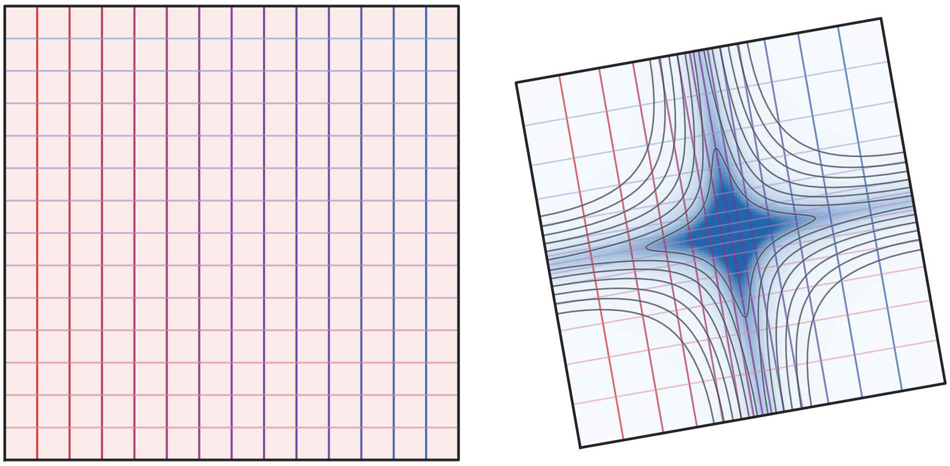
1.2 Monge Problems

Monge's formulation asks for a deterministic map transporting one law onto another. It is geometrically direct and is the natural language for generative maps, but it is not convex and can be infeasible because maps cannot split mass.

1.2.1 General Measures

Given $\alpha \in \mathcal{P}(\mathcal{X})$, $\beta \in \mathcal{P}(\mathcal{Y})$ and a cost $c : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}_+$, the Monge problem is

$$\mathcal{M}_c(\alpha,\beta) := \inf_T \left\{ \int_{\mathcal{X}} c(x, T(x))\,\mathrm{d}\alpha(x)\ ;\ T_\sharp\alpha = \beta \right\}. \tag{1.5}$$



uniform source grid pushed density and deformed grid

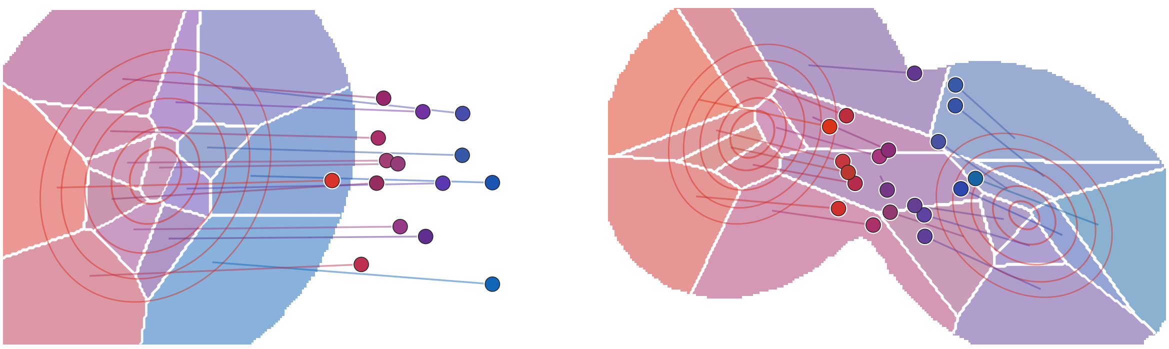
Figure 1.1: Jacobian determinant in the density push-forward formula. A smooth diffeomorphism compresses the middle of a colored square grid and applies a slight rotation. The source density is uniform, but the target density forms a central bump where deformed grid cells have smaller area. This is the geometric content of the factor $|\det \nabla T|^{-1}$ in (1.4), and later becomes the divergence structure of continuity equations.

The constraint $T_\sharp \alpha = \beta$ means that the endpoint law is imposed exactly. If $\alpha = \delta_x$ and β is not a Dirac mass, the feasible set is empty; this is the simplest mass-splitting obstruction.

Proposition 1.4 (Existence of transport maps from atomless sources)**.** *Let α and β be Borel probability measures on Polish spaces, and assume that α is atomless. Then there exists a measurable map T such that $T_\sharp \alpha = \beta$.*

Proof. Use a standard measure-isomorphism theorem to identify the atomless probability space generated by α with Lebesgue measure on $[0, 1]$, modulo null sets. A generalized quantile map sends Lebesgue measure on $[0, 1]$ to β after a Borel parametrization of the target space. Composing both maps gives a transport map. □

A useful asymmetric case is semi-discrete transport: the source has a density, the target is finitely supported, and a Monge map partitions space into cells. Figure 1.2 gives the cell picture before the fully discrete assignment problem. It also explains why reversing source and target would usually be impossible: finitely many source atoms cannot be mapped deterministically onto a continuous density.



Gaussian source to 15 atoms two-mode source to 20 atoms

Figure 1.2: Semi-discrete Monge maps. The red contours show the continuous source density. The colored regions are numerical Laguerre cells with approximately equal source mass, and the circular atoms form the discrete target. Faint segments connect cell barycenters to their images under the piecewise-constant Monge map, making the map-as-partition viewpoint explicit.

1.2.2 Discrete Monge Problem

For equal-size empirical measures

$$\alpha = \frac{1}{n}\sum_{i=1}^{n} \delta_{x_i}, \qquad \beta = \frac{1}{n}\sum_{j=1}^{n} \delta_{y_j},$$

and distinct target points, a feasible Monge map is a permutation. The discrete problem is therefore the assignment problem

$$\min_{\sigma \in \mathrm{Perm}(n)} \frac{1}{n}\sum_{i=1}^{n} C_{i,\sigma(i)}, \qquad C_{ij} = c(x_i, y_j). \tag{1.6}$$

Proposition 1.5 (Empirical Monge maps and matchings)**.** *Assume that the source atoms x_i are distinct and that $\alpha = n^{-1}\sum_i \delta_{x_i}$, $\beta = n^{-1}\sum_j \delta_{y_j}$. If the y_j are distinct and $T_\sharp\alpha = \beta$, then there exists $\sigma \in \mathrm{Perm}(n)$ such that $T(x_i) = y_{\sigma(i)}$ for all i.*

Proof. For any target atom z, the identity $T_\sharp\alpha = \beta$ gives

$$\beta(\{z\}) = \alpha(T^{-1}(\{z\})) = \frac{1}{n}\#\{i : T(x_i) = z\}.$$

Each target atom has mass $1/n$, so it receives exactly one source atom. □

In one dimension, convex costs have a special monotonicity structure: after sorting the source and target points, the equal-rank matching is optimal. This order structure is exceptional and does not survive in higher-dimensional point clouds.

The next two figures isolate the role of order on the line. Figure 1.3 shows equal-rank matching for convex costs, while Figure 1.4 contrasts it with a concave-cost regime where crossings become favorable.

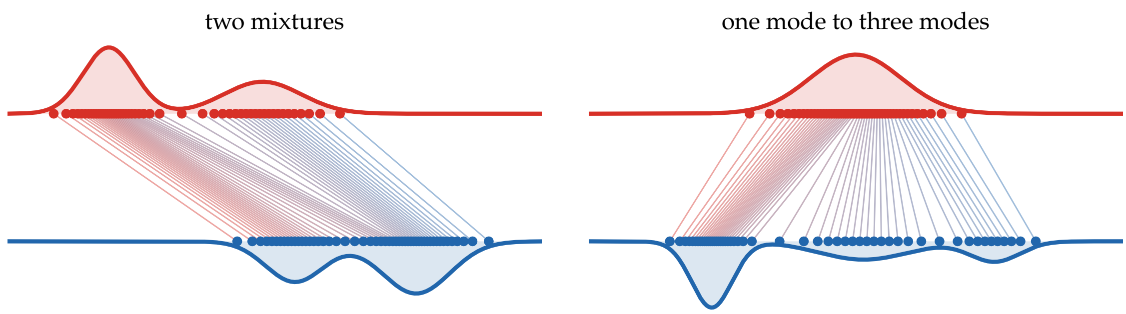


Figure 1.3: One-dimensional optimal matching by quantile sorting. The red and blue curves are smooth laws used to generate equal-weight empirical measures; the dots are inverse-CDF samples at common quantile levels. For convex costs on the line, the monotone assignment connects equal ranks.

The line also has a useful periodic variant. On the circle, one first chooses a cut, unfolds both measures into an interval and then applies the monotone one-dimensional assignment. Optimizing over the cut restores the circular geometry; see Figure 1.5. This is still an ordered one-dimensional construction, but the order is cyclic rather than linear.

In two dimensions there is no global sorting rule. Figure 1.6 keeps the same two point clouds and changes only the exponent in the cost, showing that the optimal permutation is already a genuinely geometric object.

The permutation viewpoint also assumes equal cardinalities and equal weights. Figure 1.7 previews the Kantorovich matrix formulation: as soon as masses or resolutions differ, transport must allow splitting and merging.

Rational weights give a precise bridge between assignments and transport matrices. If $a_i = k_i/N$ and $b_j = \ell_j/N$, one may duplicate x_i exactly k_i times and y_j exactly ℓ_j times, solve a uniform assignment with N particles on each side, and collapse the result back to the original supports. Figure 1.8 shows how repeated duplicated matches appear as several transport segments attached to the same displayed atom.

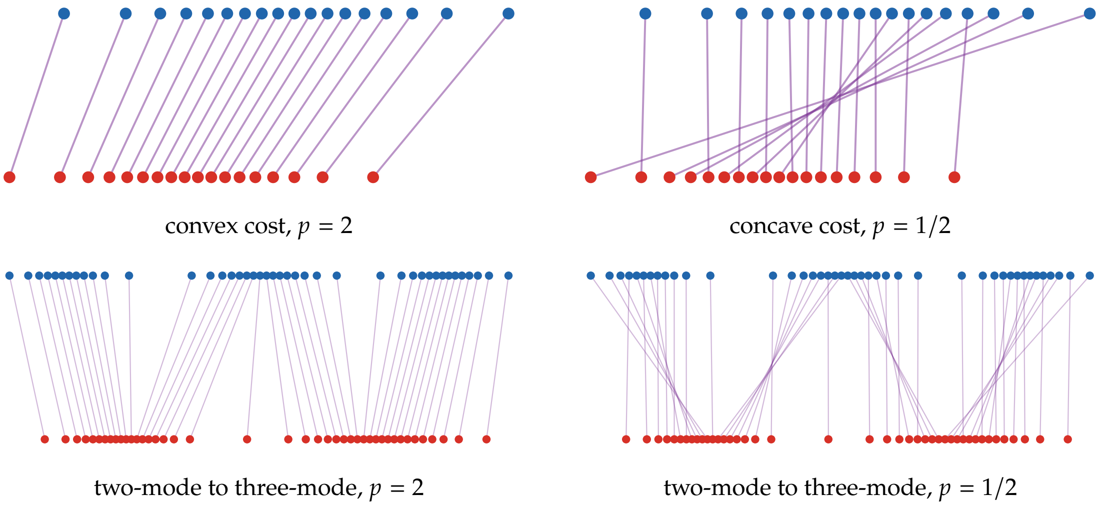


Figure 1.4: One-dimensional assignments for costs $c_p(x, y) = |x - y|^p$. For the convex quadratic cost, equal ranks are matched and the segments do not cross. For the concave cost, the optimum creates long crossing exchanges. Thus monotone rearrangement is a convex-cost phenomenon, not a generic property of all transport costs.

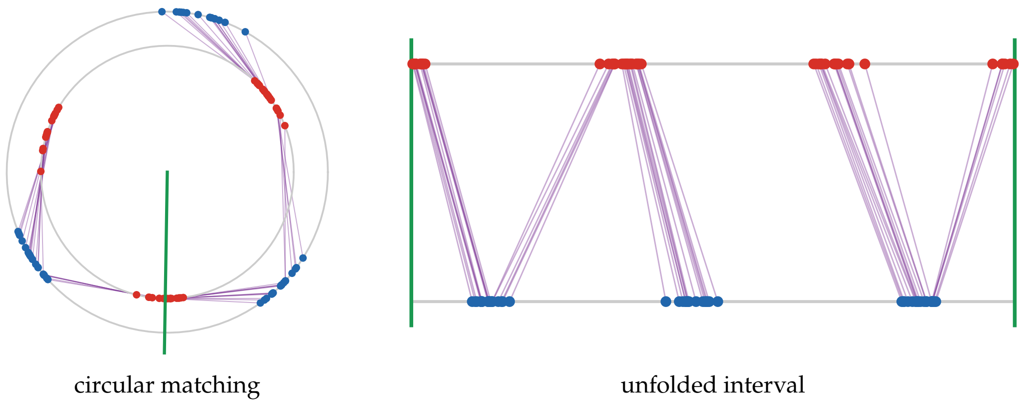


Figure 1.5: Optimal transport on the circle by cutting and unfolding. Red and blue atoms live on two copies of the circle, purple segments show the optimal cyclic matching, and the green radius marks the chosen cut. Once the circle is opened at this angle, the same assignment becomes a monotone one-dimensional matching on the interval.

1.3 Kantorovich Problems

Kantorovich's relaxation replaces deterministic maps by joint probability measures. It is the convex formulation used throughout the rest of the review, and it is the object whose dynamic counterpart becomes the Benamou–Brenier PDE.

1.3.1 General Measures

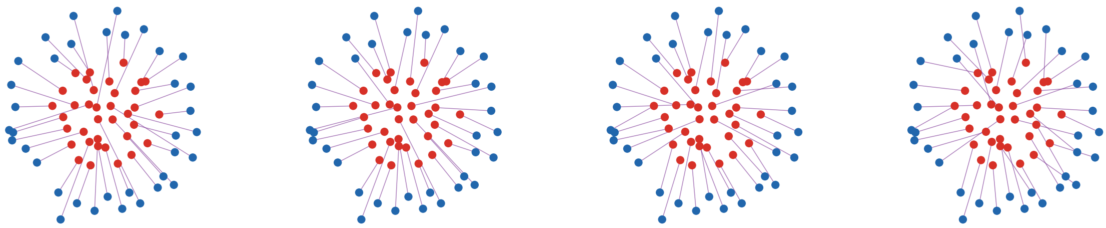
Definition 1.6 (Marginals of a joint measure). Let $\pi \in \mathcal{P}(\mathcal{X} \times \mathcal{Y})$ and let $P_{\mathcal{X}}(x, y) = x$, $P_{\mathcal{Y}}(x, y) = y$. Its marginals are $(P_{\mathcal{X}})_\sharp \pi$ and $(P_{\mathcal{Y}})_\sharp \pi$.

Definition 1.7 (Couplings). The set of couplings between $\alpha \in \mathcal{P}(\mathcal{X})$ and $\beta \in \mathcal{P}(\mathcal{Y})$ is

$$\mathcal{U}(\alpha, \beta) := \{\pi \in \mathcal{P}(\mathcal{X} \times \mathcal{Y})\,;\, (P_{\mathcal{X}})_\sharp \pi = \alpha,\ (P_{\mathcal{Y}})_\sharp \pi = \beta\}. \tag{1.7}$$

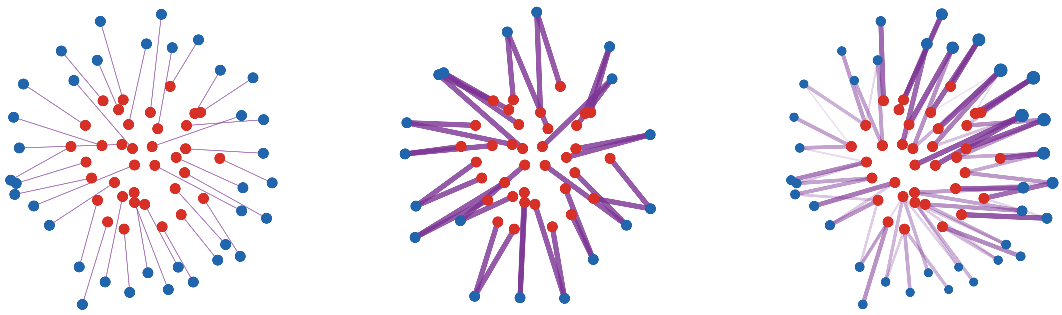
The set is never empty, since it contains the tensor product $\alpha \otimes \beta$. The Kantorovich problem is

$$\mathcal{L}_c(\alpha, \beta) := \inf_{\pi \in \mathcal{U}(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{Y}} c(x, y)\, \mathrm{d}\pi(x, y). \tag{1.8}$$



$\|x-y\|^{1/2}$ $\|x-y\|$ $\|x-y\|^2$ $\|x-y\|^6$

Figure 1.6: Optimal assignments between the same two planar point clouds for four powers of the Euclidean distance. The feasible permutations are unchanged, but the selected matching changes with the cost geometry: small powers tolerate long exchanges, whereas larger powers suppress the longest edges.



$n = m$ $m < n$ nonuniform target weights

Figure 1.7: From assignments to transport plans, using the same disk-to-annulus geometry as Figure 1.6. In the balanced equal-weight case, each source atom is matched to one target atom. With fewer target atoms, or with strongly nonuniform target weights, the coupling must merge or split mass; segment thickness and opacity encode transported mass.

If $T_\sharp\alpha = \beta$, then $\pi = (\mathrm{Id}, T)_\sharp\alpha$ is a coupling, so Kantorovich is a relaxation of Monge.

Figure 1.9 gives a first visual dictionary for couplings: a graph coupling is deterministic, the tensor product is independent and diffuse, and an optimal relaxed plan selects the dependence that lowers the cost. This is the finite-dimensional picture to keep in mind when the same symbol π later denotes a dynamic plan or a law on paths.

Proposition 1.8 (Existence on compact spaces)**.** *If $\mathcal{X}$ and $\mathcal{Y}$ are compact metric spaces and c is continuous, then the infimum in* (1.8) *is attained.*

Proof. The set $\mathcal{U}(\alpha,\beta)$ is tight and closed for weak convergence, hence compact. The functional $\pi \mapsto \int c\,\mathrm{d}\pi$ is continuous on this compact set. □

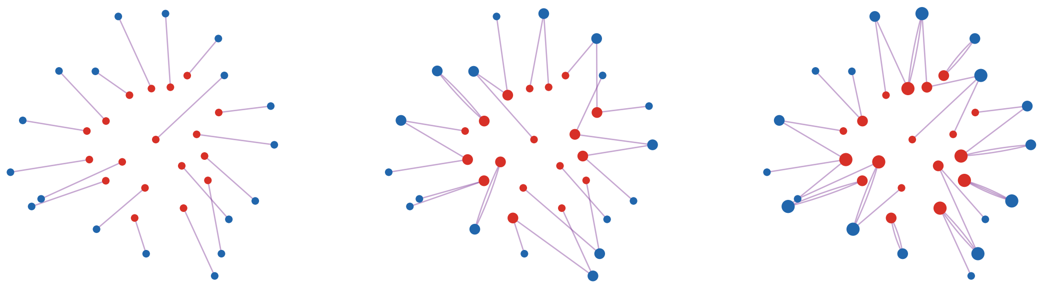
1.3.2 Discrete Kantorovich Problem

For discrete measures $\alpha = \sum_i a_i \delta_{x_i}$ and $\beta = \sum_j b_j \delta_{y_j}$, a coupling is a nonnegative matrix with prescribed row and column sums.

Definition 1.9 (Discrete couplings and mass conservation)**.** For weights $\mathrm{a} \in \Sigma_n$ and $\mathrm{b} \in \Sigma_m$,

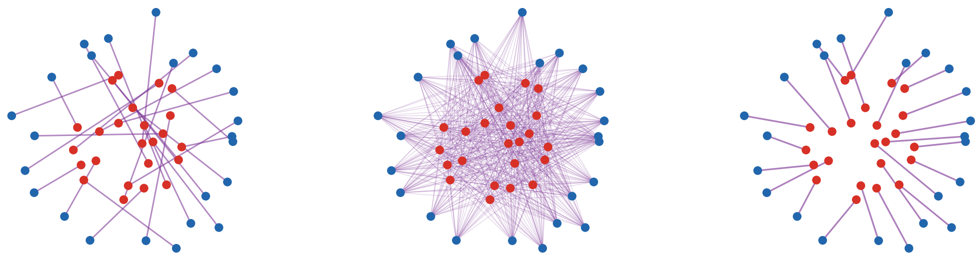
$$\mathrm{U}(\mathrm{a},\mathrm{b}) := \left\{\mathrm{P} \in \mathbb{R}_+^{n\times m}\ ;\ \mathrm{P}\mathbb{1}_m = \mathrm{a},\ \mathrm{P}^\top\mathbb{1}_n = \mathrm{b}\right\}. \tag{1.9}$$

Definition 1.10 (Discrete product coupling)**.** The product, or trivial, coupling is $(\mathrm{a}\otimes\mathrm{b})_{ij} = a_i b_j$.



uniform multiplicities | binary multiplicities | ternary multiplicities

Figure 1.8: Rational weights as duplicated uniform matchings. The red and blue locations are displayed once, while disk areas encode integer multiplicities. After duplication, the uniform assignment may contain several matched copies of the same displayed point; collapsing these copies gives the integer count matrix underlying a rational-weight Kantorovich plan.



deterministic graph | product plan $\alpha \otimes \beta$ | optimal plan

Figure 1.9: Discrete couplings represented as straight transport segments. A deterministic graph coupling sends each source to one destination, the product plan spreads every source mass over all targets, and the optimal Kantorovich plan minimizes the quadratic cost while being allowed to split mass. Line width and opacity encode the transported mass.

Given $C_{ij} = c(x_i, y_j)$, the discrete Kantorovich value is the linear program

$$\mathrm{L_C}(\mathrm{a}, \mathrm{b}) := \min_{\mathrm{P} \in \mathrm{U}(\mathrm{a},\mathrm{b})} \sum_{i,j} C_{ij} P_{ij}. \tag{1.10}$$

When $m = n$ and $\mathrm{a} = \mathrm{b} = \mathbb{1}_n/n$, the Birkhoff–von Neumann theorem implies that this relaxation admits an optimal permutation plan, so it is tight for the balanced assignment problem. For nonuniform weights or unequal cardinalities, mass splitting is essential.

The Birkhoff theorem is also a useful mental model for linear programming over couplings: a bistochastic matrix can be moved along cycles in its positive-support bipartite graph until a permutation component is exposed. Figure 1.10 displays this cycle mechanism in the finite case.

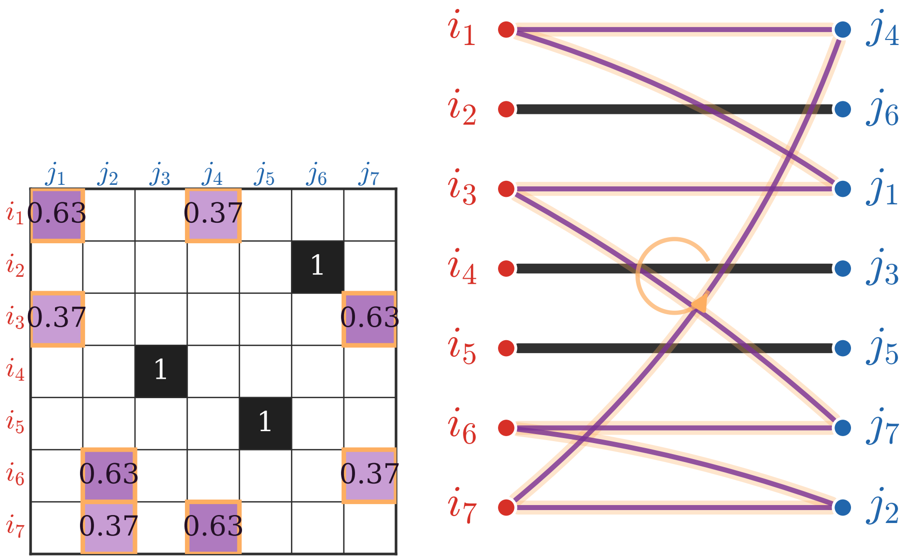
Two complementary pictures of the discrete relaxation are useful. Figure 1.11 keeps the physical segments visible, while Figure 1.12 shows the same idea as a matrix with prescribed row and column sums.

1.4 Brenier Maps and Quadratic OT

The quadratic Euclidean cost is the central case for Wasserstein PDEs. Brenier's theorem says that the optimal coupling is induced by a gradient of a convex potential whenever the source is sufficiently diffuse.

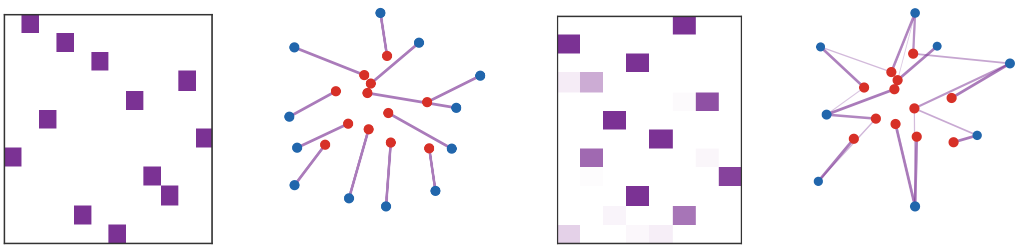
Theorem 1.11 (Brenier). *Let $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d)$ and assume that α is absolutely continuous with respect to Lebesgue measure. For the cost $c(x, y) = \|x - y\|^2$, there exists a convex function φ such that*

$$T = \nabla\varphi, \qquad T_\sharp \alpha = \beta,$$



bistochastic matrix support graph and cycle

Figure 1.10: Cycle certificate in the Birkhoff–von Neumann proof. The matrix is bistochastic but not a permutation matrix. Its positive support defines a bipartite graph; a fractional cycle permits an alternating add-subtract perturbation that preserves row and column sums and pushes at least one entry to zero. Repeating this exposes permutation matrices as extreme points.



permutation matrix assignment segments splitting matrix splitting segments

Figure 1.11: From permutation matrices to splitting couplings. When two empirical measures have the same number of atoms and uniform weights, an optimal plan can be a permutation matrix. Once target masses are nonuniform, admissible couplings may split one source among several targets or merge several sources into one target.

and T is the unique optimal Monge map α-almost everywhere. The associated optimal Kantorovich plan is unique and equals $(\mathrm{Id}, T)_\sharp\alpha$.

Thus quadratic OT is not merely a linear program over couplings; under absolute continuity it is a nonlinear elliptic problem for a convex potential. When $\alpha = \rho_\alpha \mathrm{d}x$ and $\beta = \rho_\beta \mathrm{d}y$ are smooth positive densities and $T = \nabla\varphi$ is smooth, the change-of-variables formula gives the Monge–Ampère equation

$$\rho_\alpha(x) = \rho_\beta(\nabla\varphi(x))\det(\nabla^2\varphi(x)). \tag{1.11}$$

Figure 1.13 should be read as a geometric stress test rather than a regularity theorem. It displays the particle interpolation induced by a quadratic assignment from a convex disk to a connected nonconvex target, making visible why regularity assumptions on the target geometry matter.

The Gaussian case is the main closed-form model reused later for Gaussian closures of flows.

Proposition 1.12 (Gaussian $\mathcal{W}_2$ formula and Bures covariance term)**.** *Let $\alpha = \mathcal{N}(m_\alpha, \Sigma_\alpha)$ and $\beta = \mathcal{N}(m_\beta, \Sigma_\beta)$ with positive definite covariance matrices. Then the quadratic optimal map is affine,*

$$T(x) = m_\beta + A(x - m_\alpha), \qquad A = \Sigma_\alpha^{-1/2}(\Sigma_\alpha^{1/2}\Sigma_\beta\Sigma_\alpha^{1/2})^{1/2}\Sigma_\alpha^{-1/2},$$

and

$$\mathcal{W}_2^2(\alpha, \beta) = \|m_\alpha - m_\beta\|^2 + \mathrm{tr}\left(\Sigma_\alpha + \Sigma_\beta - 2(\Sigma_\alpha^{1/2}\Sigma_\beta\Sigma_\alpha^{1/2})^{1/2}\right). \tag{1.12}$$

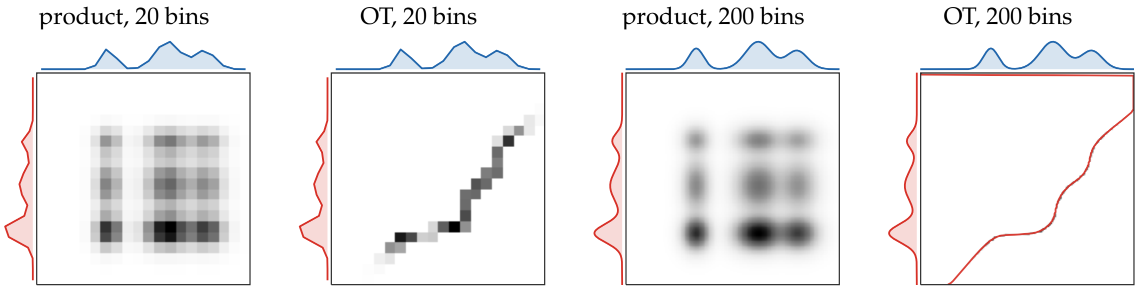


Figure 1.12: Coupling matrices with their prescribed marginals. The central grayscale image displays P_{ij}; the red curve on the left is the source marginal a, and the blue curve on top is the target marginal b. The independent product plan is diffuse, whereas the one-dimensional optimal plan concentrates near the monotone quantile correspondence.

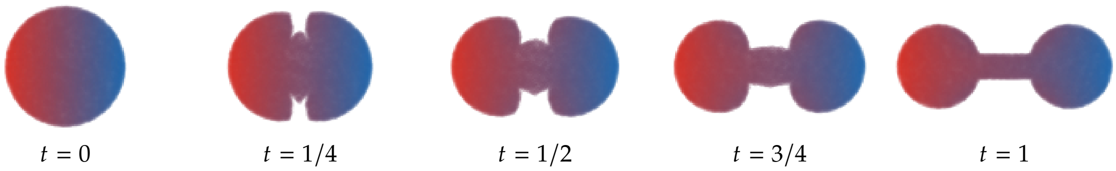


Figure 1.13: Empirical quadratic OT interpolation from a disk to a connected nonconvex two-disk domain. A dense farthest-point sample of the disk is matched to a dense farthest-point sample of two disks connected by a thin rectangle. The panels display $(1-t)x_i + tT_N(x_i)$ for the empirical optimal assignment T_N; colors are inherited from the horizontal coordinate of x_i in the initial disk, so transported material regions can be tracked.

Proof. The affine map above sends the first Gaussian to the second and has symmetric positive definite linear part. It is therefore the gradient of a convex quadratic potential, so Brenier's theorem gives optimality. Substituting the affine displacement in the quadratic transport cost gives the formula. □

Figure 1.14 turns the closed form into three complementary pictures: densities in one dimension, covariance ellipses in two dimensions, and a curve inside the cone of positive semidefinite matrices.

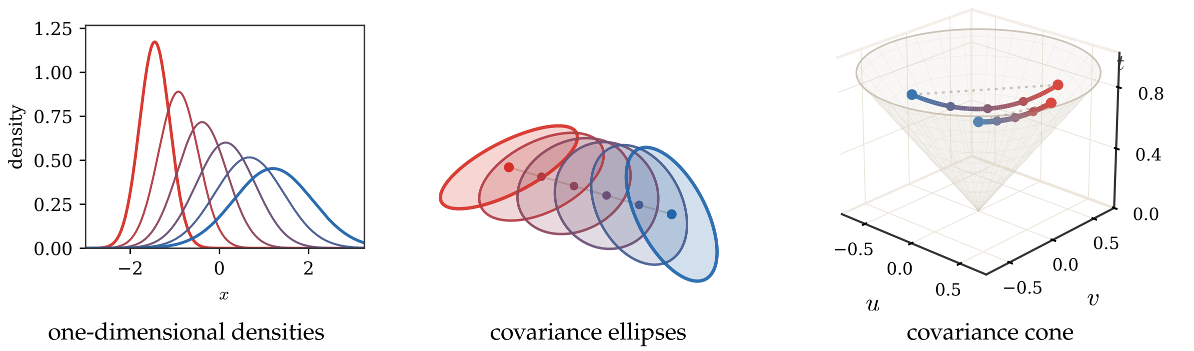


Figure 1.14: Gaussian $\mathcal{W}_2$ geodesics. In one dimension, the mean and standard deviation interpolate linearly. In two dimensions, the affine Brenier map transports covariance ellipses along the Bures covariance geometry appearing in (1.12). The cone panel represents 2×2 positive semidefinite covariances and shows that covariance interpolation is not generally Euclidean in matrix entries.

1.5 Wasserstein Distances and Geodesics

The metric structure used in the PDE sections is obtained by taking the p-th root of the Kantorovich cost for a ground metric.

Definition 1.13 (Wasserstein distance)**.** Let $(\mathcal{X}, d)$ be a metric space and $p \geqslant 1$. For $\alpha, \beta \in \mathcal{P}_p(\mathcal{X})$,

$$\mathcal{W}_p(\alpha, \beta) := \left(\inf_{\pi \in \mathcal{U}(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{X}} d(x,y)^p \, \mathrm{d}\pi(x,y) \right)^{1/p}. \tag{1.13}$$

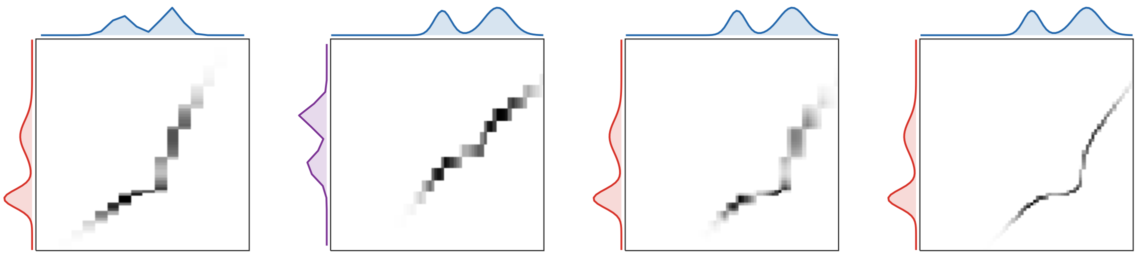
Proposition 1.14 (Metric property of the Wasserstein distance)**.** *If $(\mathcal{X}, d)$ is a metric space, then $\mathcal{W}_p$ is a distance on $\mathcal{P}_p(\mathcal{X})$.*

Proof. Nonnegativity and symmetry are immediate. If $\mathcal{W}_p(\alpha, \beta) = 0$, the optimal mass can only lie on the diagonal, hence $\alpha = \beta$. The triangle inequality follows from the gluing lemma: glue an optimal coupling between α and β with one between β and γ to obtain a joint law of (X, Y, Z); then apply Minkowski's inequality to $d(X, Z) \leqslant d(X, Y) + d(Y, Z)$. □

The gluing step is abstract but elementary in the discrete case. If $P \in \mathrm{U}(\mathrm{a}, \mathrm{b})$ and $Q \in \mathrm{U}(\mathrm{b}, \mathrm{c})$, then

$$R = P \,\mathrm{diag}(\mathrm{b})^{-1} Q$$

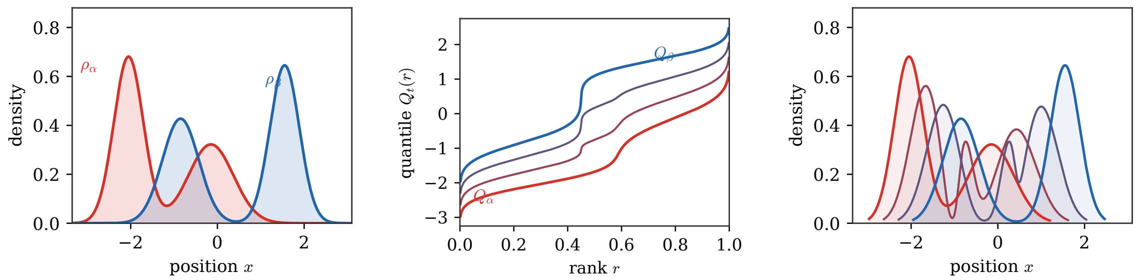
is a feasible coupling between a and c whenever all entries of b are positive. Figure 1.15 visualizes this composition of plans through an intermediate marginal.



a to b b to c glued a to c direct a to c

Figure 1.15: Discrete gluing lemma in matrix form. The first two coupling matrices share the intermediate marginal b. Multiplying through this common marginal gives a feasible glued plan between a and c, which is the coupling used in the triangle-inequality proof. The last panel shows the direct optimal coupling for comparison.

In one dimension, the geodesic can be read directly in quantile coordinates. Figure 1.16 is the static prototype for the Lagrangian views of one-dimensional PDEs used later in the dynamic and gradient-flow sections.



endpoint densities quantile interpolation geodesic densities

Figure 1.16: One-dimensional Wasserstein geodesic by quantiles. The optimal coupling matches equal quantile levels, so the interpolation is linear in inverse-CDF coordinates: $Q_t = (1-t)Q_0 + tQ_1$. The right panel shows the corresponding density path.

Interpolation induced by an optimal plan. For $\mathcal{X} = \mathbb{R}^d$ and $p = 2$, an optimal plan also defines a constant-speed geodesic. If π is optimal between α_0 and α_1, set

Definition 1.15 ($\mathcal{W}_2$ geodesic induced by an optimal plan)**.**

$$\alpha_t = ((1-t)P_0 + tP_1)_\sharp \pi, \qquad P_0(x,y) = x, \quad P_1(x,y) = y. \tag{1.14}$$

When $\pi = (\mathrm{Id}, T)_\sharp \alpha_0$ is induced by a Brenier map, this reduces to McCann's displacement interpolation

$$\alpha_t = ((1-t)\mathrm{Id} + tT)_\sharp \alpha_0.$$

Proposition 1.16 (Optimal-plan interpolation is a $\mathcal{W}_2$ geodesic)**.** *Let π be an optimal coupling between $\alpha_0, \alpha_1 \in \mathcal{P}_2(\mathbb{R}^d)$ and define α_t by (1.14). Then*

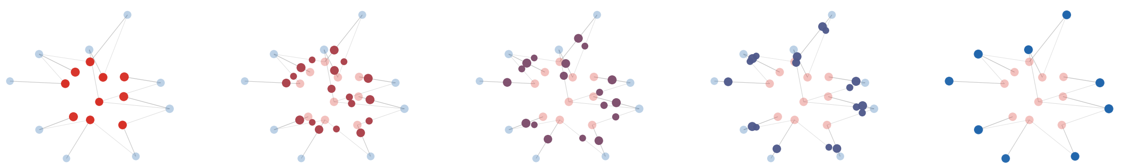
$$\mathcal{W}_2(\alpha_s, \alpha_t) = |t-s|\, \mathcal{W}_2(\alpha_0, \alpha_1), \qquad 0 \leqslant s, t \leqslant 1.$$

Proof. Coupling the particles at times s and t through the same endpoint pair $(x,y) \sim \pi$ gives

$$\mathcal{W}_2^2(\alpha_s, \alpha_t) \leqslant (t-s)^2 \int \|x-y\|^2 \,\mathrm{d}\pi = (t-s)^2\, \mathcal{W}_2^2(\alpha_0, \alpha_1).$$

The reverse inequality follows from the triangle inequality applied to the three pieces $[0,s]$, $[s,t]$ and $[t,1]$. □

The last two primer figures should be read together. Figure 1.17 illustrates the definition when the plan is a genuine coupling rather than a single-valued map: each active pair (x,y) follows a straight segment. Figure 1.18 then shows the Monge/Brenier special case on two silhouettes.



$t = 0$ $\quad$ $t = 1/4$ $\quad$ $t = 1/2$ $\quad$ $t = 3/4$ $\quad$ $t = 1$

Figure 1.17: Interpolation induced by a Kantorovich plan. Each nonzero entry of an optimal discrete coupling transports mass along the segment $(1-t)x_i + ty_j$. This construction is the static plan-level ancestor of the dynamic Benamou–Brenier viewpoint developed in Section 2.

This geodesic viewpoint is the static counterpart of the Benamou–Brenier formulation in Section 2. It is also the geometry behind the Wasserstein gradient flows of Section 3.

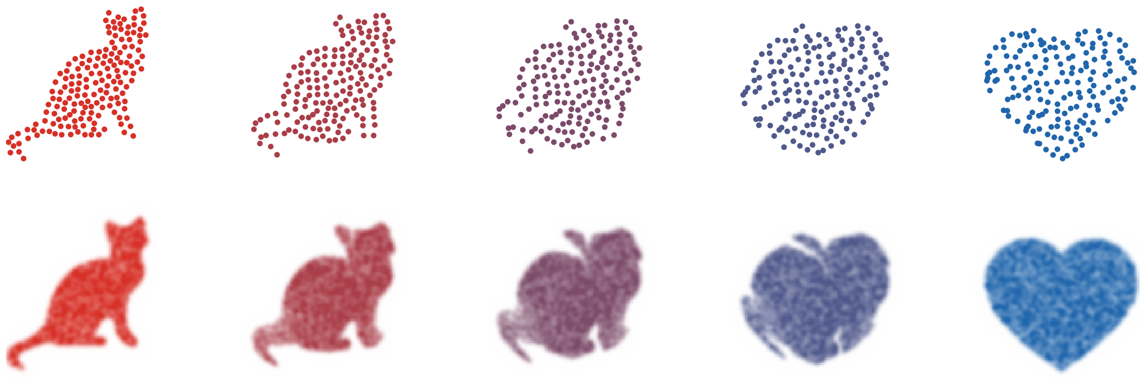


Figure 1.18: McCann displacement interpolation between two silhouette measures. The first row displays transported particles along $T_t(x) = (1-t)x + tT(x)$; the second row renders kernel-smoothed densities from a denser transported cloud. This is the map-induced version of the plan interpolation above and the geometric path used later by Wasserstein gradient-flow arguments.

SECTION 2

Dynamic Optimal Transport

Optimal transport becomes especially powerful once distances between measures are seen as actions of moving mass. This section first develops the dynamic language: continuity equations describe admissible measure evolutions, while the Benamou–Brenier formula identifies $\mathcal{W}_2$ with a least-action principle. These ideas prepare the gradient-flow and generative-model sections that follow.

2.1 Evolutions over the Space of Measures

We start with the continuity equation because it is the common language for particles, densities and weak measure evolutions. It also makes precise which velocity fields actually move mass.

Lagrangian and Eulerian descriptions. We consider the evolution $t \mapsto \alpha_t \in \mathcal{P}(\mathbb{R}^d)$. Such an evolution can be described in a "Lagrangian" way as the advection of particles along a (time-dependent) vector field $v_t(x)$ in $\mathbb{R}^d$. At the particle level, this advection is governed by

$$\frac{\mathrm{d}x(t)}{\mathrm{d}t} = v_t(x(t)), \tag{2.1}$$

and we write T_t for the associated flow map, so that $T_t(x(0)) = x(t)$. The advected measure is then $\alpha_t = (T_t)_\sharp \alpha_0$. For discrete measures, $\alpha_t = \frac{1}{n}\sum_{i=1}^n \delta_{x_i(t)}$, meaning each $x_i(t)$ solves (2.1).

In the Eulerian description, the same motion is written directly on the evolving measure: the particle ODE becomes the PDE

$$\frac{\partial \alpha_t}{\partial t} + \mathrm{div}(v_t \alpha_t) = 0. \tag{2.2}$$

This PDE is often referred to as the advection equation, the continuity equation, or Liouville's equation when operating over a phase space. It is only a classical PDE when α_t has a smooth density. For general measures, and in particular for empirical measures, it is understood in the weak sense: for any smooth test function $(t, x) \mapsto \varphi(t, x)$ compactly supported in time,

$$\int_0^1 \int_{\mathbb{R}^d} \left(\partial_t \varphi(t,x) + \langle v_t(x), \nabla_x \varphi(t,x)\rangle\right) \mathrm{d}\alpha_t(x)\mathrm{d}t = 0. \tag{2.3}$$

This equation is obtained from (2.2) by integration by parts. Hence, for smooth positive densities, the classical and weak formulations are equivalent; the weak formulation is useful because it still makes sense for discrete measures whose particles evolve according to (2.1).

Proposition 2.1 (Lagrangian flows solve the continuity equation)**.** *Consider a smooth flow* $T_t : \mathbb{R}^d \to \mathbb{R}^d$ *and define* $\alpha_t = (T_t)_\sharp \alpha_0$*. Define the Eulerian velocity field by*

$$v_t(T_t(y)) = \partial_t T_t(y).$$

Then (α_t, v_t) *solves the continuity equation in the weak sense* (2.3)*. In particular, if* $\alpha_0 = \frac{1}{n}\sum_i \delta_{x_i(0)}$ *is empirical, then* $\alpha_t = \frac{1}{n}\sum_i \delta_{x_i(t)}$ *is empirical as well, with particle velocities* $\dot{x}_i(t) = v_t(x_i(t))$*.*

Proof. Let $\varphi(t, x)$ be a smooth test function vanishing at $t = 0$ and $t = 1$. Since $\alpha_t = (T_t)_\sharp \alpha_0$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(t, x)\mathrm{d}\alpha_t(x) = \frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(t, T_t(y))\mathrm{d}\alpha_0(y).$$

The chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(t, T_t(y))\mathrm{d}\alpha_0(y) = \int \left(\partial_t\varphi(t, T_t(y)) + \langle \nabla_x \varphi(t, T_t(y)), \partial_t T_t(y)\rangle\right)\mathrm{d}\alpha_0(y).$$

Using the definition of v_t and the push-forward relation, this equals

$$\int \left(\partial_t\varphi(t, x) + \langle \nabla_x \varphi(t, x), v_t(x)\rangle\right)\mathrm{d}\alpha_t(x).$$

Integrating in time and using the boundary values of φ gives (2.3). □

From measure evolutions to vector fields. For a given evolution $(\alpha_t)_t$, there are typically infinitely many velocity fields v_t satisfying

$$\partial_t \alpha_t + \mathrm{div}(\alpha_t v_t) = 0. \tag{2.4}$$

This non-uniqueness comes from the kernel of the weighted divergence. The linear space of vector fields that leave a measure α invariant is

$$\mathcal{H}_\alpha = \{v \,;\, \mathrm{div}(\alpha v) = 0\}\,.$$

It is usually non-trivial: for instance, if α is an isotropic Gaussian, $\mathcal{H}_\alpha$ contains rotational vector fields generated by anti-symmetric matrices.

Dacorogna–Moser inversion. Reconstructing particles from an observed density evolution is therefore ill-posed. A simple choice, introduced by Dacorogna and Moser [27], is to impose that the flux $\alpha_t v_t$ is a gradient field, leading formally to

$$v_t = -\frac{1}{\alpha_t}\nabla\Delta^{-1}(\partial_t \alpha_t), \tag{2.5}$$

with suitable boundary conditions, for instance vanishing at infinity. This formula is useful conceptually but delicate when α_t vanishes, and it does not generally produce a gradient velocity field.

Least-square inversion and gradient structure. A more robust choice, used implicitly in flow matching, optimal transport and Wasserstein gradient flows, is to select among all admissible velocities the one with smallest kinetic energy:

$$\min_v \frac{1}{2}\int_0^1\int_{\mathbb{R}^d} \|v_t(x)\|^2\,\mathrm{d}\alpha_t(x)\mathrm{d}t \quad \text{subject to} \quad \partial_t\alpha_t + \mathrm{div}(\alpha_t v_t) = 0. \tag{2.6}$$

Proposition 2.2 (Least-square velocities are gradients)**.** *Assume that* $\alpha_t = \rho_t\,\mathrm{d}x$ *is a smooth positive density curve and that boundary terms vanish. The minimizer of* (2.6)*, if it exists, is a gradient field*

$$v_t = \nabla\varphi_t,$$

where φ_t*, unique up to an additive constant, solves the weighted Poisson equation*

$$-\mathrm{div}(\rho_t \nabla\varphi_t) = \partial_t\rho_t, \qquad v_t = -\nabla\Delta_{\alpha_t}^{-1}(\partial_t\alpha_t), \quad \Delta_{\alpha_t}\varphi = \mathrm{div}(\alpha_t\nabla\varphi). \tag{2.7}$$

Proof. Introduce a Lagrange multiplier φ_t for the continuity equation. The constrained problem has the formal saddle formulation

$$\min_{v}\max_{\varphi}\int_0^1\left[\frac{1}{2}\int_{\mathbb{R}^d}\|v_t(x)\|^2\,\mathrm{d}\alpha_t(x)+\int_{\mathbb{R}^d}\varphi_t(x)\big(\operatorname{div}(\alpha_t v_t)(x)+\partial_t\alpha_t(x)\big)\mathrm{d}x\right]\mathrm{d}t.$$

Integrating by parts in the divergence term gives, for each t,

$$\int\left(\frac{1}{2}\|v_t\|^2-\langle\nabla\varphi_t,\,v_t\rangle\right)\mathrm{d}\alpha_t+\int\varphi_t\,\partial_t\alpha_t.$$

The pointwise minimizer in v_t is therefore $v_t=\nabla\varphi_t$. Substituting this into the constraint $\partial_t\rho_t+\operatorname{div}(\rho_t v_t)=0$ gives the weighted Poisson equation in (2.7). The inverse notation is just a shorthand for solving this equation on zero-mean right-hand sides, modulo additive constants. □

In general this inversion is still computationally demanding, but special choices of $(\alpha_t)_t$ lead to simpler formulas; this is the mechanism exploited later by flow matching.

Algorithm 2.1 Least-square velocity reconstruction

Input: Smooth positive density curve $(\rho_t)_{t\in[0,1]}$ and boundary conditions.
Output: Minimal-energy velocity field v_t realizing the curve.
For each time t **do**:
  Compute $\partial_t\rho_t$.
  Solve weighted Poisson equation: $-\operatorname{div}(\rho_t\nabla\varphi_t)=\partial_t\rho_t,\qquad\int\varphi_t\rho_t=0.$
  Set $v_t=\nabla\varphi_t$.
Return $(v_t)_{t\in[0,1]}$.

2.2 Benamou–Brenier dynamic formulation of OT

The dynamic formulation identifies $\mathcal{W}_2$ with the kinetic energy of the cheapest continuity-equation path. It is the point where OT becomes a least-action principle.

Instead of assuming that a whole curve $(\alpha_t)_{t\in[0,1]}$ is prescribed, one only fixes its endpoints α_0 and α_1 and minimizes the least-square energy (2.6). The theorem of Benamou and Brenier states that this geodesic energy is exactly the squared Wasserstein distance [4].

Theorem 2.3 (Benamou–Brenier). *For probability measures $\alpha_0,\alpha_1\in\mathcal{P}_2(\mathbb{R}^d)$,*

$$\mathcal{W}_2^2(\alpha_0,\alpha_1)=\inf_{(\alpha_t,v_t)}\int_0^1\int_{\mathbb{R}^d}\|v_t(x)\|^2\,\mathrm{d}\alpha_t(x)\mathrm{d}t,\tag{2.8}$$

where the infimum is over (α_t,v_t) solving $\partial_t\alpha_t+\nabla\cdot(\alpha_t v_t)=0$ with $\alpha_{t=0}=\alpha_0$ and $\alpha_{t=1}=\alpha_1$. If α_0 has a density and T is the optimal Monge map $T_\sharp\alpha_0=\alpha_1$, the minimizer is

$$\alpha_t=((1-t)\mathrm{Id}+tT)_\sharp\alpha_0,\qquad v_t\big((1-t)x+tT(x)\big)=T(x)-x.\tag{2.9}$$

Proof. For the inequality "dynamic ⩽ static", assume first that a Monge map T exists and define (α_t,v_t) by (2.9). Since the Lagrangian velocity $T(x)-x$ is independent of t,

$$\int_0^1\int\|v_t\|^2\,\mathrm{d}\alpha_t\mathrm{d}t=\int\|T(x)-x\|^2\,\mathrm{d}\alpha_0(x),$$

so the dynamic cost is no larger than the static Monge cost. Without a Monge map, the same construction is made with an optimal coupling π: sample $(X,Y)\sim\pi$ and move along the straight path $\gamma_{X,Y}(t)=(1-t)X+tY$. This path measure has action $\int\|x-y\|^2\mathrm{d}\pi(x,y)$; projecting path velocities onto their conditional mean at time t gives an admissible Eulerian velocity with no larger action, so the dynamic value is no larger than the Kantorovich value.

Conversely, for a smooth deterministic path, take the flow T_t defined by $\dot{T}_t = v_t \circ T_t$ and $T_0 = \mathrm{Id}$. Then $\alpha_t = (T_t)_\sharp \alpha_0$ and $(T_1)_\sharp \alpha_0 = \alpha_1$. Jensen's inequality gives

$$\|T_1(x) - x\|^2 \leqslant \int_0^1 \|v_t(T_t(x))\|^2 \mathrm{d}t.$$

After integration with respect to α_0, the Monge cost is bounded above by the dynamic action. For general finite-energy solutions of the continuity equation, the superposition principle lifts the curve to a probability measure on absolutely continuous paths; applying Jensen's inequality pathwise gives a coupling of the endpoints whose quadratic cost is no larger than the action. Thus the Kantorovich value is bounded above by the dynamic value. □

2.2.1 Convex moment-based reformulation

Although (2.8) is not jointly convex in (α_t, v_t), it becomes convex after replacing velocities by the momentum measure $m_t = v_t \alpha_t$ and using the perspective action. In the absolutely continuous case $\alpha_t = \rho_t \, \mathrm{d}x$ and $m_t(x) = \rho_t(x) v_t(x)$, this reads

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \inf_{\substack{\partial_t \rho_t + \operatorname{div} m_t = 0 \\ \rho_{t=0}\mathrm{d}x = \alpha_0,\ \rho_{t=1}\mathrm{d}x = \alpha_1}} \int_0^1 \int_{\mathbb{R}^d} \frac{\|m_t(x)\|^2}{\rho_t(x)} \, \mathrm{d}x \, \mathrm{d}t, \tag{2.10}$$

with the usual convention that the integrand is 0 when $(\rho_t, m_t) = (0, 0)$ and $+\infty$ when $\rho_t = 0$ but $m_t \neq 0$. For singular endpoints or curves, the same statement is interpreted with vector-valued momentum measures and the corresponding recession convention. This convex reformulation enables geodesic interpolation by convex optimization once the domain is discretized.

Dual Hamilton–Jacobi formulation. The momentum formulation also has a useful dual. It turns the least-action problem into a Hamilton–Jacobi subsolution inequality for a scalar potential, with equality on the part of space-time actually visited by the optimal curve. This is the dual dynamic formulation already present in the original Benamou–Brenier perspective [4]; see also the standard accounts [98, 84]. The constants below follow the convention of (2.10), where the kinetic energy has no factor $1/2$.

Proposition 2.4 (Dual Benamou–Brenier problem)**.** *Assume, for simplicity, that the densities are smooth, compactly supported, and that boundary terms vanish. Then the convex dynamic value* (2.10) *has the dual formulation*

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \sup_{\varphi} \left\{ \int_{\mathbb{R}^d} \varphi_1 \, \mathrm{d}\alpha_1 - \int_{\mathbb{R}^d} \varphi_0 \, \mathrm{d}\alpha_0 \, ; \, \partial_t \varphi_t + \frac{1}{4} \|\nabla \varphi_t\|^2 \leqslant 0 \right\}, \tag{2.11}$$

where the supremum is over smooth test potentials $\varphi : [0,1] \times \mathbb{R}^d \to \mathbb{R}$. In the general metric-measure statement, the same formula is understood after relaxation: φ is a Hamilton–Jacobi subsolution, for instance in the viscosity sense, and finite-dimensional discretizations follow directly from Fenchel–Rockafellar duality.

If (ρ, m) and φ are smooth primal and dual optimizers, then

$$m_t = \frac{\rho_t}{2} \nabla \varphi_t, \qquad \partial_t \varphi_t + \frac{1}{4} \|\nabla \varphi_t\|^2 = 0 \quad \text{on } \{\rho_t > 0\}. \tag{2.12}$$

Equivalently, the optimal Eulerian velocity is $v_t = m_t / \rho_t = \nabla \varphi_t / 2$.

Proof. Let (ρ, m) satisfy the continuity equation and let φ be smooth. Multiplying the constraint by φ and integrating by parts gives

$$\int_{\mathbb{R}^d} \varphi_1 \, \mathrm{d}\alpha_1 - \int_{\mathbb{R}^d} \varphi_0 \, \mathrm{d}\alpha_0 = \int_0^1 \int_{\mathbb{R}^d} (\rho_t \, \partial_t \varphi_t + \langle m_t, \nabla \varphi_t \rangle) \, \mathrm{d}x \mathrm{d}t.$$

If $\partial_t \varphi_t + \|\nabla \varphi_t\|^2/4 \leqslant 0$, Young's inequality yields, pointwise,

$$\rho \, \partial_t \varphi + \langle m, \nabla \varphi \rangle \leqslant -\frac{\rho}{4} \|\nabla \varphi\|^2 + \langle m, \nabla \varphi \rangle \leqslant \frac{\|m\|^2}{\rho},$$

with the same perspective convention when $\rho = 0$. Hence the dual objective of every feasible φ is no larger than the action of every feasible primal pair.

Conversely, introduce φ as a Lagrange multiplier for $\partial_t \rho + \operatorname{div} m = 0$. After integration by parts, and up to the fixed endpoint contribution in (2.11), the pointwise minimization over (ρ, m) contains

$$\frac{\|m\|^2}{\rho} - \rho\, \partial_t \varphi - \langle m, \nabla \varphi \rangle.$$

For fixed $\rho > 0$, its minimum over m is attained at $m = \rho \nabla \varphi / 2$ and equals $-\rho(\partial_t \varphi + \|\nabla \varphi\|^2/4)$. The infimum over $\rho \geqslant 0$ is finite exactly when the Hamilton–Jacobi inequality holds, in which case the pointwise infimum is 0. Fenchel–Rockafellar duality then gives no duality gap. Equality in the two pointwise inequalities gives (2.12). □

The inequality in (2.11) is the subsolution form of the Hamilton–Jacobi equation associated with the Hamiltonian $H(p) = \|p\|^2/4$. On the transported mass, equality holds and the characteristics satisfy $\dot{x} = \nabla_p H(\nabla \varphi) = \nabla \varphi / 2$. Since this Hamiltonian is independent of x in Euclidean space, the momentum along characteristics is constant and the trajectories are straight lines. On a Riemannian manifold, the same duality uses the Hamiltonian $H_x(p) = \|p\|^2_{g_x^{-1}}/4$; the characteristics are then geodesics for the underlying metric. In this sense the Hamilton–Jacobi constraint is the dynamic counterpart of the metric, much as an eikonal equation encodes shortest paths.

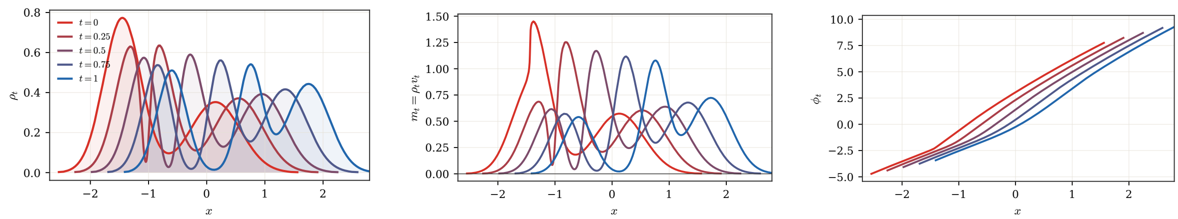
This also recovers the static Kantorovich inequality from a dynamic principle. If γ is any smooth curve with $\gamma(0) = x$ and $\gamma(1) = y$, then

$$\frac{\mathrm{d}}{\mathrm{d}t} \varphi_t(\gamma(t)) = \partial_t \varphi_t(\gamma(t)) + \langle \nabla \varphi_t(\gamma(t)), \dot{\gamma}(t) \rangle \leqslant \|\dot{\gamma}(t)\|^2,$$

where the last inequality uses $\partial_t \varphi + \|\nabla \varphi\|^2/4 \leqslant 0$ and Young's inequality. Integrating and minimizing over curves gives

$$\varphi_1(y) - \varphi_0(x) \leqslant \|x - y\|^2.$$

Thus (φ_0, φ_1) is a feasible static Kantorovich dual pair for the quadratic cost. At optimality the inequality is saturated on the endpoint pairs connected by the primal characteristics, which is the Eulerian version of complementary slackness.



density ρ_t momentum m_t dual potential φ_t

Figure 2.1: One-dimensional Benamou–Brenier primal and dual solutions. The endpoints are Gaussian mixtures and the solution is computed from monotone quantile interpolation. The left and middle panels show the primal density and momentum $m_t = \rho_t v_t$. The right panel shows the dual Hamilton–Jacobi potential, normalized up to an additive constant. Along the active transported mass, the notebook checks the primal–dual identities $m_t = \rho_t \partial_x \varphi_t / 2$ and $\partial_t \varphi_t + |\partial_x \varphi_t|^2/4 = 0$.

Proximal splitting. The convex momentum formulation also explains the original Benamou–Brenier solver. Papadakis, Peyré and Oudet [70] showed that, after discretization, the ALG2 scheme is a Douglas–Rachford splitting, equivalently ADMM on the Fenchel–Rockafellar dual. Suppressing discretization indices, write $U = (\rho, m)$ and introduce the perspective integrand

$$J(\rho, m) = \begin{cases} \|m\|^2/\rho, & \rho > 0, \\ 0, & (\rho, m) = (0, 0), \\ +\infty, & \text{otherwise.} \end{cases}$$

Let

$$\mathcal{F}(U) = \int_0^1 \int_{\mathbb{R}^d} J(\rho_t(x), m_t(x))\, \mathrm{d}x\, \mathrm{d}t, \qquad C = \{(\rho, m)\,;\, \partial_t \rho + \operatorname{div} m = 0,\ \rho_0 \mathrm{d}x = \alpha_0,\ \rho_1 \mathrm{d}x = \alpha_1\},$$

and let $\mathcal{G} = \iota_C$ be the indicator of this affine continuity constraint. The convex Benamou–Brenier problem is therefore

$$\min_U \mathcal{F}(U) + \mathcal{G}(U).$$

On the resulting Hilbert space of unknowns, or formally for an L^2-type product structure, the two proximal operators are

$$\operatorname{prox}_{\tau\mathcal{F}}(\bar{U}) = \operatorname{argmin}_U \frac{1}{2}\|U - \bar{U}\|_{\mathrm{H}}^2 + \tau\mathcal{F}(U), \qquad \operatorname{prox}_{\tau\mathcal{G}}(\bar{U}) = \operatorname{argmin}_{U\in C} \frac{1}{2}\|U - \bar{U}\|_{\mathrm{H}}^2.$$

For the standard product metric, the first map is local in (t, x): it is the proximal operator of the convex perspective J. The second map is the orthogonal projection onto the affine set defined by the divergence equation and endpoint constraints. Douglas–Rachford then alternates these two simple operations:

$$\begin{aligned} U^{k+1} &= \operatorname{prox}_{\tau\mathcal{F}}(Z^k), \\ \widetilde{U}^{k+1} &= \operatorname{prox}_{\tau\mathcal{G}}(2U^{k+1} - Z^k), \\ Z^{k+1} &= Z^k + \widetilde{U}^{k+1} - U^{k+1}. \end{aligned}$$

Equivalently one may swap the roles of $\mathcal{F}$ and $\mathcal{G}$. At convergence, the two shadow sequences U^k and $\widetilde{U}^k$ agree and give a minimizer of the convex dynamic problem. This viewpoint is useful because it separates the nonlinear but pointwise perspective proximal step from the global but linear projection onto the continuity equation [70, 31].

Algorithm 2.2 Douglas–Rachford for dynamic Benamou–Brenier

Input: Functionals $\mathcal{F}, \mathcal{G} = \iota_C$, proximal parameter $\tau > 0$, initial field Z^0.
Output: Discrete density-momentum field $U^\star$.
For $k = 0, 1, \dots$ **do**:
  $U^{k+1} = \operatorname{prox}_{\tau\mathcal{F}}(Z^k)$.
  Project reflected point: $\widetilde{U}^{k+1} = \operatorname{prox}_{\tau\mathcal{G}}(2U^{k+1} - Z^k) = \operatorname{Proj}_C(2U^{k+1} - Z^k)$.
  Update $Z^{k+1} = Z^k + \widetilde{U}^{k+1} - U^{k+1}$.
  If $\|U^{k+1} - \widetilde{U}^{k+1}\| \leqslant \mathrm{tol}$ **then**:
    Return U^{k+1}.

density sequence midpoint velocities

Figure 2.2: Benamou–Brenier geodesic between two sampled silhouettes. A discrete quadratic OT plan between finely subsampled cat and two-disks point clouds induces the McCann interpolation $Z_t = (1-t)X + tY$, which is the Lagrangian realization of the least-action solution. The left panel renders local color images of the smaller-bandwidth kernel-smoothed densities with enough padding to include the full silhouettes. The right panel overlays shortened velocity arrows centered at evenly subsampled midpoint particles $Z_{1/2}$; each displayed arrow runs in data coordinates from a source-side tail to a target-side head along the matched characteristic direction $Y - X$, but is not drawn as the full endpoint segment from X to Y.

Path-space formulation. Let $\mathcal{S} = C([0,1]; \mathbb{R}^d)$ be the space of continuous paths endowed with the uniform topology. For $t \in [0,1]$ define the evaluation map

$$P_t : \mathcal{S} \to \mathbb{R}^d, \qquad P_t(\gamma) = \gamma(t).$$

The Benamou–Brenier cost admits the equivalent formulation

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \inf_{M\in\mathcal{P}(\mathcal{S})} \left\{ \int_{\mathcal{S}} \int_0^1 \|\dot{\gamma}(t)\|^2 \mathrm{d}t \, \mathrm{d}M(\gamma) \; ; \; (P_0)_\sharp M = \alpha_0, \; (P_1)_\sharp M = \alpha_1 \right\}.$$

If α_0 has a density, the minimizer M^* is unique. Its time marginals reproduce the optimal curve: $\alpha_t = (P_t)_\sharp M^*$ for all t. Furthermore, for a.e. t, denoting $Q_t(\gamma) = \dot\gamma(t)$ on absolutely continuous paths, the conditional law of the velocity is deterministic:

$$(P_t, Q_t)_\sharp M^*(\mathrm{d}x, \mathrm{d}q) = \alpha_t(\mathrm{d}x)\delta_{v_t^*(x)}(\mathrm{d}q),$$

where v_t^* is the optimal velocity field in the Benamou–Brenier formulation. Hence M^* concentrates on straight-line geodesics and, for a.e. t, assigns exactly one direction at α_t-a.e. spatial point.

2.3 Extensions of the dynamic formulation

The same variational grammar extends beyond the quadratic Wasserstein distance. One changes either the kinetic exponent, the mobility or the balance equation, while keeping a continuity-type constraint and a convex perspective action.

Generalized Benamou–Brenier distances. The dynamic formulation is not specific to $\mathcal{W}_2$. For measures with finite p-th moments and $p > 1$, one has the analogous action formula

$$\mathcal{W}_p^p(\alpha_0, \alpha_1) = \inf_{\substack{\partial_t\alpha_t + \nabla\cdot(\alpha_t v_t) = 0\\ \alpha_{t=0}=\alpha_0,\ \alpha_{t=1}=\alpha_1}} \int_0^1 \int_{\mathbb{R}^d} |v_t(x)|^p \, \mathrm{d}\alpha_t(x)\, \mathrm{d}t.$$

When $\alpha_t = \rho_t \, \mathrm{d}x$ and $m_t = \rho_t v_t$, this becomes the convex perspective action

$$\int_0^1 \int_{\mathbb{R}^d} \frac{|m_t(x)|^p}{\rho_t(x)^{p-1}} \, \mathrm{d}x\, \mathrm{d}t, \qquad \partial_t\rho_t + \nabla\cdot m_t = 0,$$

with the usual convention that the integrand is 0 if $(\rho, m) = (0, 0)$ and $+\infty$ if $\rho = 0$ but $m \neq 0$.

A second class of variants changes the mobility of the medium: the quadratic action $|m|^2/\rho$ is replaced by $|m|^2/\theta(\rho)$ for a suitable concave mobility θ. Under appropriate structural assumptions, this produces transport metrics adapted to nonlinear diffusions and finite-volume discretizations [29]. The finite-state version, where the mobility is dictated by a reversible Markov chain, is developed in Section 2.4. These extensions keep the same variational grammar as Benamou–Brenier: a continuity-type constraint, an action density, and geodesics obtained by minimizing an integrated kinetic cost.

Dynamic unbalanced OT. Unbalanced dynamic transport is obtained by allowing mass to be created and destroyed along the path. The continuity equation is replaced by a balance equation, and the action penalizes both spatial motion and growth. This dynamic formulation underlies the Hellinger–Kantorovich and Wasserstein–Fisher–Rao metrics [54, 21]; its equivalence with static entropy-transport and cone formulations is developed in [55, 20]. A representative quadratic action is

$$\partial_t\rho_t + \nabla\cdot m_t = s_t, \qquad \int_0^1 \int \left(\frac{|m_t|^2}{\rho_t} + \kappa^2 \frac{s_t^2}{\rho_t} \right) \mathrm{d}x\, \mathrm{d}t,$$

with the same perspective convention as above. Equivalently, writing $m_t = \rho_t v_t$ and $s_t = \rho_t g_t$, one minimizes $\int_0^1 \int (|v_t|^2 + \kappa^2 g_t^2) \mathrm{d}\rho_t \mathrm{d}t$ under $\partial_t\rho_t + \nabla\cdot(\rho_t v_t) = g_t\rho_t$. The parameter κ fixes the relative cost of reaction and transport; changing it rescales the radial/angular balance in the associated cone metric. For measure-valued triples, the action is understood in the lower-semicontinuous perspective sense

$$\mathcal{A}_\kappa(\rho, m, s) := \int \left(\frac{\|\dot m\|^2}{\dot\rho} + \kappa^2 \frac{\dot s^2}{\dot\rho} \right) \mathrm{d}\lambda, \qquad (\dot\rho, \dot m, \dot s) = \left(\frac{\mathrm{d}\rho}{\mathrm{d}\lambda}, \frac{\mathrm{d}m}{\mathrm{d}\lambda}, \frac{\mathrm{d}s}{\mathrm{d}\lambda} \right),$$

where λ dominates ρ and the total variations of m and s, and the value is independent of this choice. The convention is $0/0 = 0$ and $a/0 = +\infty$ for $a > 0$, so finite action forces both the flux and source to be absolutely continuous with respect to the transported mass.

Proposition 2.5 (Static/dynamic equivalence for unbalanced OT)**.** *Fix the action above and let* CW_κ *be the static cone value of unbalanced optimal transport, with the cone metric normalized to the same growth scale* κ*. For nonnegative finite measures* α_0, α_1 *on* $\mathbb{R}^d$*, the dynamic value*

$$\mathrm{WFR}_\kappa^2(\alpha_0, \alpha_1) := \inf_{\substack{\partial_t\rho_t + \nabla\cdot m_t = s_t\\ \rho_0=\alpha_0,\ \rho_1=\alpha_1}} \int_0^1 \mathcal{A}_\kappa(\rho_t, m_t, s_t)\, \mathrm{d}t \tag{2.13}$$

equals the static cone formulation $\mathrm{CW}_\kappa(\alpha_0, \alpha_1)$. *Hence the static unbalanced problem and the balance-equation least-action problem define the same geodesic distance.*

Proof. The cone construction turns variation of mass into radial motion and spatial transport into angular motion on $\mathfrak{C}[\mathbb{R}^d]$. Applying the Benamou–Brenier theorem on the cone to the lifted endpoint measures gives a dynamic least-action problem on $\mathfrak{C}[\mathbb{R}^d]$ whose static value is the cone value CW_κ. This is the standard static/dynamic identification for the Hellinger–Kantorovich and Wasserstein–Fisher–Rao metrics [54, 55, 21, 20].

Projecting a cone curve back to the base space with the weight r^2 produces a measure curve ρ_t, a spatial flux m_t and a source term s_t satisfying the balance equation. With the matching normalization of the cone metric, the cone kinetic energy decomposes exactly into the perspective action $\mathcal{A}_\kappa$ in (2.13). Conversely, any finite-action triple (ρ_t, m_t, s_t) can be lifted to a cone curve whose radial velocity realizes the growth term and whose spatial velocity realizes the transport term, with the same action after relaxation. The two infima are therefore equal; lower semicontinuity gives the general finite-measure statement from the smooth positive case. □

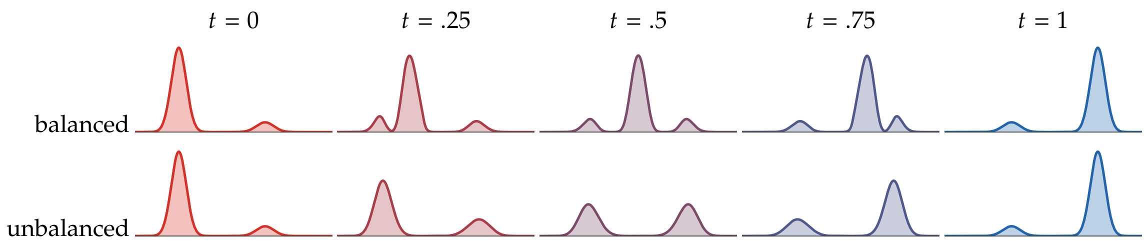


Figure 2.3: Balanced and unbalanced Sinkhorn-barycenter interpolations between two one-dimensional Gaussian mixtures with swapped modal masses. The balanced row conserves total mass, so excess mass from the dominant left mode must move along the line toward the dominant right target mode, producing transient mass in the middle. The unbalanced row uses KL-relaxed marginal constraints; mass can be attenuated near overrepresented modes and recreated near underrepresented modes, giving a reaction–transport interpolation closer to the Wasserstein–Fisher–Rao intuition.

2.4 Discrete Wasserstein Geometries on Markov Chains

The Benamou–Brenier construction also has a finite-state analogue, but not the naive one obtained by putting the Euclidean metric on the simplex. The key idea, introduced by Maas and independently developed in related forms by Mielke and by Chow–Huang–Li–Zhou, is to use the transition graph of a reversible Markov chain to define the admissible directions and the mobility of the mass [60, 64, 22]. The resulting distance turns the heat flow of the chain into the gradient flow of the discrete Shannon entropy. This is the finite-dimensional counterpart of the fact that the heat equation on $\mathbb{R}^d$ is the Wasserstein gradient flow of entropy, and it is also the geometric structure used in numerical work on discrete metric-measure spaces [33, 34].

Let $\mathcal{X} = \{1, \dots, n\}$ and let $K = (K_{ij})$ denote the off-diagonal transition rates of an irreducible continuous-time Markov chain which is reversible with respect to a probability vector π, so that $\pi_i K_{ij} = \pi_j K_{ji}$ for $i \neq j$. We write a probability $p \in \Sigma_n$ in density form $p_i = \pi_i \rho_i$, where

$$\Sigma_n := \left\{ p \in \mathbb{R}^n_+ : \sum_i p_i = 1 \right\}.$$

The entropy relative to π is

$$\mathrm{Ent}_\pi(\rho) := \sum_i \pi_i \rho_i \log \rho_i .$$

The logarithmic mean

$$\theta(a, b) := \begin{cases} \dfrac{a - b}{\log a - \log b}, & a \neq b, \\ a, & a = b, \end{cases}$$

is the specific mobility which makes the entropy gradient exactly coincide with the Markov evolution. For a potential $\psi \in \mathbb{R}^n$, define the Onsager operator

$$(\mathcal{K}_\rho \psi)_i := \sum_j K_{ij} \theta(\rho_i, \rho_j)(\psi_i - \psi_j). \tag{2.14}$$

Its associated action is

$$\mathcal{A}(\rho, \psi) := \frac{1}{2} \sum_{i,j} \pi_i K_{ij} \theta(\rho_i, \rho_j)(\psi_i - \psi_j)^2. \tag{2.15}$$

The discrete transport distance is the least action

$$\mathcal{W}_K^2(\rho^0, \rho^1) := \inf_{\rho_t, \psi_t} \int_0^1 \mathcal{A}(\rho_t, \psi_t) \, \mathrm{d}t, \qquad \dot{\rho}_t + \mathcal{K}_{\rho_t} \psi_t = 0, \tag{2.16}$$

with endpoints $\rho_0 = \rho^0$, $\rho_1 = \rho^1$. Equivalently, one can write the same formula in edge-flux variables, exactly as in a finite-volume discretization: the flux is only allowed along edges where $K_{ij} > 0$, and the denominator in the kinetic energy is the logarithmic mean of the two endpoint densities.

Although the minimizing-movement construction is introduced only in Section 3, it is useful to keep its intuition in mind before the next statement. A metric gradient flow can be read as the continuous-time limit of the implicit Euler, or JKO, step

$$\rho^{k+1} \in \operatorname{argmin}_\rho \frac{1}{2\tau} \mathcal{W}_K^2(\rho, \rho^k) + \mathrm{Ent}_\pi(\rho). \tag{2.17}$$

For the distance $\mathcal{W}_K$, the first-order optimality condition of this step gives, for small τ,

$$\frac{\rho^{k+1} - \rho^k}{\tau} = -\mathcal{K}_{\rho^k} \log \rho^k + O(\tau) = K\rho^k + O(\tau),$$

where $(K\rho)_i = \sum_j K_{ij}(\rho_j - \rho_i)$. Thus the discrete Wasserstein geometry is engineered so that the entropy gradient flow is the Markov semigroup; the proposition below makes this identity explicit.

Proposition 2.6 (Entropy gradient flow of a reversible Markov chain)**.** *Let K be reversible with invariant law π. The gradient flow of* Ent_π *for the metric $\mathcal{W}_K$ is the forward equation of the Markov chain,*

$$\dot{\rho}_i(t) = \sum_j K_{ij} \big(\rho_j(t) - \rho_i(t)\big). \tag{2.18}$$

Equivalently, for the masses $p_i(t) = \pi_i \rho_i(t)$, this is $\dot{p}_i(t) = \sum_j (p_j(t) K_{ji} - p_i(t) K_{ij})$.

Proof. The first variation of the entropy, with respect to the weighted pairing $\sum_i \pi_i \xi_i \varphi_i$, is $\log \rho_i + 1$. Constants do not contribute to $\mathcal{K}_\rho$, hence the metric gradient-flow equation is

$$\dot{\rho} = -\mathcal{K}_\rho \log \rho.$$

Using the identity

$$\theta(a, b)(\log a - \log b) = a - b,$$

one obtains, componentwise,

$$\dot{\rho}_i = -\sum_j K_{ij} \theta(\rho_i, \rho_j)(\log \rho_i - \log \rho_j) = \sum_j K_{ij}(\rho_j - \rho_i),$$

which is the density form of the Kolmogorov forward equation. Multiplying by π_i and using detailed balance gives the equation for the probability masses. □

Closed forms in dimensions two and three. For the two- and three-state examples below, take the uniform random walk on the complete neighbor graph, i.e. every pair of distinct states is declared neighboring. Thus $\pi_i = 1/n$, and $K_{ij} = 1/(n-1)$ for $i \neq j$. On Σ_2, write $p = (r, 1-r)$ and $q = (s, 1-s)$. Since there is only one edge, the distance reduces to the scalar Riemannian length

$$\mathcal{W}_K(p,q) = \left| \int_s^r \frac{\mathrm{d}u}{\sqrt{\theta(u, 1-u)}} \right|, \qquad 0 < r, s < 1. \tag{2.19}$$

This formula is closed but not Euclidean: the logarithmic mean changes the cost of moving mass depending on the current split between the two states.

On Σ_3, the complete-neighbor graph is a triangle. For $p \in \mathrm{int}(\Sigma_3)$, set

$$a_{ij}(p) := \frac{1}{2}\theta(p_i, p_j), \qquad 1 \leqslant i < j \leqslant 3.$$

For a tangent vector $u \in \mathbb{R}^3$ with $u_1 + u_2 + u_3 = 0$, orient the edges as $1 \to 2$, $1 \to 3$, $2 \to 3$. The squared norm induced by the discrete Wasserstein metric is

$$\|u\|_p^2 = \min_{q_{12}, q_{13}, q_{23}} \left\{ \frac{q_{12}^2}{a_{12}} + \frac{q_{13}^2}{a_{13}} + \frac{q_{23}^2}{a_{23}} \right\}, \tag{2.20}$$

subject to

$$u_1 + q_{12} + q_{13} = 0, \qquad u_2 - q_{12} + q_{23} = 0, \qquad u_3 - q_{13} - q_{23} = 0.$$

Eliminating the three edge fluxes gives an explicit formula. With $D = a_{12}^{-1} + a_{13}^{-1} + a_{23}^{-1}$,

$$q_{12}^* = \frac{u_2/a_{23} - u_1/a_{13}}{D}, \qquad q_{13}^* = -u_1 - q_{12}^*, \qquad q_{23}^* = q_{12}^* - u_2,$$

and $\|u\|_p^2$ is obtained by inserting these values in (2.20). Therefore

$$\mathcal{W}_K^2(p^0, p^1) = \inf_{p_t \in \mathrm{int}(\Sigma_3)} \int_0^1 \|\dot{p}_t\|_{p_t}^2 \, \mathrm{d}t, \qquad p_0 = p^0, \quad p_1 = p^1. \tag{2.21}$$

Thus the three-state distance is an explicit two-dimensional Riemannian geodesic problem on the open triangle. The formula is simple enough to compute directly, but it already shows the main difference with Euclidean geometry on the simplex: the local metric depends nonlinearly on the current density through logarithmic edge mobilities. Figure 2.4 visualizes these small-dimensional geometries and compares them with the ordinary Wasserstein distance associated with the 0/1 ground metric, for which W_2^2 is exactly the total variation distance.

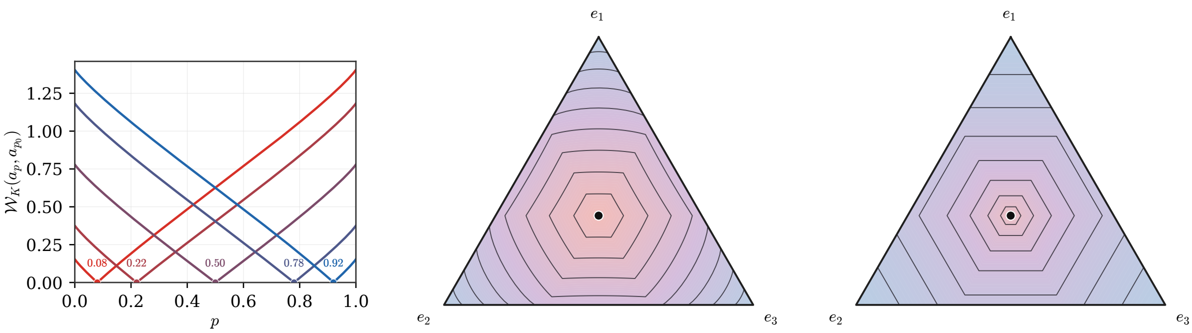


Figure 2.4: Discrete Wasserstein distances on small Markov-chain simplices. Left: the closed-form profiles $p \mapsto \mathcal{W}_K(a_p, a_{p_0})$, with $a_p = (p, 1-p)$, for several anchors p_0 on Σ_2. Middle: numerical level sets of $\mathcal{W}_K(a, \bar{a})$ on Σ_3, where $\bar{a} = (1/3, 1/3, 1/3)$, computed from the local Riemannian norm induced by the complete-neighbor Markov chain. Right: the corresponding level sets for the ordinary W_2 distance with $d(i,j) = 1$ for $i \neq j$, i.e. $W_2^2(a, \bar{a}) = \|a - \bar{a}\|_{\mathrm{TV}}$.

SECTION 3

Wasserstein Gradient Flows

Once $\mathcal{W}_2$ is a dynamic metric, one can run gradient descent directly on the space of measures. This section derives the formal Wasserstein gradient, explains the JKO minimizing-movement scheme, records the role of geodesic convexity in convergence, and then applies the same calculus to mean-field neural-network training.

3.1 Minimizing Movements and Wasserstein Gradients

This first section explains how a variational implicit-Euler step on measures gives rise, in the small-step limit, to a continuity equation driven by the Wasserstein gradient of the energy.

We now consider a function $f(\alpha)$ and seek a minimizing evolution $(\alpha_t)_t$. The general strategy of minimizing movement over a metric space is to construct a discrete-time evolution using an implicit Euler scheme:

$$\alpha_{t+\tau} := \arg\min_{\alpha} \frac{1}{2\tau}\mathcal{W}_2(\alpha_t, \alpha)^2 + f(\alpha). \tag{3.1}$$

Algorithm 3.1 JKO minimizing movement

Input: Energy f, initial measure α^0, time step $\tau > 0$, number of steps K.
Output: Discrete gradient-flow trajectory $(\alpha^k)_{k=0}^K$.
For $k = 0, \ldots, K-1$ **do**:
$\alpha^{k+1} \in \operatorname{argmin}_{\alpha\in\mathcal{P}_2(\mathbb{R}^d)} \frac{1}{2\tau}\mathcal{W}_2^2(\alpha^k, \alpha) + f(\alpha)$.
Set $\alpha_t^\tau = \alpha^k$ for $t \in [k\tau, (k+1)\tau)$.
Return $(\alpha^k)_{k=0}^K$ and α_t^τ.

Euclidean gradient flows. If we restrict (3.1) to finite dimensions and assume $\alpha_t = \delta_{x(t)}$ and $\alpha = \delta_x$ (single Dirac measures), this matches the implicit Euler scheme:

$$x(t+\tau) := \arg\min_x \frac{1}{2\tau}\|x - x(t)\|^2 + h(x),$$

where $h(x) = f(\delta_x)$. Its solution is formally given by the implicit Euler formula:

$$x(t+\tau) = (\mathrm{Id} + \tau\nabla h)^{-1}(x(t)).$$

In contrast, the explicit Euler scheme is:

$$x(t+\tau) = (\mathrm{Id} - \tau\nabla h)(x(t)) = x(t) - \tau\nabla h(x(t)).$$

Both schemes converge as $\tau \to 0$ to:

$$\dot{x}(t) = -\nabla h(x(t)). \tag{3.2}$$

Wasserstein gradient formula. The implicit Euler scheme has the advantage that it does not require h or f to be smooth. For f, this is crucial to handle evolutions over measures that may have densities, atoms or other singular parts.

As $\tau \to 0$, under certain conditions on f, (3.1) defines a continuous evolution $t \mapsto \alpha_t$. As discussed earlier, this evolution can be described as a Lagrangian evolution (2.1). We use the following first-variation convention: for any $\beta \in \mathcal{P}(\mathbb{R}^d)$ and the signed zero-mass perturbation $\rho = \beta - \alpha$,

$$f((1-\tau)\alpha + \tau\beta) = f(\alpha + \tau\rho) = f(\alpha) + \tau\int [\delta f(\alpha)](x)\mathrm{d}\rho(x) + o(\tau).$$

The key infinitesimal object is the vector field that represents this differential in the Wasserstein metric.

Definition 3.1 (Wasserstein gradient)**.** Assume that f admits a smooth first variation $\delta f(\alpha)$. In the smooth formal calculus on $\mathcal{P}_2(\mathbb{R}^d)$, the Wasserstein gradient of f at α is the gradient vector field

$$\nabla_{\mathcal{W}} f(\alpha) = \nabla_x \delta f(\alpha).$$

The associated formal gradient flow is the continuity equation

$$\frac{\partial \alpha_t}{\partial t} + \mathrm{div}(-\nabla_{\mathcal{W}} f(\alpha_t)\alpha_t) = 0. \tag{3.3}$$

The following proposition explains why this vector field is the Riemannian gradient for the $L^2(\alpha)$ metric on velocities.

Proposition 3.2 (Formal Wasserstein gradient)**.** *Assume that f admits a smooth first variation $\delta f(\alpha)$ and that α has a smooth positive density. For infinitesimal perturbations generated by a velocity field v through $(\mathrm{Id} + \tau v)_\sharp \alpha$, the differential of f is*

$$\frac{\mathrm{d}}{\mathrm{d}\tau}_{|\tau=0} f\left((\mathrm{Id} + \tau v)_\sharp \alpha\right) = \int \langle \nabla \delta f(\alpha)(x), v(x)\rangle \mathrm{d}\alpha(x).$$

Hence, for the Riemannian metric $\|v\|^2_{L^2(\alpha)} = \int \|v\|^2 \mathrm{d}\alpha$, the Wasserstein gradient is the vector field

$$\nabla_{\mathcal{W}} f(\alpha) = \nabla \delta f(\alpha).$$

Proof. The push-forward expansion gives, in the sense of distributions,

$$(\mathrm{Id} + \tau v)_\sharp \alpha = \alpha - \tau\, \mathrm{div}(\alpha v) + o(\tau).$$

Using the definition of the first variation,

$$f((\mathrm{Id} + \tau v)_\sharp \alpha) = f(\alpha) - \tau \int \delta f(\alpha)\, \mathrm{div}(\alpha v)\mathrm{d}x + o(\tau).$$

An integration by parts, with either compact support or vanishing boundary flux, gives

$$-\int \delta f(\alpha)\, \mathrm{div}(\alpha v)\mathrm{d}x = \int \langle \nabla \delta f(\alpha), v\rangle \mathrm{d}\alpha.$$

By definition of the Riesz representative for the $L^2(\alpha)$ metric, this representative is $\nabla \delta f(\alpha)$. □

The Wasserstein gradient-flow viewpoint already appears in John D. Lafferty's PhD work, published as "The Density Manifold and Configuration Space Quantization", under the name "density manifold". It was then systematically developed by Otto, who exposed the formal Riemannian structure of this space [68]. Rigorous metric-space treatments and numerical JKO schemes can be found in [2, 5, 72, 37].

From the JKO step to the velocity field. A first-order expansion of the JKO step explains why (3.3) uses the vector field $\nabla_{\mathcal{W}} f(\alpha)$. Write (3.1) as a minimization over displacement fields v such that $\alpha = (\mathrm{Id} + \tau v)_\sharp \alpha_t$:

$$\min_v \frac{1}{2\tau} \tau^2 \|v\|^2_{L^2(\alpha_t)} + f((\mathrm{Id} + \tau v)_\sharp \alpha_t).$$

Then we perform a first-order Taylor expansion of this formulation using

$$(\mathrm{Id} + \tau v)_\sharp \alpha_t = \alpha_t - \tau\, \mathrm{div}(v\alpha_t) + o(\tau)$$

$$\begin{aligned} f((\mathrm{Id} + \tau v)_\sharp \alpha_t) &= f(\alpha_t) - \tau \int \delta f(\alpha_t)\, \mathrm{div}(v\alpha_t)\mathrm{d}x + o(\tau) \\ &= f(\alpha_t) + \tau \int \langle \nabla_x \delta f(\alpha_t)(x), v(x)\rangle \mathrm{d}\alpha_t(x) + o(\tau) \end{aligned}$$

to obtain the following first-order expansion in τ of the problem minimized in (3.1)

$$\min_v f(\alpha_t) + \tau \int \left[\frac{1}{2}\|v(x)\|^2 + \langle \nabla_{\mathcal{W}} f(\alpha_t)(x), v(x)\rangle\right] \mathrm{d}\alpha_t(x) + o(\tau).$$

The pointwise minimizer is $v = -\nabla_W f(\alpha_t)$, which gives the velocity in the continuity equation. We now detail examples of such Wasserstein gradient flows.

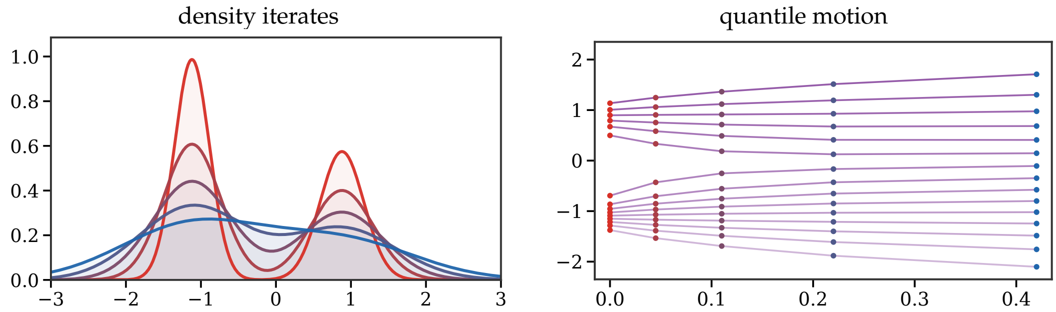


Figure 3.1: JKO minimizing movements for the entropy flow in one dimension. The left panel displays successive implicit-Euler minimizers for the heat equation, colored from red to blue. The right panel tracks inverse CDF values $Q_t(s) = F_t^{-1}(s)$ for selected probability levels s, giving a Lagrangian view of the proximal movement in Wasserstein space.

Discrete evolutions. If $f(\alpha)$ can be evaluated on discrete distributions and ∇_W is continuous in this case, the flow (3.3) maintains the number of Dirac masses, $\alpha_t = \frac{1}{n}\sum_i \delta_{x_i(t)}$. The particles $X(t) := (x_i(t))_i$ evolve according to a system of coupled ODEs:

$$\dot{x}_i(t) = -n\nabla_{x_i} F(X(t)), \tag{3.4}$$

where $F(X) := f\left(\frac{1}{n}\sum_i \delta_{x_i}\right)$ and the factor n comes from the empirical Wasserstein metric $\frac{1}{n}\sum_i \|\dot{x}_i\|^2$.

Algorithm 3.2 Empirical Wasserstein particle descent

Input: Particles $X^0 = (x_1^0, \dots, x_n^0)$, functional f, step size h, tolerance tol.
Output: Particle trajectory $(X^k)_k$ and empirical measures.
Define $F(X) = f\left(\frac{1}{n}\sum_{i=1}^n \delta_{x_i}\right)$.
For $k = 0, 1, \dots$ **do**:
  For $i = 1, \dots, n$ **do**
    $g_i^k = n\nabla_{x_i} F(X^k), \qquad x_i^{k+1} = x_i^k - h\, g_i^k.$
  If $\max_i \|x_i^{k+1} - x_i^k\| \leqslant$ tol **then**:
    Return $\frac{1}{n}\sum_i \delta_{x_i^{k+1}}$.

Linear Functionals. The first benchmark is a functional whose first variation does not depend on the current measure.

Example 3.3 (Linear potentials generate independent particles). Take a linear functional

$$f(\alpha) = \int h(x)\mathrm{d}\alpha(x). \tag{3.5}$$

Then $\delta f(\alpha) = h$ is independent of α, and the flow (3.3) becomes

$$\frac{\partial \alpha_t}{\partial t} + \mathrm{div}(-\nabla h \alpha_t) = 0.$$

This implies particles move independently according to the usual gradient flow (3.2).

Shannon Neg-Entropy. Entropy gives the opposite benchmark: the flow is no longer a deterministic push-forward of particles, but a diffusion of density.

Example 3.4 (Entropy generates heat and porous-medium flows). The canonical density-dependent functional is the Shannon neg-entropy

$$f(\alpha) = \int \log\left(\frac{\mathrm{d}\alpha}{\mathrm{d}x}(x)\right) \mathrm{d}\alpha(x). \tag{3.6}$$

Here, $\delta f(\alpha) = \log(\mathrm{d}\alpha/\mathrm{d}x)$ up to an additive constant, so $\nabla_W f(\alpha) = \nabla\alpha/\alpha$ (often called the score). The flow (3.3) becomes the heat equation

$$\partial_t \alpha_t = \Delta\alpha_t.$$

Other entropy functionals lead to nonlinear diffusion equations; finite-volume and particle discretizations are discussed in [12, 40, 60, 32]. For a generalized entropy

$$f(\alpha) = \int g\left(\frac{\mathrm{d}\alpha}{\mathrm{d}x}\right)\mathrm{d}x, \tag{3.7}$$

with a scalar convex function g, one obtains nonlinear diffusions in the smooth-density regime:

$$\frac{\partial\alpha_t}{\partial t} = \Delta(P(\alpha_t)),$$

where the pressure P satisfies $P'(s) = sg''(s)$. For example, $g(s) = s\log(s)$ gives $P(s) = s$ and recovers (3.6), while $g(s) = s^m/(m-1)$, $m > 1$, gives $P(s) = s^m$ up to an additive constant and yields the porous-medium equation.

The preceding examples are also governed by a precise geodesic-convexity criterion. A celebrated theorem by McCann [62] states that an internal energy of the form (3.7), for $g : \mathbb{R}^+ \to \mathbb{R}\cup\{+\infty\}$ with $g(0) = 0$, is geodesically convex on $\mathcal{P}(\mathbb{R}^d)$ when g is convex and the map $r \mapsto r^d g(r^{-d})$ is convex and nonincreasing on $(0, +\infty)$. Examples of such functions are $g(s) = s^q$ for $q > 1$ and Shannon entropy $g(s) = s\log(s)$. By contrast, $g(s) = -\log(s)$, associated with the reverse KL divergence, does not satisfy this displacement-convexity criterion.

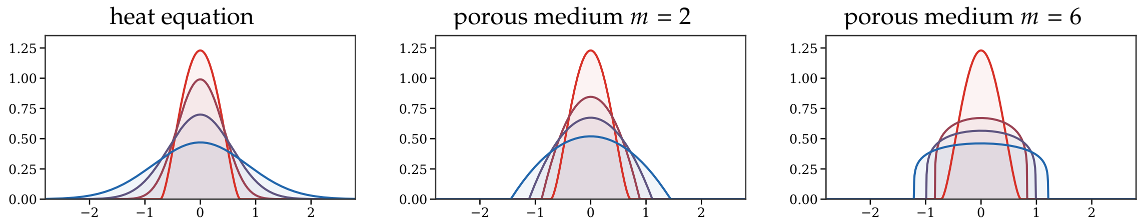


Figure 3.2: Entropy-driven Wasserstein gradient flows from the same compact initial density. The heat flow is generated by Shannon entropy $g(\rho) = \rho\log\rho$ and instantly develops Gaussian tails. The porous-medium flows use the power entropy $g(\rho) = \rho^m/(m-1)$, hence $\partial_t\rho = \Delta(\rho^m)$: the middle panel has $m = 2$, while the right panel has the stronger nonlinearity $m = 6$, i.e. $\partial_t\rho = \Delta(\rho^6)$. Larger powers diffuse mainly where the density is high, producing a flatter core and a sharper compact free boundary.

Interaction Energies. In a similar spirit, to obtain nonlinear evolutions without requiring the measure to have density, one can consider

$$f(\alpha) := \iint k(x, y)\mathrm{d}\alpha(x)\mathrm{d}\alpha(y). \tag{3.8}$$

For a symmetric kernel k:

$$\delta f(\alpha)(x) = 2\int k(x, y)\mathrm{d}\alpha(y), \quad \nabla_W f(\alpha)(x) = 2\int \nabla_x k(x, y)\mathrm{d}\alpha(y).$$

For $\alpha_0 = \frac{1}{n}\sum_i \delta_{x_i}$, the flow (3.3) implies particles $(x_i(t))_i$ obey:

$$\dot{x}_i(t) = -\frac{2}{n}\sum_j \nabla k(x_i(t), x_j(t)).$$

If k is positive definite, or more generally conditionally positive definite on signed measures of zero total mass as for the energy-distance kernel $k(x, y) = -\|x - y\|$, and one minimizes the squared kernel discrepancy to a teacher distribution β, then

$$\|\alpha - \beta\|_k^2 = \iint k\mathrm{d}\alpha\mathrm{d}\alpha - 2\int\left(\int k(x, y)\mathrm{d}\beta(y)\right)\mathrm{d}\alpha(x) + \text{constant}.$$

Thus MMD-type training energies are exactly an interaction energy plus a linear potential; the teacher distribution appears through the potential $x \mapsto -2\int k(x,y)\mathrm{d}\beta(y)$. The corresponding empirical Wasserstein gradient flow is

$$\dot{x}_i(t) = -\frac{2}{n}\sum_j \nabla_x k(x_i(t), x_j(t)) + 2\int \nabla_x k(x_i(t), y)\mathrm{d}\beta(y).$$

The corresponding simulation loop is Algorithm 3.3.

Algorithm 3.3 MMD particle flow against a teacher law

Input: Initial particles $(x_i^0)_{i=1}^n$, teacher law β or teacher samples $(y_b)_{b=1}^B$, kernel k, step size h.
Output: Particle trajectory targeting β.
For $k = 0, 1, \ldots$ **do**:
  For $i = 1, \ldots, n$ **do**
    Set self-interaction $r_i^k = -\frac{2}{n}\sum_{j=1}^n \nabla_x k(x_i^k, x_j^k)$.
    If β is available analytically **then**:
      Set teacher attraction $a_i^k = 2\int \nabla_x k(x_i^k, y)\mathrm{d}\beta(y)$.
    If only samples $(y_b)_{b=1}^B$ are available **then**:
      Set $a_i^k = \frac{2}{B}\sum_{b=1}^B \nabla_x k(x_i^k, y_b)$.
    Set velocity $v_i^k = r_i^k + a_i^k$.
    Update $x_i^{k+1} = x_i^k + h\, v_i^k$.
Return $(x_i^k)_{i,k}$.

The first term is a kernelized self-interaction, while the second is the attraction induced by the continuous teacher kernel mean. At the continuum level, characteristic positive-definite kernels, and the Euclidean energy-distance kernel on probability measures, have β as the unique minimizer of $\|\alpha - \beta\|_k^2$. For a finite number of particles, however, the flow can only form a kernelized quadrature of β, and small particle systems may cover the target modes poorly. Figure 3.3 illustrates this finite-particle effect.

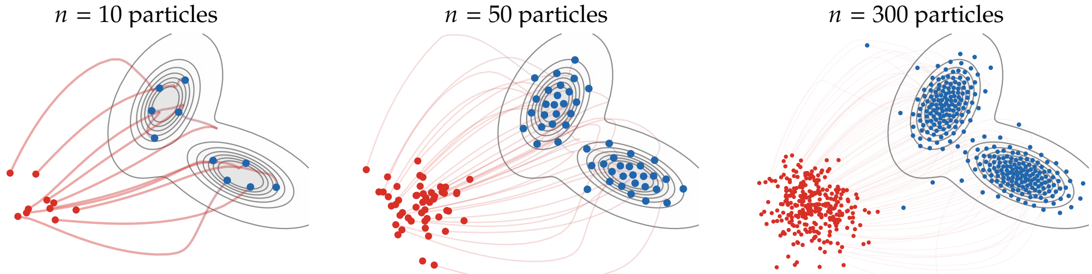


Figure 3.3: Particle count in the deterministic Wasserstein gradient flow of the squared MMD-type discrepancy to a smooth two-Gaussian teacher distribution, using here the energy-distance kernel $k(x,y) = -\|x-y\|$. The teacher itself is shown only through true density contours, while red dots are a compact shifted Gaussian initialization placed away from the target, red-to-blue curves show a thinned subset of particle trajectories, and blue dots show the stabilized long-time particles. With too few particles, the empirical measure forms a sparse kernelized quadrature and may under-cover the target modes; increasing n makes the particle cloud approximate the continuous target geometry more faithfully.

Stochastic particles and McKean–Vlasov limits. More generally, deterministic particle flows have stochastic counterparts, where Brownian noise at the particle level becomes an entropy term at the level of measures. If the drift does not depend on the empirical measure, each particle evolves independently according to

$$\mathrm{d}X_t = b(X_t)\mathrm{d}t + \sqrt{2}\sigma\mathrm{d}B_t,$$

and the one-particle law $\alpha_t = \rho_t \mathrm{d}x$ directly satisfies the linear Fokker–Planck equation

$$\partial_t \rho_t = -\operatorname{div}(b\rho_t) + \sigma^2\Delta\rho_t.$$

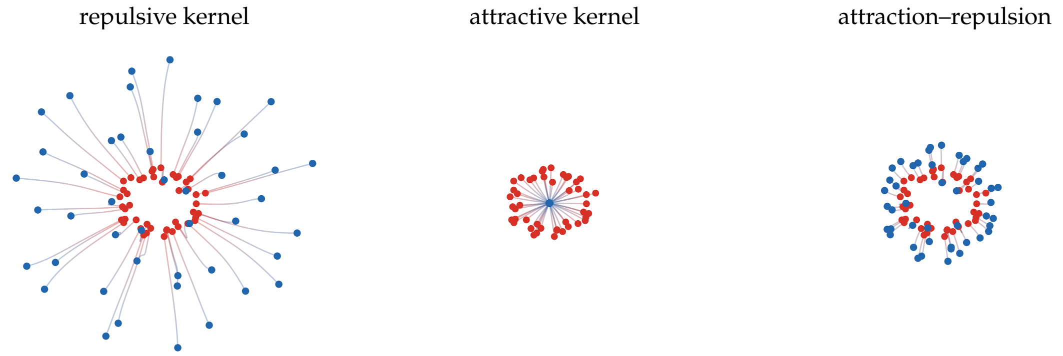


Figure 3.4: Interaction-energy particle flows for three choices of k. A positive Gaussian kernel $k(x, y) = \exp(-\|x - y\|^2/(2\sigma^2))$ produces short-range repulsion under Wasserstein descent; changing its sign produces attraction and collapse; adding a quadratic long-range attraction to the repulsive kernel yields a balanced attraction–repulsion dynamics. The curves use arclength-based red-to-blue coloring along a longer integration of the coupled particle ODE (3.4).

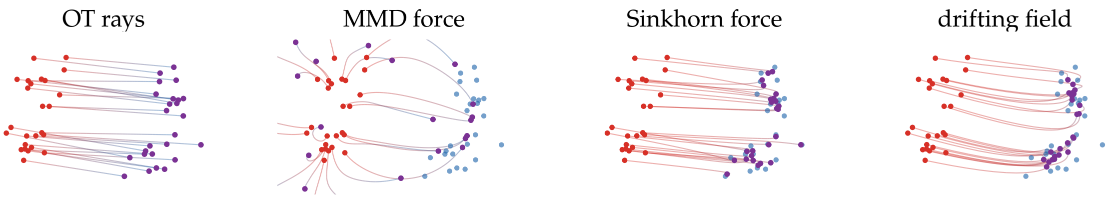


Figure 3.5: Particle trajectories induced by different discrepancy geometries. The red particles and blue target cloud are the same in all panels. Straight OT displacement produces rays from an optimal matching; an MMD-type witness field gives smoother nonlocal forces; the Sinkhorn-divergence force is an entropic, debiased transport attraction; and the normalized drifting field combines attraction to data with self-repulsion. The figure is qualitative: it compares geometric behavior, not solver performance.

Example 3.5 (Langevin drift as a free-energy flow). If $b = -\nabla V$, this linear Fokker–Planck equation is the $\mathcal{W}_2$ gradient flow of the free energy

$$\rho \mapsto \int V\rho\,\mathrm{d}x + \sigma^2 \int \rho \log \rho\,\mathrm{d}x.$$

The mean-field case is different: the drift is recomputed from the current empirical distribution of all particles,

$$\mathrm{d}X_i^n(t) = b(X_i^n(t), \mu_t^n)\mathrm{d}t + \sqrt{2}\sigma\mathrm{d}B_i(t), \qquad \mu_t^n = \frac{1}{n}\sum_{i=1}^n \delta_{X_i^n(t)}.$$

For finite n, the empirical law μ_t^n is itself random. Under suitable Lipschitz, growth and chaotic-initialization assumptions, propagation of chaos states that finitely many particles become asymptotically independent as $n \to \infty$, all with the same deterministic law $\rho_t\mathrm{d}x$; equivalently, the empirical measure μ_t^n converges in probability to this law. The limiting density solves the nonlinear Fokker–Planck, or McKean–Vlasov, equation

$$\partial_t \rho_t = -\operatorname{div}\big(b(x, \rho_t)\rho_t\big) + \sigma^2 \Delta \rho_t.$$

When the interaction drift has variational form

$$b(x, \rho) = -\nabla \frac{\delta \mathcal{E}}{\delta \rho}(x),$$

this PDE is the Wasserstein gradient flow of the entropy-regularized energy

$$\mathcal{E}(\rho) + \sigma^2 \int \rho \log \rho\,\mathrm{d}x.$$

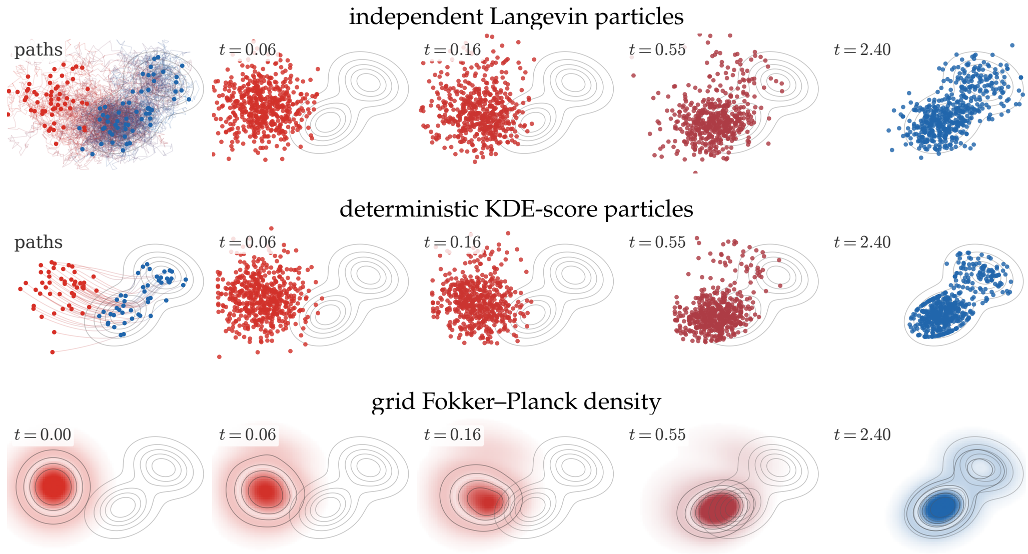


Figure 3.6: Three numerical representations of the same entropy-regularized Wasserstein gradient flow of $\mathrm{KL}(\rho|\beta)$, where β is a two-Gaussian target shifted to the right of an initially isotropic Gaussian density. The first row simulates independent Langevin particles and displays a thinned set of trajectories in the left panel. The second row evolves many deterministic particles with velocity $\tau(\nabla \log \beta - \nabla \log \rho_t)$, estimating $\nabla \log \rho_t$ by a sharper kernel-density score; only representative trajectories and particle subsets are displayed. The third row solves the corresponding Fokker–Planck equation on a grid, starting from the initial density in the left panel. The remaining columns use front-loaded times, so that the onset of the flow and the later deformation toward a bimodal law are both visible.

Gradient flows under density constraints. The same minimizing-movement formalism also handles nonsmooth energies. A prototypical example is a linear energy, generated by a potential, with a hard upper bound on the density,

$$F(\rho \mathrm{d}x) = \int_\Omega h(x)\rho(x)\mathrm{d}x + \iota_{\mathcal{K}_\kappa}(\rho \mathrm{d}x), \qquad \mathcal{K}_\kappa = \{\rho \mathrm{d}x \in \mathcal{P}_2(\Omega) \,;\, 0 \leqslant \rho \leqslant \kappa \text{ a.e.}\}\,, \tag{3.9}$$

where $\Omega \subset \mathbb{R}^d$ is a convex domain and $\mathcal{K}_\kappa$ is assumed nonempty; if Ω has finite volume, this requires $\kappa|\Omega| \geqslant 1$. The indicator term has no classical first variation; it contributes the normal cone to $\mathcal{K}_\kappa$ in the Wasserstein first-order condition. The JKO step becomes

$$\alpha^{k+1} \in \underset{\alpha \in \mathcal{K}_\kappa}{\operatorname{argmin}}\ \frac{1}{2\tau}\mathcal{W}_2^2(\alpha^k, \alpha) + \int_\Omega h \,\mathrm{d}\alpha. \tag{3.10}$$

This is the mathematical mechanism behind macroscopic crowd-motion models with congestion, where the desired velocity $-\nabla h$ is projected onto velocities that do not increase density beyond the maximal value κ [61, 85].

The constraint set is compatible with Wasserstein geometry.

Proposition 3.6 (Geodesic convexity of density caps)**.** *Assume that $\Omega \subset \mathbb{R}^d$ is convex and let $\kappa > 0$. The set $\mathcal{K}_\kappa$ in (3.9) is geodesically convex in $\mathcal{P}_2(\Omega)$: if $\alpha_0, \alpha_1 \in \mathcal{K}_\kappa$, then the constant-speed $\mathcal{W}_2$ geodesic $(\alpha_t)_{t\in[0,1]}$ between them also belongs to $\mathcal{K}_\kappa$.*

Proof. Assume first that the endpoints have smooth positive densities $\alpha_i = \rho_i \mathrm{d}x$ with $\rho_i \leqslant \kappa$, and let $T = \nabla\varphi$ be the Brenier map from α_0 to α_1. Since Ω is convex, $T_t = (1-t)\mathrm{Id} + tT$ maps Ω into Ω. The interpolation is $\alpha_t = (T_t)_\sharp \alpha_0$ and

$$\rho_t(T_t(x)) = \frac{\rho_0(x)}{\det((1-t)I + t\nabla T(x))}.$$

The concavity of $\det^{1/d}$ on positive semidefinite matrices and the identity $\rho_1(T(x))\det \nabla T(x) = \rho_0(x)$ give

$$\rho_t(T_t(x))^{-1/d} \geqslant (1-t)\rho_0(x)^{-1/d} + t\rho_1(T(x))^{-1/d} \geqslant \kappa^{-1/d}.$$

Hence $\rho_t \leqslant \kappa$ for all t. The general case follows by approximation and lower semicontinuity of the L^∞ bound, as in McCann's displacement-convexity theory [62]. □

The formal PDE contains a pressure field p_t, the Lagrange multiplier of the constraint. With no-flux boundary condition $\rho_t \nabla(h + p_t) \cdot n = 0$ on $\partial\Omega$, the constrained gradient flow is the complementarity system

$$\partial_t \rho_t = \operatorname{div}\left(\rho_t \nabla(h + p_t)\right), \qquad 0 \leqslant \rho_t \leqslant \kappa, \qquad p_t \geqslant 0, \qquad p_t(\kappa - \rho_t) = 0. \tag{3.11}$$

The pressure vanishes in the unsaturated region and acts only where $\rho_t = \kappa$, pushing mass away from congested zones so that the density cap remains satisfied.

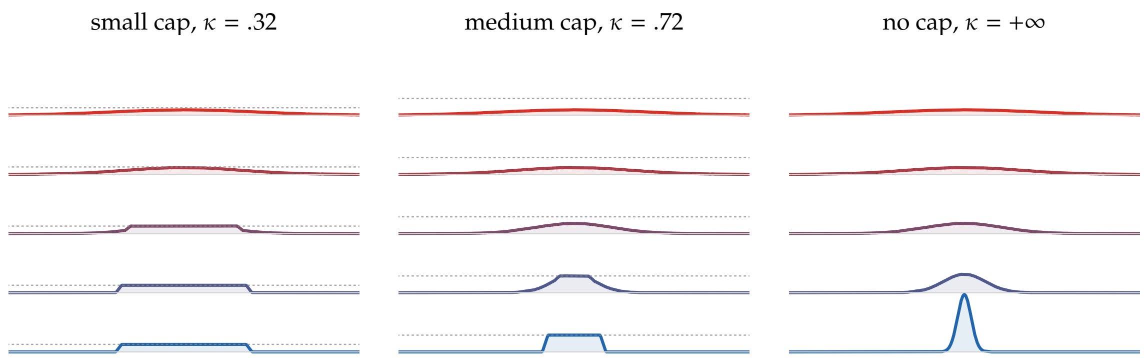


Figure 3.7: Density-constrained gradient flow for the attractive quadratic potential $h(x) = \|x\|^2/2$ in one dimension, computed in quantile variables by explicit Lagrangian descent followed by the isotonic projection enforcing $\partial_s q \geqslant 1/\kappa$. Each panel stacks five times vertically, colored from red to blue, using the same vertical density scale in all three panels. The dashed gray line marks the maximal density κ for the constrained runs. A small cap forces the mass to form a wide saturated block, a medium cap allows a narrower congested region, and the unconstrained flow concentrates freely near the minimizer of h.

Multi-species gradient flows. Several applications involve several densities living on the same physical space: chemical species, color channels, cell populations or competing phases. Let $\Omega \subset \mathbb{R}^d$ be the physical domain and assume that the component masses $\mathbf{m} = (m_1, \ldots, m_p)$ are positive and fixed. A convenient notation is

$$\mathcal{P}_{2,\mathbf{m}}(\Omega; \mathbb{R}^p_+) = \left\{ \alpha = (\alpha_1, \ldots, \alpha_p)\ ;\ \alpha_i \in \mathcal{M}_+(\Omega),\ \alpha_i(\Omega) = m_i,\ \int_\Omega \|x\|^2 \mathrm{d}\alpha_i(x) < +\infty \right\}. \tag{3.12}$$

Since the components are finite measures rather than necessarily probabilities, set

$$\mathcal{W}^2_{2,\oplus}(\alpha, \eta) = \sum_{i=1}^p m_i\ \mathcal{W}_2^2\left(\frac{\alpha_i}{m_i}, \frac{\eta_i}{m_i}\right). \tag{3.13}$$

This is the diagonal, independent-channel geometry; cross-effects may still enter through the energy or through constraints. In the broader dynamic language of vector-valued measures, this is the special case of a diagonal mobility. The corresponding JKO step is

$$\alpha^{k+1} \in \underset{\alpha \in \mathcal{P}_{2,\mathbf{m}}(\Omega; \mathbb{R}^p_+)}{\operatorname{argmin}} \ \frac{1}{2\tau} \mathcal{W}^2_{2,\oplus}(\alpha^k, \alpha) + F(\alpha). \tag{3.14}$$

For smooth densities and first variations $\varphi_i = \delta F / \delta \rho_i$, this yields

$$\partial_t \rho_i = \operatorname{div}\left(\rho_i \nabla \varphi_i\right), \qquad \varphi_i = \frac{\delta F}{\delta \rho_i}, \qquad i = 1, \ldots, p. \tag{3.15}$$

The equations are independent only if F is separable. Otherwise the coupling enters through the first variations φ_i, not through the metric tensor. Carlier, Chizat and Laborde [10] use displacement smoothness of entropic OT to prove well-posedness of Wasserstein gradient flows that include multi-species systems. Congestion and shared-density variants are closely related to crowd and traffic models [61, 11, 85].

A second useful model imposes a pointwise composition constraint. Given a fixed reference density $\beta = b\mathrm{d}x$ with total mass $\sum_i m_i$, set

$$C_\beta = \left\{ \alpha \in \mathcal{P}_{2,\mathbf{m}}(\Omega; \mathbb{R}^p_+) \; ; \; \sum_{i=1}^{p} \alpha_i = \beta \right\}. \tag{3.16}$$

The constrained JKO step is

$$\alpha^{k+1} \in \underset{\alpha \in C_\beta}{\operatorname{argmin}} \frac{1}{2\tau} \mathcal{W}^2_{2,\oplus}(\alpha^k, \alpha) + F(\alpha). \tag{3.17}$$

In the smooth-density regime, the formal constrained flow has a common pressure λ_t:

$$\partial_t \rho_i = \operatorname{div}\left(\rho_i \nabla(\varphi_i - \lambda_t)\right), \qquad \operatorname{div}(b\nabla\lambda_t) = \operatorname{div}\left(\sum_{i=1}^{p} \rho_i \nabla\varphi_i\right). \tag{3.18}$$

For the separable Shannon entropy $F(\alpha) = \sum_i \int_\Omega \rho_i \log \rho_i \mathrm{d}x$, this becomes

$$\partial_t \rho_i = \Delta\rho_i - \operatorname{div}(\rho_i \nabla\lambda_t), \qquad \operatorname{div}(b\nabla\lambda_t) = \Delta b. \tag{3.19}$$

When b is constant, the pressure can be chosen constant and the system reduces to independent heat equations which nevertheless preserve the pointwise sum $\sum_i \rho_i = b$. This product geometry should be contrasted with vector-valued Benamou–Brenier distances with non-diagonal mobilities, where the metric itself couples the components.

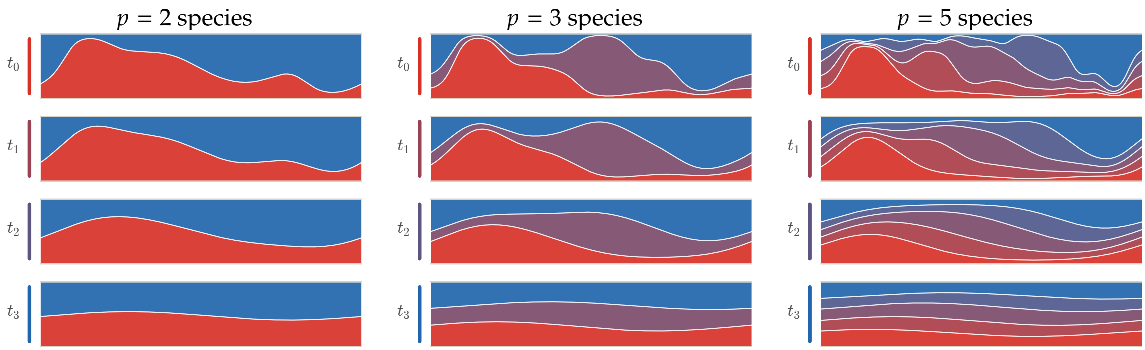


Figure 3.8: Multi-species entropy flow under the shared-density constraint $\sum_i \rho_i = 1$ on the periodic interval. Each row is a time snapshot of the stacked densities, with the species shown as colored bands and time increasing from top to bottom. For a constant total density, the pressure in (3.19) is constant, so each component follows the heat equation while the sum of the bands remains exactly flat.

3.2 Unbalanced Wasserstein–Fisher–Rao Gradient Flows

The Wasserstein gradient flows above conserve total mass, because their tangent vectors are generated by continuity equations. When the unknown measure represents an intensity, a population or a cloud of trainable particles, it is often useful to combine spatial motion with creation and destruction of mass. The dynamic unbalanced distance defined in Section 2.3, through the balance-equation formula (2.13), gives exactly this geometry.

Definition 3.7 (WFR minimizing movement)**.** Let $F : \mathcal{M}_+(\mathbb{R}^d) \to \mathbb{R} \cup \{+\infty\}$ be an energy on positive finite measures and let WFR_κ be the Wasserstein–Fisher–Rao distance with growth scale $\kappa > 0$, defined by the dynamic balance-equation formulation (2.13) and its static equivalence in Proposition 2.5. The unbalanced JKO step is

$$\alpha^{k+1} \in \underset{\alpha \in \mathcal{M}_+(\mathbb{R}^d)}{\operatorname{argmin}} \frac{1}{2\tau} \mathrm{WFR}^2_\kappa(\alpha^k, \alpha) + F(\alpha). \tag{3.20}$$

The continuous-time limit, when it exists, is called the WFR_κ gradient flow of F.

Formally, a smooth curve $\alpha_t = \rho_t \mathrm{d}x$ has tangent vectors of the form

$$\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = g_t \rho_t,$$

with squared norm $\int (\|v_t\|^2 + \kappa^2 g_t^2)\rho_t \mathrm{d}x$. This is the infinitesimal version of the balance-equation action in Section 2.

Proposition 3.8 (Formal WFR gradient-flow PDE). *Assume that F has a smooth first variation $\varphi_t = \delta F(\alpha_t)$ and that $\alpha_t = \rho_t \mathrm{d}x$ is smooth and positive. The formal* WFR_κ *gradient flow of F is*

$$\partial_t \rho_t = \operatorname{div}(\rho_t \nabla \varphi_t) - \frac{1}{\kappa^2}\varphi_t \rho_t. \tag{3.21}$$

Along such a smooth solution,

$$\frac{\mathrm{d}}{\mathrm{d}t} F(\alpha_t) = -\int \left(\|\nabla \varphi_t\|^2 + \frac{1}{\kappa^2}\varphi_t^2 \right) \rho_t \,\mathrm{d}x \leqslant 0. \tag{3.22}$$

Proof. For an infinitesimal tangent pair (v, g), the density variation is $\zeta = -\operatorname{div}(\rho v) + g\rho$. The first variation of F is

$$\int \varphi\, \zeta = \int \langle \nabla \varphi, v \rangle \rho \,\mathrm{d}x + \int \varphi g \rho \,\mathrm{d}x,$$

after integration by parts. For the inner product

$$\langle (v, g), (\tilde{v}, \tilde{g}) \rangle_\rho = \int \langle v, \tilde{v} \rangle \rho \,\mathrm{d}x + \kappa^2 \int g \tilde{g} \rho \,\mathrm{d}x,$$

the Riesz representative is $(\nabla\varphi, \varphi/\kappa^2)$. Steepest descent therefore uses $(v, g) = (-\nabla\varphi, -\varphi/\kappa^2)$, which gives (3.21). Substituting this pair in the first-variation formula gives

$$\frac{\mathrm{d}}{\mathrm{d}t} F(\alpha_t) = -\int \left(\|\nabla \varphi_t\|^2 + \frac{1}{\kappa^2}\varphi_t^2 \right) \rho_t \,\mathrm{d}x,$$

which is (3.22). □

The reaction term makes additive constants in the first variation meaningful: adding a constant to φ changes the global growth rate. This is not a defect but the infinitesimal signature of optimizing over positive finite measures, rather than on the probability simplex. The rigorous metric-space construction of Kantorovich–Fisher–Rao and WFR gradient flows is developed in [37, 21, 20]. Chizat's measure-optimization framework relates weight-changing gradient methods to mirror- and Bregman-type descents on positive measures [17, 18]. In machine learning, birth–death dynamics use the same reaction intuition to let a particle population reweight, remove and create neurons during training [80].

Example 3.9 (Relative entropy and birth–death relaxation). Let $\beta = b\mathrm{d}x$ and use the finite-measure relative entropy

$$F(\alpha) = \mathrm{KL}(\alpha|\beta) = \int \left(\rho \log \frac{\rho}{b} - \rho + b\right) \mathrm{d}x, \qquad \alpha = \rho \mathrm{d}x.$$

This is the Csiszár divergence on positive measures; it is nonnegative and is minimized, without imposing a mass constraint, at $\alpha = \beta$. Its first variation is $\delta F(\alpha) = \log(\rho/b)$. The balanced $\mathcal{W}_2$ flow, when the total mass is fixed, is the Fokker–Planck equation

$$\partial_t \rho_t = \operatorname{div}\left(\rho_t \nabla \log \frac{\rho_t}{b}\right) = \Delta \rho_t - \operatorname{div}(\rho_t \nabla \log b),$$

whereas the WFR_κ flow is

$$\partial_t \rho_t = \Delta \rho_t - \operatorname{div}(\rho_t \nabla \log b) - \frac{1}{\kappa^2} \rho_t \log \frac{\rho_t}{b}. \tag{3.23}$$

The first two terms transport and diffuse mass, while the last term kills mass where $\rho_t > b$ and creates it where $\rho_t < b$.

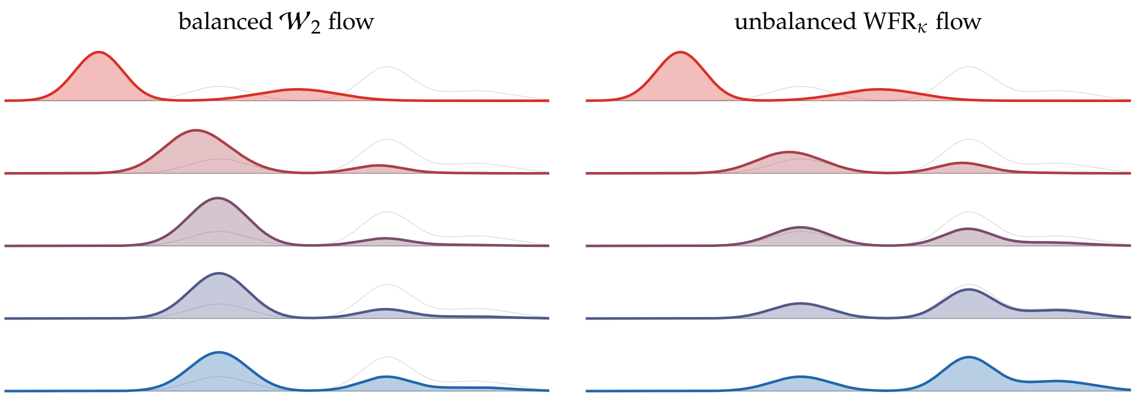


Figure 3.9: Balanced and unbalanced gradient flows of $\mathrm{KL}(\rho|\beta)$ for one-dimensional Gaussian mixtures. In each panel the colored density stacks correspond to five increasing times, from red to blue, and the faint gray curve repeated on each row is the target mixture β. The balanced Wasserstein flow is the conservative Fokker–Planck equation, so mass must move through the low-density region between modes. The WFR_κ flow adds the birth–death reaction term in (3.23); it attenuates overrepresented regions and creates missing mass near target modes, reaching the target shape much faster.

3.3 Geodesic Convexity and Convergence

Geodesic convexity is the convexity notion adapted to Wasserstein geometry. It is the condition that turns the formal gradient-flow calculus into a convergence theory.

Geodesics and convexity. A constant-speed $\mathcal{W}_2$ geodesic between α_0 and α_1 is obtained, as in Definition 1.15, from any optimal coupling $\pi^\star \in \mathcal{U}(\alpha_0, \alpha_1)$ by the McCann interpolation

$$\alpha_t = ((1-t)P_0 + tP_1)_\sharp \pi^\star, \qquad t \in [0,1],$$

where $P_0(x, y) = x$ and $P_1(x, y) = y$. If the optimal plan is induced by a Brenier map T, this reduces to $((1-t)\mathrm{Id} + tT)_\sharp \alpha_0$. The coupling formula is important because geodesics exist even when no Monge map exists, for instance when a Dirac mass must split.

Definition 3.10 (Geodesic convexity)**.** A functional f on $\mathcal{P}_2(\mathbb{R}^d)$ is geodesically convex if for every $\mathcal{W}_2$ geodesic $(\alpha_t)_t$,

$$f(\alpha_t) \leqslant (1-t) f(\alpha_0) + t f(\alpha_1).$$

It is λ-geodesically convex if the right-hand side is improved by $-\frac{\lambda}{2} t(1-t)\, \mathcal{W}_2^2(\alpha_0, \alpha_1)$.

Proposition 3.11 (Basic geodesically convex energies)**.** *The following formal statements hold on* $\mathcal{P}_2(\mathbb{R}^d)$.

1. *If h is convex, then $\alpha \mapsto \int h\mathrm{d}\alpha$ is geodesically convex; if h is λ-strongly convex, it is λ-geodesically convex.*

2. *If $W(x-y)$ is convex as a function of the displacement, then $\alpha \mapsto \frac{1}{2}\iint W(x-y)\mathrm{d}\alpha(x)\mathrm{d}\alpha(y)$ is geodesically convex.*

3. *Shannon entropy $\alpha \mapsto \int \rho \log \rho \, \mathrm{d}x$ is geodesically convex.*

4. *The relative entropy $\mathrm{KL}(\alpha|\gamma)$ with $\mathrm{d}\gamma = e^{-V}\mathrm{d}x/Z$ is λ-geodesically convex when V is λ-strongly convex.*

Proof. Along a Monge geodesic $X_t = (1-t)X_0 + tX_1$, convexity of h gives $h(X_t) \leqslant (1-t)h(X_0) + th(X_1)$, and strong convexity gives the additional quadratic term; integrating proves the first claim. The interaction claim follows similarly by applying convexity of W to pairwise differences $X_t - X_t' = (1-t)(X_0 - X_0') + t(X_1 - X_1')$ and integrating over two independent copies. The entropy claim is McCann's displacement convexity theorem; at the density level it follows from the concavity of the Jacobian determinant under the interpolation of optimal maps. Finally, $\mathrm{KL}(\alpha|\gamma) = \int \rho \log \rho \, \mathrm{d}x + \int V \mathrm{d}\alpha + \text{constant}$, so it is the sum of displacement-convex entropy and a λ-geodesically convex linear potential. □

A useful non-example. The squared distance to a fixed measure is not geodesically convex in general. This is a sharp contrast with Hilbert spaces, where $x \mapsto \|x-y\|^2$ is convex along line segments. In $\mathcal{P}_2(\mathbb{R}^d)$, $d \geqslant 2$, take

$$\beta = \frac{1}{2}\delta_{(-1/2,-1/2)} + \frac{1}{2}\delta_{(1/2,1/2)}, \quad \alpha_0 = \frac{1}{2}\delta_{(-1,0)} + \frac{1}{2}\delta_{(1,0)}, \quad \alpha_1 = \frac{1}{2}\delta_{(0,-1)} + \frac{1}{2}\delta_{(0,1)}.$$

One optimal coupling between α_0 and α_1 matches $(-1,0)$ with $(0,1)$ and $(1,0)$ with $(0,-1)$. The associated geodesic has midpoint

$$\alpha_{1/2} = \frac{1}{2}\delta_{(-1/2,1/2)} + \frac{1}{2}\delta_{(1/2,-1/2)}.$$

Then

$$\mathcal{W}_2^2(\alpha_0,\beta) = \mathcal{W}_2^2(\alpha_1,\beta) = \frac{1}{2}, \qquad \mathcal{W}_2^2(\alpha_{1/2},\beta) = 1,$$

which violates geodesic convexity. Thus objectives built from squared Wasserstein distances can be non-convex in the Wasserstein geometry itself. This is one reason why optimal quantization, which minimizes $\mathcal{W}_2^2(\alpha,\nu)$ over m-atomic measures ν, leads to non-convex Lloyd-type algorithms and convergence only to local minima.

Convergence of the flow. In general, analyzing (3.3) is delicate. The cleanest case is when f is geodesically convex in the sense above. This condition is the Wasserstein analogue of convexity in Euclidean gradient descent.

Proposition 3.12 (Energy decay for convex Wasserstein flows)**.** *Assume formally that f is geodesically convex, admits a smooth first variation, and has a minimizer $\alpha^\star$. Let $(\alpha_t)_t$ be a smooth solution of the Wasserstein gradient flow*

$$\partial_t \alpha_t + \mathrm{div}(\alpha_t v_t) = 0, \qquad v_t = -\nabla_{\mathcal{W}} f(\alpha_t).$$

Then

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\alpha_t) = -\int \|\nabla_{\mathcal{W}} f(\alpha_t)(x)\|^2 \,\mathrm{d}\alpha_t(x) \leqslant 0.$$

If T_t is the optimal map from α_t to $\alpha^\star$, then

$$f(\alpha_t) - f(\alpha^\star) \leqslant -\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\mathcal{W}_2^2(\alpha_t,\alpha^\star),$$

and consequently

$$f(\alpha_t) - f(\alpha^\star) \leqslant \frac{\mathcal{W}_2^2(\alpha_0,\alpha^\star)}{2t}.$$

If f is λ-geodesically convex with $\lambda > 0$, then

$$f(\alpha_t) - f(\alpha^\star) \leqslant e^{-2\lambda t}\big(f(\alpha_0) - f(\alpha^\star)\big).$$

Proof. The chain rule and Proposition 3.2 give

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\alpha_t) = \int \langle \nabla_{\mathcal{W}} f(\alpha_t)(x),\, v_t(x)\rangle \,\mathrm{d}\alpha_t(x) = -\int \|\nabla_{\mathcal{W}} f(\alpha_t)(x)\|^2 \,\mathrm{d}\alpha_t(x).$$

Geodesic convexity along the geodesic $((1-s)\mathrm{Id} + sT_t)_\sharp \alpha_t$ gives

$$f(\alpha^\star) - f(\alpha_t) \geqslant \int \langle \nabla_{\mathcal{W}} f(\alpha_t)(x),\, T_t(x) - x\rangle \,\mathrm{d}\alpha_t(x).$$

Since $v_t = -\nabla_{\mathcal{W}} f(\alpha_t)$, this reads

$$f(\alpha_t) - f(\alpha^\star) \leqslant \int \langle v_t(x),\, T_t(x) - x\rangle \,\mathrm{d}\alpha_t(x).$$

The standard first-variation formula for the squared Wasserstein distance gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\mathcal{W}_2^2(\alpha_t, \alpha^\star) = \int \langle x - T_t(x),\, v_t(x)\rangle \,\mathrm{d}\alpha_t(x),$$

which proves the differential inequality. Integrating it from 0 to t and using the monotonicity of $s \mapsto f(\alpha_s)$ gives

$$t\big(f(\alpha_t) - f(\alpha^\star)\big) \leqslant \int_0^t \big(f(\alpha_s) - f(\alpha^\star)\big)\,\mathrm{d}s \leqslant \frac{1}{2}\mathcal{W}_2^2(\alpha_0, \alpha^\star).$$

If f is λ-geodesically convex, the Wasserstein analogue of strong convexity gives the slope inequality

$$\int \|\nabla_{\mathcal{W}} f(\alpha_t)\|^2 \mathrm{d}\alpha_t \geqslant 2\lambda\big(f(\alpha_t) - f(\alpha^\star)\big).$$

Combining it with the energy dissipation identity yields

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(f(\alpha_t) - f(\alpha^\star)\big) \leqslant -2\lambda\big(f(\alpha_t) - f(\alpha^\star)\big),$$

and Gronwall's lemma gives the exponential rate. □

Proposition 3.13 (Convex examples covered by the theory)**.** *The hypotheses of Proposition 3.12 are satisfied in the following standard cases, at least at the formal smooth level used in this section.*

1. *For the linear energy $f(\alpha) = \int h \mathrm{d}\alpha$, geodesic convexity holds when h is convex. If h is λ-strongly convex, then f is λ-geodesically convex and the flow enjoys the exponential rate of Proposition 3.12.*

2. *For the interaction energy $f(\alpha) = \frac{1}{2}\iint W(x-y)\mathrm{d}\alpha(x)\mathrm{d}\alpha(y)$, geodesic convexity holds when W is convex and even. This covers repulsive or attractive pairwise models whose displacement cost has no non-convex wells.*

3. *The Shannon entropy $f(\alpha) = \int \rho \log \rho \,\mathrm{d}x$ and, more generally, McCann displacement-convex internal energies generate diffusion-type Wasserstein gradient flows.*

4. *If $\gamma = Z^{-1}e^{-V}\mathrm{d}x$ and V is λ-strongly convex, then the relative entropy $\mathrm{KL}(\alpha|\gamma)$ is λ-geodesically convex. Its flow is the Fokker–Planck equation with invariant law γ.*

Proof. Let $(\alpha_t)_t$ be the McCann interpolation between α_0 and α_1, written with an optimal coupling as $X_t = (1-t)X_0 + tX_1$. For a linear energy, Jensen's inequality gives

$$h(X_t) \leqslant (1-t)h(X_0) + t h(X_1),$$

and the strong convexity version gives the additional term $-\frac{\lambda}{2}t(1-t)\|X_0 - X_1\|^2$. Integrating over the optimal coupling proves geodesic convexity and λ-geodesic convexity.

For interaction energies, use two independent copies of the optimal coupling. The pairwise displacement evolves as

$$X_t - X_t' = (1-t)(X_0 - X_0') + t(X_1 - X_1').$$

Convexity of W gives the convexity inequality after integration over the product coupling. Evenness of W ensures that the interaction is symmetric in the two particles and matches the usual factor $1/2$ in (3.8).

The entropy claim is McCann's displacement-convexity theorem. For smooth positive densities and Brenier maps, it follows from the change-of-variables formula and the concavity of the determinant along positive matrices; the general statement is obtained by approximation. Finally,

$$\mathrm{KL}(\alpha|\gamma) = \int \rho \log \rho \, \mathrm{d}x + \int V \mathrm{d}\alpha + \log Z,$$

so it is the sum of the displacement-convex entropy and the λ-geodesically convex linear potential generated by V. Proposition 3.12 then applies to all four cases. □

Convexity and curvature. The same language is not restricted to subsets of $\mathbb{R}^d$. If $(\mathcal{X}, d, \mathfrak{m})$ is a geodesic metric-measure space, $\mathcal{W}_2$ geodesics can be defined by transporting each pair of endpoints along metric geodesics, or more intrinsically by dynamical optimal plans on path space, as discussed in Section 1.5. Given a reference measure $\mathfrak{m}$, the entropy relative to $\mathfrak{m}$ is

$$\mathrm{Ent}_{\mathfrak{m}}(\alpha) := \begin{cases} \int_{\mathcal{X}} \rho \log \rho \, \mathrm{d}\mathfrak{m}, & \text{if } \alpha = \rho\, \mathfrak{m}, \\ +\infty, & \text{otherwise.} \end{cases}$$

On a smooth Riemannian manifold (M, g), the Ricci curvature tensor Ric_g is the trace of the Riemann curvature tensor; the lower bound $\mathrm{Ric}_g \geqslant \lambda g$ means that $\mathrm{Ric}_g(v, v) \geqslant \lambda |v|_g^2$ for every tangent vector v. The fundamental link between curvature and optimal transport is that this tensor lower bound is exactly encoded by geodesic convexity of entropy.

Theorem 3.14 (Ricci curvature and entropy convexity)**.** *Let (M, g) be a smooth compact connected Riemannian manifold without boundary, and let $\mathfrak{m} = \mathrm{vol}_g$. For $\lambda \in \mathbb{R}$, the lower Ricci bound $\mathrm{Ric}_g \geqslant \lambda g$ holds if and only if $\mathrm{Ent}_{\mathfrak{m}}$ is λ-geodesically convex on $(\mathcal{P}_2(M), \mathcal{W}_2)$.*

This equivalence was developed in the smooth Riemannian setting by Cordero-Erausquin, McCann and Schmuckenschläger and by von Renesse and Sturm [25, 101]; it is a central theme of the optimal-transport approach to curvature in Villani's monograph [99]. Lott–Villani and Sturm then used the same entropy-convexity principle to define synthetic lower Ricci curvature bounds on metric-measure spaces [59, 90, 91]. Outside this convex, curvature-controlled regime, such as in the mean-field neural-network example below, the flow may still be informative but its convergence analysis requires problem-specific arguments.

3.4 Training Two-Layer MLPs as Wasserstein Flows

Mean-field limits recast the training of wide neural networks as transport of a distribution of neurons. This section shows how the particle ODE of gradient descent becomes a Wasserstein flow in parameter space. This viewpoint, often summarized as treating parameters as interacting particles, was developed in closely related forms by Mei, Montanari and Nguyen [63], Rotskoff and Vanden-Eijnden [81], and Chizat and Bach [19]. These works pass from the finite-width empirical neuron dynamics to a limiting nonlinear PDE for the neuron law.

We use $z \in \mathbb{R}^d$ for the input data and $y \in \mathbb{R}^{d'}$ for the label. A neuron is a particle

$$x = (u, v) \in \mathbb{R}^d \times \mathbb{R}^{d'},$$

where u is the inner weight and v is the outer vector weight. For a scalar nonlinearity σ, define the vector-valued feature

$$\psi(x, z) = v\, \sigma(\langle u, z \rangle) \in \mathbb{R}^{d'}.$$

The width-n network and its mean-field version are

$$G_X(z) = \frac{1}{n} \sum_{i=1}^{n} \psi(x_i, z), \qquad G_\alpha(z) = \int \psi(x, z) \mathrm{d}\alpha(x), \qquad \alpha = \frac{1}{n} \sum_i \delta_{x_i}.$$

This formulation removes the artificial ordering of neurons and allows α to be a continuous distribution of infinitely many neurons.

Let ρ be a probability distribution on data-label pairs $(z, y) \in \mathbb{R}^d \times \mathbb{R}^{d'}$. The population risk is

$$f(\alpha) = \int \ell(G_\alpha(z), y) \mathrm{d}\rho(z, y),$$

and the empirical risk is the special case $\rho = \rho_N := N^{-1} \sum_{k=1}^N \delta_{(z_k, y_k)}$. Since $\alpha \mapsto G_\alpha$ is linear, f is convex as a function of α whenever $\ell(\cdot, y)$ is convex. For the empirical neuron law $\alpha_X = n^{-1} \sum_i \delta_{x_i}$, the Wasserstein metric induces on particles the rescaled metric $n^{-1} \sum_i \|\dot{x}_i\|^2$. The corresponding particle flow is

$$\dot{x}_i = -n \nabla_{x_i} F(X), \qquad F(X) = f\left(\frac{1}{n} \sum_i \delta_{x_i}\right),$$

which is the gradient flow of $F(X) = f(\alpha_X)$ for this Wasserstein particle metric, equivalently Euclidean gradient descent with the time scale multiplied by n. It gives a particle discretization of (3.3).

Assume that ℓ is differentiable in its first variable. The first variation is

$$\delta f(\alpha)(x) = \int \langle \nabla_1 \ell(G_\alpha(z), y),\ \psi(x, z) \rangle \mathrm{d}\rho(z, y), \tag{3.24}$$

and the Wasserstein gradient in parameter space is

$$\nabla_{\mathcal{W}} f(\alpha)(x) = \nabla_x \delta f(\alpha)(x) = \int [D_x \psi(x, z)]^\top \nabla_1 \ell(G_\alpha(z), y) \mathrm{d}\rho(z, y).$$

For the squared Euclidean loss $\ell(s, y) = \frac{1}{2}\|s - y\|^2$, the energy is the sum of a quadratic interaction and a linear potential:

$$f(\alpha) = \frac{1}{2} \iint k(x, x') \mathrm{d}\alpha(x) \mathrm{d}\alpha(x') + \int g(x) \mathrm{d}\alpha(x) + \frac{1}{2} \int \|y\|^2 \mathrm{d}\rho(z, y), \tag{3.25}$$

with

$$k(x, x') = \int \langle \psi(x, z),\ \psi(x', z) \rangle \mathrm{d}\rho(z, y), \qquad g(x) = -\int \langle y,\ \psi(x, z) \rangle \mathrm{d}\rho(z, y). \tag{3.26}$$

Thus

$$\delta f(\alpha)(x) = \int k(x, x') \mathrm{d}\alpha(x') + g(x), \qquad \nabla_{\mathcal{W}} f(\alpha)(x) = \int \nabla_x k(x, x') \mathrm{d}\alpha(x') + \nabla_x g(x).$$

These kernels are generally not convex in the particle variable, so the geodesic-convex convergence theory above does not apply directly.

Classical convexity and stationarity. Before using the specific homogeneity mechanism of Chizat and Bach, it is useful to isolate a simpler convex-analytic principle behind many mean-field arguments. Consider an energy of the form

$$F(\alpha) = \frac{1}{2} \iint k(x, x') \mathrm{d}\alpha(x) \mathrm{d}\alpha(x') + \int V(x) \mathrm{d}\alpha(x) + C,$$

on probability measures over a parameter domain. Assume that the quadratic part is convex in the classical sense, namely convex for the affine structure of measures:

$$Q((1-s)\alpha + s\beta) \leqslant (1-s) Q(\alpha) + s Q(\beta), \qquad Q(\alpha) = \frac{1}{2} \iint k \mathrm{d}\alpha \mathrm{d}\alpha.$$

This is ordinary convexity of the functional on the convex set of measures, not displacement convexity along $\mathcal{W}_2$ geodesics.

Proposition 3.15 (Affine convexity and stationary densities). *Let $F = Q + \int V \,\mathrm{d}\alpha + C$ be as above, and assume that Q is classically convex. Suppose that a Wasserstein gradient flow for F converges to a measure $\alpha_\infty = \rho_\infty \mathrm{d}x$. Assume also the standard regularity needed to pass to the limit in the first variation, and assume that the support and positivity of ρ_∞ allow the stationary condition to be tested against all admissible zero-mass density perturbations. In the form needed here, assume that this stationarity yields, for every competitor β, the variational inequality*

$$\int \delta F(\alpha_\infty)(x) \mathrm{d}(\beta - \alpha_\infty)(x) \geqslant 0.$$

Then α_∞ is a global minimizer of F.

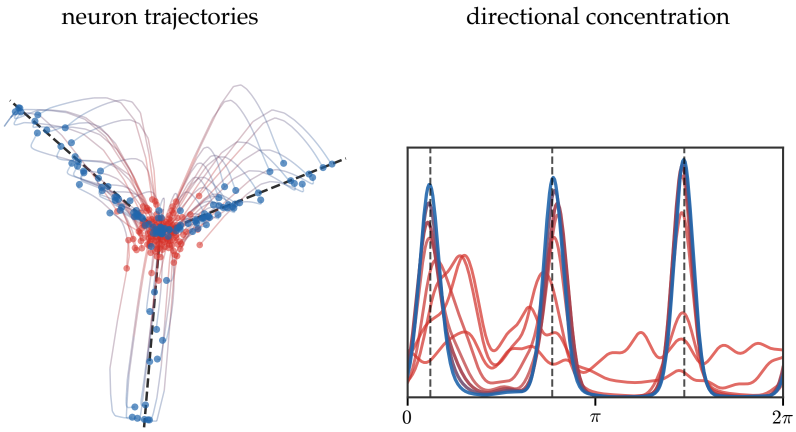


Figure 3.10: Mean-field training of a homogeneous two-layer model as transport in neuron space. The left panel shows the Wasserstein particle gradient flow in the reduced homogeneous coordinates $(|u|v_1, |u|v_2)$, with black dashed rays marking the teacher directions. The right panel shows the weighted angular density along a front-loaded sequence of times, colored from red to blue, so that the early concentration of neuron directions is visible. The display follows the rendering of the auxiliary MLP experiment but keeps only the W_2 flow, not the spectral-flow comparison.

Proof sketch. The dissipation identity for the gradient flow gives stationarity of the limit: formally, after passing to the limit,

$$\int \|\nabla \delta F(\alpha_\infty)\|^2 \mathrm{d}\alpha_\infty = 0.$$

Without such a support and positivity assumption, this identity only controls the first variation on the region explored by the limit. The density hypothesis allows one to test against sufficiently many signed density perturbations of total mass zero. By approximation and the assumed regularity, this yields the displayed first-order variational inequality for arbitrary competitors β. Classical convexity of F in the affine variable α then gives the usual subgradient inequality

$$F(\beta) \geqslant F(\alpha_\infty) + \int \delta F(\alpha_\infty) \mathrm{d}(\beta - \alpha_\infty) \geqslant F(\alpha_\infty).$$

Thus no competitor has smaller energy. For square-loss two-layer mean-field models, (3.25) is exactly of this quadratic-plus-linear form, and positive semidefiniteness of the induced kernel k is the classical convexity assumption. □

The convergence theory then separates two regimes. Chizat and Bach prove global convergence for the unregularized, noiseless Wasserstein flow of positively homogeneous models [19]. Mei, Montanari and Nguyen prove convergence for the noisy, regularized dynamics [63]: at the PDE level the noise adds a diffusion, or Laplacian, term, so an initially singular neuron law is immediately smoothed and acquires a density. Rotskoff and Vanden-Eijnden emphasize the parameters-as-particles interpretation, long-time convergence and asymptotic error scaling with the network width [81]. The following formal statement isolates the noiseless Chizat–Bach mechanism and ignores the technical issues due to ReLU non-smoothness, support propagation and compactness.

Proposition 3.16 (Formal global optimality for two-homogeneous mean-field flows)**.** *Assume that the feature is positively two-homogeneous in the neuron variable,*

$$\psi(\lambda x, z) = \lambda^2 \psi(x, z) \qquad (\lambda > 0),$$

and that $f(\alpha) = J(G_\alpha)$ with J convex and differentiable as a functional of the predictor. Let α be a smooth stationary point of the Wasserstein flow, so that $\nabla_x \delta f(\alpha)(x) = 0$ on $\operatorname{supp}(\alpha)$. Assume also full directional support: for every nonzero direction ω, the support of α intersects the ray $\{\lambda\omega : \lambda > 0\}$. Then α is a global minimizer of f over the mean-field model class.

Proof. Write

$$h_\alpha(x) = \delta f(\alpha)(x) = \langle \nabla J(G_\alpha), \psi(x, \cdot) \rangle_\rho .$$

By two-homogeneity of ψ, one has $h_\alpha(\lambda x) = \lambda^2 h_\alpha(x)$. Normalize a nonzero direction ω and choose $r_\omega > 0$ with $r_\omega \omega \in \operatorname{supp}(\alpha)$. Stationarity gives a zero radial derivative at this point:

$$0 = \frac{\mathrm{d}}{\mathrm{d}r} h_\alpha(r\omega)\Big|_{r=r_\omega} = 2r_\omega h_\alpha(\omega).$$

Hence $h_\alpha(\omega) = 0$ for every direction ω, and by homogeneity $h_\alpha(x) = 0$ for every x.

For any competitor β, convexity of J gives

$$f(\beta) - f(\alpha) \geqslant \int h_\alpha(x)\mathrm{d}(\beta - \alpha)(x) = 0.$$

Thus no competitor has smaller risk. The rigorous theorem replaces the full directional support assumption by propagation and overparameterization hypotheses ensuring that a negative descent direction would be present in the support and would contradict stationarity. □

3.5 Functional Inequalities via Optimal Transport

Optimal transport is also a proof technique for sharp inequalities. The common mechanism is simple: push one density to another by a monotone or Brenier map, interpolate along the transport rays, and convert convexity of the determinant, entropy, or the cost into an integral inequality. This section records a few representative examples. To keep the presentation readable, the proofs are written in the smooth Euclidean setting; the usual regularization and approximation arguments give the standard Borel or Sobolev versions. The transport viewpoint is developed systematically in McCann's displacement convexity principle, the Riemannian interpolation inequalities of Cordero-Erausquin, McCann and Schmuckenschläger, and Villani's account of concentration and functional inequalities [62, 25, 99].

Brunn–Minkowski and isoperimetry. The most geometric use of OT is to prove that transporting two uniform measures and looking at the interpolated support increases volume at least concavely. The perimeter inequality is then obtained by differentiating this volume estimate after adding a small ball.

Proposition 3.17 (Brunn–Minkowski and Euclidean isoperimetry)**.** *Let $A, B \subset \mathbb{R}^d$ be bounded Borel sets with positive Lebesgue measure. For $t \in [0, 1]$, set*

$$(1-t)A + tB := \{(1-t)x + ty \ : \ x \in A,\ y \in B\}.$$

Then

$$\operatorname{Vol}((1-t)A + tB)^{1/d} \geqslant (1-t)\operatorname{Vol}(A)^{1/d} + t\operatorname{Vol}(B)^{1/d}.$$

Consequently, if A is smooth and bounded and $\omega_d = \operatorname{Vol}(B_1)$, then

$$\operatorname{Per}(A) \geqslant d\,\omega_d^{1/d}\operatorname{Vol}(A)^{(d-1)/d}.$$

Proof. We first give the argument for regular sets, the general Borel case following by approximation from inside and outside. Let α and β be the uniform probability measures on A and B, and let $T = \nabla\varphi$ be the Brenier map with $T_\sharp\alpha = \beta$. The interpolating map $T_t = (1-t)\mathrm{Id} + tT$ sends α to a measure α_t supported in $(1-t)A + tB$. At points where T is differentiable,

$$\det DT_t = \det((1-t)\mathrm{Id} + tDT).$$

Since DT is symmetric positive semidefinite and $\det^{1/d}$ is concave on the positive semidefinite cone,

$$\det(DT_t)^{1/d} \geqslant (1-t) + t\det(DT)^{1/d}.$$

The change-of-variables identity for the uniform measures gives $\det(DT) = \operatorname{Vol}(B)/\operatorname{Vol}(A)$ almost everywhere. Hence the density ρ_t of α_t satisfies

$$\rho_t^{-1/d} \geqslant (1-t)\operatorname{Vol}(A)^{1/d} + t\operatorname{Vol}(B)^{1/d}.$$

This means that $\rho_t \leqslant c_t^{-d}$ on its support, where $c_t = (1-t)\operatorname{Vol}(A)^{1/d} + t\operatorname{Vol}(B)^{1/d}$. Since α_t has total mass one and is supported in $(1-t)A + tB$, this gives $\operatorname{Vol}((1-t)A + tB) \geqslant c_t^d$, which is the displayed Brunn–Minkowski inequality.

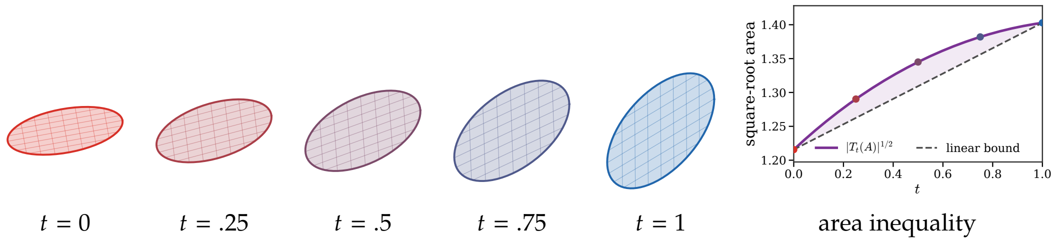


Figure 3.11: Brunn–Minkowski through an affine optimal-transport interpolation between two ellipses. For ellipses, the Brenier map is affine, so the transported support $T_t(A)$ remains an ellipse and its area is computed exactly from the determinant of $(1-t)\mathrm{Id} + tDT$. The right panel shows that $|T_t(A)|^{1/2}$ lies above the linear interpolation of the endpoint square-root areas, which is the two-dimensional determinant concavity behind the proof of Brunn–Minkowski.

For isoperimetry, apply Brunn–Minkowski with $B = B_1$ and write $A + \varepsilon B_1$ for the parallel set. Then

$$\mathrm{Vol}(A + \varepsilon B_1)^{1/d} \geqslant \mathrm{Vol}(A)^{1/d} + \varepsilon \omega_d^{1/d}.$$

Using $\mathrm{Vol}(A + \varepsilon B_1) = \mathrm{Vol}(A) + \varepsilon \, \mathrm{Per}(A) + o(\varepsilon)$ and differentiating at $\varepsilon = 0$ yields the claimed perimeter bound. □

Figure 3.11 isolates the determinant-concavity step in the special case where the optimal map is affine.

Prékopa–Leindler. The functional analogue of Brunn–Minkowski replaces sets by densities. The transport proof is the same argument with the Jacobian no longer constant.

Proposition 3.18 (Prékopa–Leindler inequality)**.** *Let $u, v, w : \mathbb{R}^d \to [0, +\infty)$ be integrable, and let $t \in [0,1]$. Assume that*

$$w((1-t)x + ty) \geqslant u(x)^{1-t} v(y)^t \qquad \textit{for all } x, y \in \mathbb{R}^d.$$

Then

$$\int_{\mathbb{R}^d} w \geqslant \left(\int_{\mathbb{R}^d} u\right)^{1-t} \left(\int_{\mathbb{R}^d} v\right)^t.$$

Proof. Set $U = \int u$ and $V = \int v$, and assume $U, V > 0$. Let $\alpha = (u/U)\mathrm{d}x$ and $\beta = (v/V)\mathrm{d}x$, and let T be the Brenier map from α to β. Put $T_t = (1-t)\mathrm{Id} + tT$. The change of variables gives

$$\int w \geqslant \int w(T_t(x)) \det(DT_t(x))\mathrm{d}x.$$

By the assumption on w and by $\det((1-t)\mathrm{Id} + tDT) \geqslant \det(DT)^t$ for positive semidefinite DT,

$$\int w \geqslant \int u(x)^{1-t} v(T(x))^t \det(DT(x))^t \mathrm{d}x.$$

Since $T_\sharp \alpha = \beta$, the Monge–Ampère identity gives

$$\det(DT(x)) = \frac{u(x)V}{v(T(x))U}$$

almost everywhere. Substitution yields

$$\int w \geqslant \left(\frac{V}{U}\right)^t \int u(x)\mathrm{d}x = U^{1-t}V^t.$$

Approximation removes the smoothness and positivity assumptions. □

Gaussian logarithmic Sobolev inequality. Transport also gives a short proof of the sharp Gaussian logarithmic Sobolev inequality. The key point is that the Jacobian equation for the Brenier map can be bounded by $\log\det A \leqslant \operatorname{tr}(A - \mathrm{Id})$, and Gaussian integration by parts converts the trace term into a Fisher information.

Proposition 3.19 (Gaussian logarithmic Sobolev inequality)**.** *Let γ_d be the standard Gaussian measure on $\mathbb{R}^d$. If f is smooth, positive, has sufficient decay, and $\int f\,\mathrm{d}\gamma_d = 1$, then*

$$\mathrm{Ent}_{\gamma_d}(f) := \int f \log f \,\mathrm{d}\gamma_d \leqslant \frac{1}{2}\int \frac{\|\nabla f\|^2}{f}\,\mathrm{d}\gamma_d.$$

Proof. Let $\mu = f\gamma_d$, and let $T = \nabla\varphi$ be the Brenier map with $T_\sharp\mu = \gamma_d$. Write $T(x) = x + \theta(x)$. The Monge–Ampère equation reads

$$f(x)e^{-\|x\|^2/2} = e^{-\|T(x)\|^2/2}\det(DT(x)).$$

Thus

$$\log f(x) = -\langle x,\, \theta(x)\rangle - \frac{1}{2}\|\theta(x)\|^2 + \log\det(\mathrm{Id} + D\theta(x)).$$

Since DT is positive semidefinite, $\log\det(\mathrm{Id} + D\theta) \leqslant \operatorname{tr}(D\theta) = \operatorname{div}\theta$. Multiplying by f and integrating with respect to γ_d gives

$$\mathrm{Ent}_{\gamma_d}(f) \leqslant \int f(\operatorname{div}\theta - \langle x,\, \theta\rangle)\,\mathrm{d}\gamma_d - \frac{1}{2}\int f\|\theta\|^2\,\mathrm{d}\gamma_d.$$

Gaussian integration by parts gives

$$\int f(\operatorname{div}\theta - \langle x,\, \theta\rangle)\,\mathrm{d}\gamma_d = -\int \langle \nabla f,\, \theta\rangle\,\mathrm{d}\gamma_d.$$

Completing the square pointwise,

$$-\langle \nabla f,\, \theta\rangle - \frac{1}{2}f\|\theta\|^2 \leqslant \frac{1}{2}\frac{\|\nabla f\|^2}{f}.$$

This proves the inequality. The general Sobolev-class statement follows by approximation. □

Talagrand's transport-entropy inequality. The same Jacobian estimate gives the sharp quadratic transport-entropy inequality for the Gaussian. This result is due to Talagrand [94]; the link with logarithmic Sobolev inequalities and the Otto calculus is one of the starting points of the Otto–Villani theory [69, 99].

Proposition 3.20 (Gaussian T_2 inequality)**.** *Let $\mu = f\gamma_d$ be a probability measure with finite second moment. Then*

$$\frac{1}{2}\mathcal{W}_2^2(\mu, \gamma_d) \leqslant \mathrm{KL}(\mu|\gamma_d) = \int f\log f\,\mathrm{d}\gamma_d.$$

Proof. Assume first that f is smooth and positive. Let $T = \nabla\varphi$ be the Brenier map with $T_\sharp\gamma_d = \mu$, and write $T(x) = x + \theta(x)$. The change of variables gives

$$f(T(x))e^{-\|T(x)\|^2/2}\det(DT(x)) = e^{-\|x\|^2/2}.$$

Hence

$$\log f(T(x)) = \langle x,\, \theta(x)\rangle + \frac{1}{2}\|\theta(x)\|^2 - \log\det(\mathrm{Id} + D\theta(x)).$$

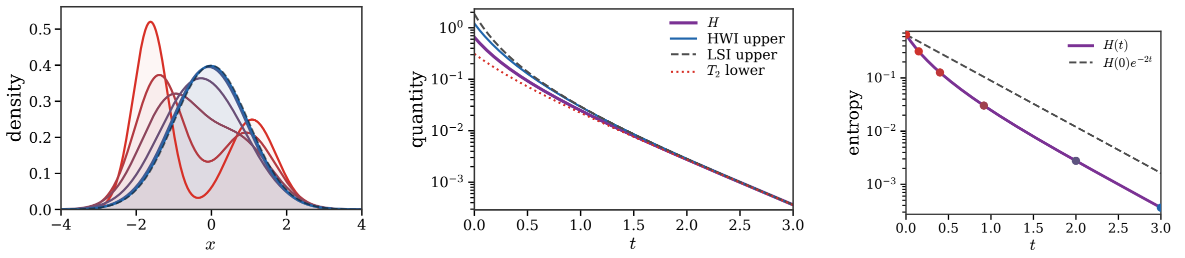
Using again $\log\det(\mathrm{Id} + D\theta) \leqslant \operatorname{div}\theta$ and integrating with respect to γ_d,

$$\mathrm{KL}(\mu|\gamma_d) \geqslant \frac{1}{2}\int \|\theta\|^2\,\mathrm{d}\gamma_d + \int (\langle x,\, \theta\rangle - \operatorname{div}\theta)\,\mathrm{d}\gamma_d.$$

The last integral vanishes by Gaussian integration by parts. Since T is optimal,

$$\int \|\theta\|^2\,\mathrm{d}\gamma_d = \mathcal{W}_2^2(\gamma_d, \mu),$$

which proves the claim. Approximation gives the general case. □



Ornstein–Uhlenbeck densities static inequalities entropy decay

Figure 3.12: Functional inequalities along the Ornstein–Uhlenbeck flow from a one-dimensional Gaussian mixture to the standard Gaussian. The left panel shows the density relaxation, with the Gaussian target as a dashed curve. The middle panel compares the entropy $H = \mathrm{KL}(\alpha_t|\gamma)$ with the HWI upper bound $W\sqrt{I} - W^2/2$, the logarithmic-Sobolev upper bound $I/2$, and the Talagrand lower bound $W^2/2$. The right panel displays the dynamic consequence of log-Sobolev, namely exponential entropy decay below $H(0)e^{-2t}$.

The HWI bridge. The previous two Gaussian inequalities are not isolated facts. The HWI inequality of Otto and Villani [69] compares three quantities at once: the entropy H, the Wasserstein distance W, and the Fisher information I. It is one of the cleanest examples where displacement convexity turns into a functional inequality.

Proposition 3.21 (Otto–Villani HWI inequality). *Let $\beta = \rho_\beta \mathrm{d}x = e^{-V}\mathrm{d}x/Z$ be a smooth probability measure in $\mathcal{P}_2(\mathbb{R}^d)$, and assume that $\nabla^2 V \geq \lambda \mathrm{Id}$ for some $\lambda \in \mathbb{R}$. For an absolutely continuous probability measure $\alpha = \rho_\alpha \mathrm{d}x$ in $\mathcal{P}_2(\mathbb{R}^d)$, define the relative Fisher information*

$$\mathcal{I}_\beta(\alpha) := \int \|\nabla \log \frac{\rho_\alpha}{\rho_\beta}\|^2 \, \mathrm{d}\alpha,$$

with the convention $\mathcal{I}_\beta(\alpha) = +\infty$ if α is not absolutely continuous or if the expression is not defined. Then

$$\mathrm{KL}(\alpha|\beta) \leqslant \mathcal{W}_2(\alpha,\beta)\sqrt{\mathcal{I}_\beta(\alpha)} - \frac{\lambda}{2}\mathcal{W}_2^2(\alpha,\beta).$$

Proof. If $\mathcal{I}_\beta(\alpha) = +\infty$, there is nothing to prove. Assume first that ρ_α is smooth and positive. Let T be the Brenier map from α to β, set $v = T - \mathrm{Id}$, and let $\alpha_t = ((1-t)\mathrm{Id} + tT)_\sharp \alpha$. By Proposition 3.11, $F(\eta) = \mathrm{KL}(\eta|\beta)$ is λ-geodesically convex. Thus the one-dimensional function $f(t) = F(\alpha_t)$ satisfies

$$f(1) \geqslant f(0) + f'(0) + \frac{\lambda}{2}\mathcal{W}_2^2(\alpha,\beta).$$

Since $f(1) = F(\beta) = 0$, it remains to compute the initial derivative. The continuity equation for the geodesic gives $\partial_t \alpha_t + \nabla \cdot (\alpha_t v_t) = 0$ and $v_0 = v$. Hence

$$f'(0) = \int \langle \nabla \log \frac{\rho_\alpha}{\rho_\beta}, v \rangle \, \mathrm{d}\alpha.$$

Therefore

$$\mathrm{KL}(\alpha|\beta) \leqslant -\int \langle \nabla \log \frac{\rho_\alpha}{\rho_\beta}, T - \mathrm{Id} \rangle \, \mathrm{d}\alpha - \frac{\lambda}{2}\mathcal{W}_2^2(\alpha,\beta).$$

Cauchy's inequality and $\int \|T - \mathrm{Id}\|^2 \mathrm{d}\alpha = \mathcal{W}_2^2(\alpha,\beta)$ give the announced bound. The general case follows by the standard approximation and lower-semicontinuity of entropy, Fisher information, and $\mathcal{W}_2$. □

When $\lambda > 0$, Young's inequality $rs - \lambda r^2/2 \leqslant s^2/(2\lambda)$ gives the logarithmic Sobolev estimate $\mathrm{KL}(\alpha|\beta) \leqslant \mathcal{I}_\beta(\alpha)/(2\lambda)$. Thus HWI explains why the same curvature assumption controls both the static inequality and the exponential decay of the associated Fokker–Planck flow below.

Figure 3.12 shows these inequalities on an exact one-dimensional Ornstein–Uhlenbeck relaxation, where all quantities can be evaluated accurately by grid quadrature and quantile inversion.

Back to gradient flows. Functional inequalities become convergence rates once they are combined with the energy-dissipation identity of a Wasserstein gradient flow. This is why the previous inequalities are not merely static estimates: they quantify relaxation of the Fokker–Planck equation.

The logarithmic Sobolev assumption below is the same curvature-controlled mechanism as in the HWI discussion. With the convention used here, the standard Gaussian $\gamma = \mathcal{N}(0, \mathrm{Id})$ satisfies it with $\lambda = 1$, exactly as in Proposition 3.19. More generally, if $\beta = Z^{-1}e^{-V}\mathrm{d}x$ and $\nabla^2 V \geq \lambda \mathrm{Id}$, then the Bakry–Emery criterion gives the logarithmic Sobolev inequality with constant λ. Thus the hypothesis covers strongly log-concave targets, and it is the functional-inequality counterpart of the λ-geodesic convexity of $\mathrm{KL}(\cdot|\beta)$ discussed in Section 3.3.

Proposition 3.22 (Log-Sobolev inequality implies entropy decay)**.** *Let $\beta = e^{-V}\mathrm{d}x/Z$ be a smooth probability measure on $\mathbb{R}^d$. Assume that β satisfies the logarithmic Sobolev inequality with constant $\lambda > 0$,*

$$\mathrm{KL}(\alpha|\beta) \leqslant \frac{1}{2\lambda}\int \|\nabla \log \frac{\mathrm{d}\alpha}{\mathrm{d}\beta}\|^2 \,\mathrm{d}\alpha.$$

Let α_t solve the Wasserstein gradient flow of $F(\alpha) = \mathrm{KL}(\alpha|\beta)$, namely

$$\partial_t \rho_t = \nabla \cdot (\rho_t \nabla V) + \Delta \rho_t, \qquad \alpha_t = \rho_t \mathrm{d}x.$$

Then

$$\mathrm{KL}(\alpha_t|\beta) \leqslant e^{-2\lambda t}\,\mathrm{KL}(\alpha_0|\beta).$$

Proof. The first variation of F is $\log(\rho/\rho_\beta)$ up to an additive constant. Along the smooth Wasserstein gradient flow,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{KL}(\alpha_t|\beta) = -\int \|\nabla \log \frac{\mathrm{d}\alpha_t}{\mathrm{d}\beta}\|^2 \,\mathrm{d}\alpha_t.$$

The logarithmic Sobolev inequality bounds the Fisher information below by $2\lambda\,\mathrm{KL}(\alpha_t|\beta)$. Hence

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{KL}(\alpha_t|\beta) \leqslant -2\lambda\,\mathrm{KL}(\alpha_t|\beta),$$

and Gronwall's lemma gives the result. □

3.6 Second-Order Momentum Flows

The minimizing-movement viewpoint is intrinsically first order: a JKO step produces the next measure by trading off proximity to the previous measure and decrease of the energy. Momentum adds a memory of velocity. The state is therefore no longer only a measure, but a measure together with a tangent, or equivalently phase-space, variable. This makes a direct implicit JKO construction less natural than for first-order flows. We instead use an explicit construction, in the same spirit as optimization and machine-learning practice: momentum reduces zig-zagging across stiff directions, helps iterates keep a coherent direction through shallow valleys, and can accelerate convergence when the geometry is favorable [77, 93]. The goal is to write this inertial idea in a form compatible with Wasserstein particle flows.

Finite-dimensional momentum. Let $F : \mathbb{R}^N \to \mathbb{R}$ be a smooth finite-dimensional objective. Polyak's heavy-ball method [76] augments gradient descent with a velocity variable. With step size $h > 0$ and damping $\gamma > 0$, one convenient velocity form is

$$S^{k+1} = (1-\gamma h)S^k - h\nabla F(X^k), \qquad X^{k+1} = X^k + hS^{k+1}. \tag{3.27}$$

If S^k approximates $\dot{X}(kh)$, this is the explicit Euler discretization of the first-order system

$$\dot{X}(t) = S(t), \qquad \dot{S}(t) = -\gamma S(t) - \nabla F(X(t)), \tag{3.28}$$

or equivalently the damped second-order equation

$$\ddot{X}(t) + \gamma \dot{X}(t) + \nabla F(X(t)) = 0. \tag{3.29}$$

Nesterov's accelerated method [67] uses a closely related extrapolation but evaluates the gradient at a look-ahead point,

$$Y^k = X^k + \theta_k(X^k - X^{k-1}), \qquad X^{k+1} = Y^k - h\nabla F(Y^k), \tag{3.30}$$

with $\theta_k = (k-1)/(k+r-1)$ for the classical convex schedule $r = 3$. Its scaling limit is the time-dependent damping equation

$$\ddot{X}(t) + \frac{r}{t}\dot{X}(t) + \nabla F(X(t)) = 0. \tag{3.31}$$

This ODE interpretation is due to Su, Boyd and Candès [92]; see also the variational perspective of Wibisono, Wilson and Jordan [103], and the inertial dynamics with Hessian-driven damping studied by Attouch, Peypouquet and Redont [3]. High-resolution ODE limits refine this picture by keeping terms that distinguish heavy-ball and Nesterov dynamics beyond their leading continuous-time behavior [86]. For the Wasserstein extension below, the autonomous heavy-ball form (3.28) is the cleanest template.

Empirical Wasserstein lift. We now take a functional f on measures and lift it to particle configurations. For $X = (x_1, \dots, x_n) \in (\mathbb{R}^d)^n$, write

$$\mu_X := \frac{1}{n}\sum_{i=1}^{n} \delta_{x_i}, \qquad F(X) := f(\mu_X). \tag{3.32}$$

This is the same empirical lift as in (3.4), and is the particle viewpoint underlying mean-field analyses of wide neural networks and Wasserstein gradient methods [19, 63, 81]. The important point is that the configuration space is not used with the plain Euclidean metric, but with the empirical metric

$$\|\dot{X}\|_n^2 := \frac{1}{n}\sum_{i=1}^{n} \|\dot{x}_i\|^2,$$

which is the metric induced by $\mathcal{W}_2$ on equally weighted empirical measures with fixed labels. Moving only x_i to $x_i + \varepsilon\xi$ gives

$$\left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\right|_{\varepsilon=0} F(x_1, \dots, x_i + \varepsilon\xi, \dots, x_n) = \frac{1}{n}\langle \nabla\delta f(\mu_X)(x_i),\, \xi\rangle.$$

Thus, if $\nabla^{(n)}F$ denotes the gradient for the metric $\|\cdot\|_n$, then

$$(\nabla^{(n)}F(X))_i = n\nabla_{x_i}F(X) = \nabla\delta f(\mu_X)(x_i) = \nabla_{\mathcal{W}} f(\mu_X)(x_i). \tag{3.33}$$

The first-order Wasserstein particle descent is therefore $\dot{x}_i = v_{\mu_X}(x_i)$ with

$$v_\mu(x) := -\nabla_{\mathcal{W}} f(\mu)(x) = -\nabla\delta f(\mu)(x). \tag{3.34}$$

Applying the heavy-ball equation to the empirical metric gives the second-order Wasserstein momentum system

$$\dot{x}_i(t) = s_i(t), \qquad \dot{s}_i(t) = -\gamma s_i(t) + v_{\mu_t}(x_i(t)), \qquad \mu_t = \frac{1}{n}\sum_{i=1}^{n} \delta_{x_i(t)}. \tag{3.35}$$

Equivalently,

$$\ddot{x}_i(t) + \gamma\dot{x}_i(t) = v_{\mu_t}(x_i(t)) = -\nabla_{\mathcal{W}} f(\mu_t)(x_i(t)).$$

This is an explicit inertial particle method for the measure energy f: the Wasserstein steepest-descent field acts as an acceleration rather than as an instantaneous velocity. The statement is deliberately formal at this level; rigorous mean-field limits require stability and regularity assumptions on v_μ, as in propagation-of-chaos arguments for classical Vlasov dynamics and in mean-field analyses of interacting learning systems.

Proposition 3.23 (Discrete mechanical energy dissipation)**.** *Assume that f and the particle solution are smooth enough to justify differentiating the identities above along* (3.35)*. Define*

$$\mathcal{H}_n(X, S) := F(X) + \frac{1}{2n}\sum_{i=1}^{n} \|s_i\|^2, \qquad S = (s_1, \dots, s_n).$$

Then along (3.35)*,*

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{H}_n(X(t), S(t)) = -\frac{\gamma}{n}\sum_{i=1}^{n} \|s_i(t)\|^2 \leqslant 0. \tag{3.36}$$

Proof. By (3.33) and the definition $v_\mu = -\nabla_W f(\mu)$, one has $\nabla_{x_i} F(X) = -v_{\mu_X}(x_i)/n$. Hence

$$\frac{\mathrm{d}}{\mathrm{d}t} F(X(t)) = \sum_{i=1}^n \langle \nabla_{x_i} F(X(t)),\, s_i(t)\rangle = -\frac{1}{n}\sum_{i=1}^n \langle v_{\mu_t}(x_i(t)),\, s_i(t)\rangle.$$

On the other hand,

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2n}\sum_i \|s_i(t)\|^2 = \frac{1}{n}\sum_i \langle s_i(t),\, -\gamma s_i(t) + v_{\mu_t}(x_i(t))\rangle.$$

The force terms cancel, leaving (3.36). □

The identity says that friction dissipates exactly the kinetic part of the mechanical energy. In the undamped case $\gamma = 0$, the same computation gives conservation of $\mathcal{H}_n$.

Phase-space formulation. Because momentum is part of the state, the mean-field object is not merely a measure on positions. It is a law on phase space (x, s), whose spatial marginal is the current distribution of particles. Here s is a Lagrangian particle velocity; it should not be confused with the Eulerian momentum measure $m = \rho v$ used in the Benamou–Brenier convex formulation (2.10). The particle ODE then becomes a transport equation in phase space. This is the deterministic analogue of the kinetic mean-field viewpoint used for interacting particle systems; in learning, heavy-ball mean-field limits have recently been used to analyze wide networks trained with momentum [104].

Proposition 3.24 (Phase-space Liouville formulation)**.** *Let v_μ be smooth enough that (3.35) has a classical solution. Define the empirical phase-space measure*

$$\eta_t^n := \frac{1}{n}\sum_{i=1}^n \delta_{(x_i(t), s_i(t))} \in \mathcal{P}(\mathbb{R}_x^d \times \mathbb{R}_s^d), \qquad \mu_t^n = (\pi_x)_\sharp \eta_t^n.$$

Then η_t^n solves, in the sense of distributions,

$$\partial_t \eta_t^n + \nabla_x \cdot (s\, \eta_t^n) + \nabla_s \cdot \big((-\gamma s + v_{\mu_t^n}(x))\, \eta_t^n\big) = 0. \tag{3.37}$$

If, as $n \to \infty$, the empirical phase-space laws converge to a smooth density $\eta_t(x, s)$ and the nonlinear fields converge accordingly, the limiting kinetic equation is

$$\partial_t \eta_t + \nabla_x \cdot (s\, \eta_t) + \nabla_s \cdot \big((-\gamma s + v_{\mu_t}(x))\, \eta_t\big) = 0, \qquad \mu_t = (\pi_x)_\sharp \eta_t. \tag{3.38}$$

Proof. Let $\varphi \in C_c^\infty(\mathbb{R}_x^d \times \mathbb{R}_s^d)$. Along the particle system,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi\, \mathrm{d}\eta_t^n = \frac{1}{n}\sum_{i=1}^n \left[\langle \nabla_x \varphi(x_i, s_i),\, s_i\rangle + \langle \nabla_s \varphi(x_i, s_i),\, -\gamma s_i + v_{\mu_t^n}(x_i)\rangle\right].$$

Rewriting the right-hand side as an integral against η_t^n and integrating by parts gives (3.37). Passing formally to the mean-field limit gives (3.38). □

Remark 3.25 (Newtonian limit). If damping is removed, or if inertia dominates friction after rescaling, (3.35) becomes

$$\ddot{x}_i(t) = v_{\mu_t}(x_i(t)).$$

This is Newton's law with acceleration field v_{μ_t}. Thus first-order Wasserstein gradient flow reads the force as a velocity, while second-order momentum flow reads the same force as an acceleration. The price is that the state is no longer just the spatial measure μ_t; it is the phase-space law of position and velocity. In this limit the flow is Hamiltonian rather than dissipative whenever the force comes from a smooth interaction energy and $\gamma = 0$.

Example 3.26 (Quadratic energies and gravitational attraction). Consider the interaction energy

$$f(\mu) = \frac{1}{2}\iint k(x, y)\, \mathrm{d}\mu(x)\mathrm{d}\mu(y) + \int V(x)\, \mathrm{d}\mu(x), \tag{3.39}$$

with $k(x, y) = k(y, x)$. Its first variation is

$$\delta f(\mu)(x) = \int k(x, y)\,\mathrm{d}\mu(y) + V(x),$$

and therefore

$$v_\mu(x) = -\int \nabla_x k(x, y)\,\mathrm{d}\mu(y) - \nabla V(x). \tag{3.40}$$

For the attractive Newtonian kernel in three dimensions, $k(x, y) = -G/\|x - y\|$, understood with a short-distance regularization if needed, this gives

$$v_\mu(x) = G \int \frac{y - x}{\|x - y\|^3}\,\mathrm{d}\mu(y) - \nabla V(x),$$

the usual mean-field gravitational acceleration. The phase-space equation (3.38) is then a damped Vlasov-type equation. This kinetic viewpoint is classical in statistical physics: Boltzmann introduced the description of gases through phase-space distributions, the collisionless mean-field limit gives Liouville–Vlasov equations, and adding collision operators leads to the Boltzmann equation [14, 97, 7].

The numerical illustration below uses the energy-distance kernel

$$k(x, y) = -\|x - y\|, \qquad f(\mu) = \mathrm{MMD}_k^2(\mu, \beta) = \iint k(x, y)\,\mathrm{d}(\mu - \beta)(x)\mathrm{d}(\mu - \beta)(y).$$

This is the usual squared energy distance between μ and β. Away from the diagonal, its Wasserstein velocity is

$$v_\mu(x) = 2 \int \frac{x - y}{\|x - y\|}\,\mathrm{d}(\mu - \beta)(y).$$

For numerical stability, Figure 3.13 uses the smoothed distance $\sqrt{\|x - y\|^2 + \delta^2}$ with a small δ. It compares the first-order flow with the undamped Newton lift $\ddot{x} = v_{\mu_t}(x) = -\nabla_W f(\mu_t)(x)$, initialized with zero velocity. The initial measure is a single Gaussian located to the left of a two-Gaussian target mixture. Both the transported measure and the target are discretized by large empirical clouds, and the smooth density views are KDE renderings of these particles. The trajectory panels display only a representative subset of initial particles selected by farthest-point sampling.

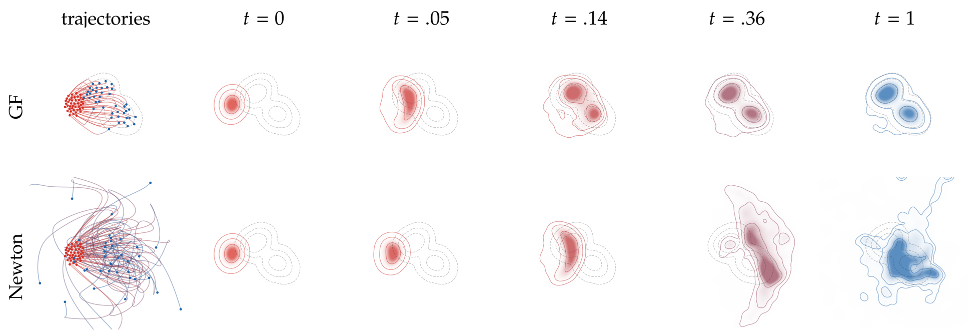


Figure 3.13: First-order and Newton Wasserstein particle flows for the squared MMD with the energy-distance kernel $k(x, y) = -\|x - y\|$. The first row follows the overdamped Wasserstein gradient flow. The second row follows the infinite-momentum Newton dynamics $\ddot{x} = -\nabla_W f(\mu_t)(x)$, so the same Wasserstein force is read as an acceleration rather than a velocity. Both rows use the same large empirical source cloud and the same empirical two-Gaussian target cloud. Dashed gray contours show a KDE of the target particles, colored panels show KDEs of the evolving transported particles, and the left panels show farthest-point-subsampled trajectories.

Entropy without an empirical energy. The previous interaction example has a genuine value on empirical measures: once the particles are known, the double sum approximates the interaction energy. Entropy gives the complementary situation. Its Wasserstein gradient flow is the simplest diffusion equation, but the entropy itself is finite only on absolutely continuous measures. A particle method must therefore introduce a density, or score, closure before the inertial template can be applied.

Example 3.27 (Entropy-driven inertial flow). Let

$$f(\mu) = \begin{cases} \int_{\mathbb{R}^d} \rho(x) \log \rho(x) \, \mathrm{d}x, & \text{if } \mu = \rho \mathrm{d}x, \\ +\infty, & \text{otherwise.} \end{cases}$$

Thus $f(\mu_X) = +\infty$ for an empirical measure $\mu_X = n^{-1} \sum_i \delta_{x_i}$, and the finite-dimensional lift $F(X) = f(\mu_X)$ is not meaningful. For $\mu = \rho \mathrm{d}x$ with a smooth positive density, however,

$$\delta f(\mu)(x) = \log \rho(x) + 1, \qquad v_\mu(x) = -\nabla \log \rho(x).$$

The first-order Wasserstein gradient flow is therefore the heat equation $\partial_t \rho_t = \Delta \rho_t$. If the spatial marginal remains smooth and positive, the corresponding formal second-order phase-space equation is

$$\partial_t \eta_t + \nabla_x \cdot (s \eta_t) + \nabla_s \cdot \big((-\gamma s - \nabla \log \rho_t(x)) \eta_t\big) = 0, \qquad \rho_t(x) = \int_{\mathbb{R}^d} \eta_t(x, s) \, \mathrm{d}s. \tag{3.41}$$

This is a kinetic, inertial version of entropy diffusion: the entropy score acts as an acceleration rather than as an instantaneous velocity.

For the numerical illustration, we add the quadratic confinement $V(x) = \|x\|^2/2$. In one dimension the overdamped equation becomes the Ornstein–Uhlenbeck Fokker–Planck equation

$$\partial_t \rho_t = \partial_{xx} \rho_t + \partial_x (x \rho_t),$$

whose stationary law is the centered Gaussian $\mathcal{N}(0, 1)$. The Newton lift is discretized directly in phase space: for a density $\eta_t(x, s)$ with spatial marginal $\rho_t(x) = \int \eta_t(x, s) \, \mathrm{d}s$, it reads

$$\partial_t \eta_t + \partial_x (s \eta_t) + \partial_s \big((-\partial_x \log \rho_t(x) - x) \eta_t\big) = 0.$$

Both equations are solved by finite differences on fixed grids, with a mild grid smoothing only to evaluate the logarithmic derivative in the Newton row. The first row of Figure 3.14 shows only the spatial density of the Wasserstein gradient flow. The second and third rows show, for the Newton equation, respectively the spatial marginal and the full phase-space density; the latter uses white for zero mass and black for high mass.

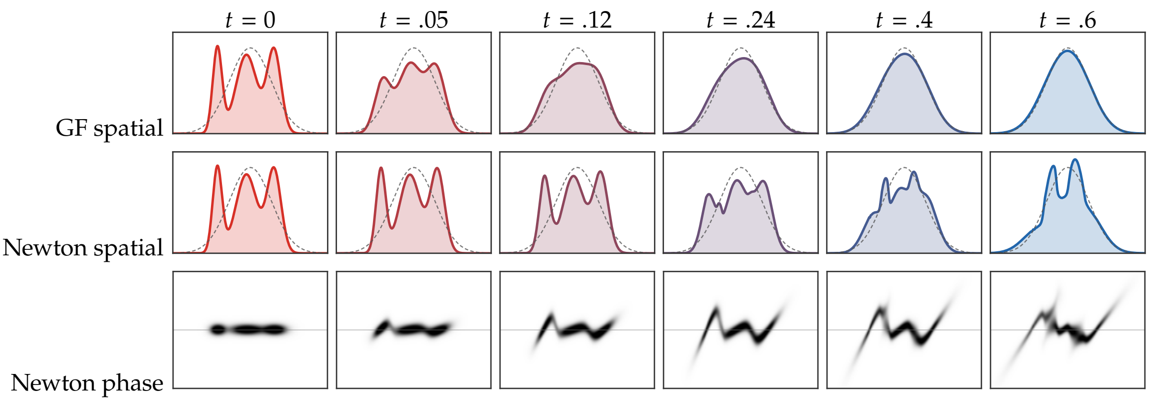


Figure 3.14: Finite-difference entropy and inertial entropy flows in one dimension. The initial spatial density is a three-Gaussian mixture with different widths and overlapping components. The top row solves the confined entropy Wasserstein gradient flow, i.e. the OU/Fokker–Planck equation, whose stationary density is the dashed centered Gaussian. The two lower rows solve the undamped Newton lift on a grid in (x, s), initialized with a narrow centered speed distribution: its spatial marginal is shown in color and the full phase-space density below in grayscale.

3.7 Normalized Spectral Wasserstein Dynamics

The preceding section added inertia to the usual Wasserstein gradient flow. A different modification, closer to many large-scale optimizers, is to normalize the descent direction itself. Normalized gradient methods replace the Euclidean steepest-descent direction by the solution of a norm-constrained linear minimization problem [71]; this point of view is useful for ordinary normalized SGD [66, 26], for tensor-aware optimizers such as Shampoo [42], and for Muon-type spectral normalizations used in large-scale training [49, 57]. We now describe the mean-field version developed in [74]. The trace gauge gives the classical $\mathcal{W}_2$ flow, while the operator gauge $\gamma(M) = \lambda_{\max}(M)$ gives the idealized Muon geometry.

Spectral tangent norm. For this survey, a monotone spectral gauge is a convex, positively 1-homogeneous map $\gamma : \mathbb{S}^d_+ \to \mathbb{R}_+$, vanishing only at 0, invariant under orthogonal conjugation and increasing for the Loewner order. Its polar set is

$$\mathcal{B}_\gamma := \{A \succeq 0\,;\ \mathrm{tr}(AM) \leqslant \gamma(M) \quad \text{for all } M \succeq 0\}\,,$$

so that $\gamma(M) = \sup_{A\in\mathcal{B}_\gamma} \mathrm{tr}(AM)$ for $M \succeq 0$. For a probability measure α and a velocity field $v \in L^2(\alpha; \mathbb{R}^d)$, define the spectral tangent norm

$$\mathcal{N}_{\gamma,\alpha}(v)^2 := \gamma\left(\int v(x)v(x)^\top \mathrm{d}\alpha(x)\right). \tag{3.42}$$

It measures the covariance of the velocity field before applying the gauge. Thus the geometry is global in direction: two velocities with the same pointwise L^2 norm may have different costs if their velocity covariance is distributed differently across directions.

Definition 3.28 (Spectral steepest-descent direction)**.** For $g \in L^2(\alpha; \mathbb{R}^d)$, the spectral steepest-descent direction is any minimizer

$$\mathcal{J}^\alpha_\gamma(g) \in \operatorname*{argmin}_{v\in L^2(\alpha;\mathbb{R}^d)} \left\{\int \langle g(x), v(x)\rangle \mathrm{d}\alpha(x) + \frac{1}{2}\mathcal{N}_{\gamma,\alpha}(v)^2\right\}. \tag{3.43}$$

The field g should be read as the Wasserstein gradient $g = \nabla_{\mathcal{W}} f(\alpha) = \nabla\delta f(\alpha)$. Since $\mathcal{N}_{\gamma,\alpha}$ is 1-homogeneous in v, (3.43) is the squared-norm steepest-descent step associated with the spectral tangent norm. Its ray is the norm-constrained linear minimization oracle: after rescaling, $\mathcal{J}^\alpha_\gamma(g)$ minimizes $\int\langle g, w\rangle \mathrm{d}\alpha$ over $\mathcal{N}_{\gamma,\alpha}(w) \leqslant 1$. Thus fixed-speed normalized algorithms keep the same direction but choose a different scalar normalization. In the trace case, $\mathcal{N}_{\mathrm{tr},\alpha}(v)^2 = \int \|v\|^2 \mathrm{d}\alpha$ and (3.43) gives $\mathcal{J}^\alpha_{\mathrm{tr}}(g) = -g$. For other gauges, the velocity is globally preconditioned by the covariance of g under α.

Proposition 3.29 (Polar formula for the spectral direction)**.** *Let*

$$S_\alpha(g) := \int g(x)g(x)^\top \mathrm{d}\alpha(x).$$

Assume that the inverse-trace problem admits a positive definite minimizer

$$A^\star_\alpha \in \operatorname*{argmin}_{A\in\mathcal{B}_\gamma\cap\mathbb{S}^d_{++}} \mathrm{tr}\left(A^{-1}S_\alpha(g)\right).$$

Then

$$\mathcal{J}^\alpha_\gamma(g)(x) = -(A^\star_\alpha)^{-1}g(x) \tag{3.44}$$

is a spectral steepest-descent direction. If the minimizer lies on the boundary of $\mathcal{B}_\gamma$, the same formula is understood by a limiting argument along positive definite minimizers, or equivalently by replacing the inverse by the corresponding inverse on the range of $S_\alpha(g)$.

Proof. Using the polar formula $\gamma(M) = \sup_{A\in\mathcal{B}_\gamma} \mathrm{tr}(AM)$, the objective in (3.43) can be written as

$$\sup_{A\in\mathcal{B}_\gamma} \left\{\int \langle g, v\rangle \mathrm{d}\alpha + \frac{1}{2}\int v^\top Av \mathrm{d}\alpha\right\}.$$

For a fixed $A \succ 0$, the quadratic lower bound

$$\int \langle g, v\rangle \mathrm{d}\alpha + \frac{1}{2}\int v^\top Av \mathrm{d}\alpha$$

is minimized by $v_A = -A^{-1}g$, with value $-\frac{1}{2}\operatorname{tr}(A^{-1}S_\alpha(g))$. Let $v_\star = -(A^\star_\alpha)^{-1}g$ and

$$M_\star = \int v_\star(x)v_\star(x)^\top \mathrm{d}\alpha(x) = (A^\star_\alpha)^{-1}S_\alpha(g)(A^\star_\alpha)^{-1}.$$

Since $A^\star_\alpha$ minimizes $A \mapsto \operatorname{tr}(A^{-1}S_\alpha(g))$ over the convex polar set, its first-order optimality condition gives

$$\operatorname{tr}(AM_\star) \leqslant \operatorname{tr}(A^\star_\alpha M_\star) \qquad \text{for all } A \in \mathcal{B}_\gamma.$$

Thus $A^\star_\alpha$ is active in the polar representation of $\gamma(M_\star)$, and

$$\gamma(M_\star) = \operatorname{tr}(A^\star_\alpha M_\star) = \operatorname{tr}((A^\star_\alpha)^{-1}S_\alpha(g)).$$

For any v, the polar formula gives

$$\int \langle g, v\rangle \mathrm{d}\alpha + \frac{1}{2}\mathcal{N}_{\gamma,\alpha}(v)^2 \geqslant \int \langle g, v\rangle \mathrm{d}\alpha + \frac{1}{2}\int v^\top A^\star_\alpha v \,\mathrm{d}\alpha,$$

and the right-hand side is minimized at $v_\star$. The activity identity above shows that equality holds at $v_\star$, hence $v_\star$ minimizes (3.43). The boundary case follows by approximation with positive definite matrices in the polar set. □

For the trace gauge, $\mathcal{B}_\gamma = \{A\ ;\ 0 \preceq A \preceq \mathrm{Id}\}$ and the minimizer is $A^\star_\alpha = \mathrm{Id}$. For the operator gauge $\gamma(M) = \lambda_{\max}(M)$, one has $\mathcal{B}_\gamma = \{A \succeq 0\ ;\ \operatorname{tr}(A) \leqslant 1\}$; if $S_\alpha(g) \succ 0$, then

$$A^\star_\alpha = \frac{S_\alpha(g)^{1/2}}{\operatorname{tr}(S_\alpha(g)^{1/2})}, \qquad \mathcal{J}^\alpha_\gamma(g)(x) = -\operatorname{tr}(S_\alpha(g)^{1/2})\, S_\alpha(g)^{-1/2}g(x).$$

This is the continuum covariance-normalization formula that becomes the Muon polar factor for empirical measures.

JKO interpretation. Let $\mathcal{W}_\gamma$ denote the spectral Wasserstein distance generated by the same gauge,

$$\mathcal{W}_\gamma(\alpha,\beta)^2 := \inf_{\pi\in\mathcal{U}(\alpha,\beta)} \gamma\left(\int (x-y)(x-y)^\top \mathrm{d}\pi(x,y)\right).$$

The normalized dynamics can be obtained formally from the JKO scheme

$$\alpha^{k+1} \in \operatorname*{argmin}_{\alpha\in\mathcal{P}_2(\mathbb{R}^d)} \frac{1}{2\tau}\mathcal{W}_\gamma(\alpha^k,\alpha)^2 + f(\alpha). \tag{3.45}$$

For a small displacement $\alpha = (\mathrm{Id}+\tau v)_\sharp \alpha^k$, the dynamic interpretation of $\mathcal{W}_\gamma$ developed in [74] gives the tangent expansion

$$\mathcal{W}_\gamma(\alpha^k, (\mathrm{Id}+\tau v)_\sharp\alpha^k)^2 = \tau^2\mathcal{N}_{\gamma,\alpha^k}(v)^2 + o(\tau^2).$$

Combining this with the first-variation expansion of f yields the infinitesimal minimization problem (3.43).

Proposition 3.30 (Formal normalized spectral gradient flow)**.** *Assume that f has a smooth first variation and that the tangent expansion of $\mathcal{W}_\gamma$ above is valid along smooth perturbations. Then the formal small-step limit of* (3.45) *solves*

$$\partial_t\alpha_t + \operatorname{div}\left(\alpha_t\, \mathcal{J}^{\alpha_t}_\gamma(\nabla_W f(\alpha_t))\right) = 0. \tag{3.46}$$

Equivalently, if $A^\star_t$ solves the inverse-trace problem for

$$S_t = \int \nabla_W f(\alpha_t)(x)\nabla_W f(\alpha_t)(x)^\top \mathrm{d}\alpha_t(x),$$

then the velocity is

$$v_t(x) = -(A^\star_t)^{-1}\nabla_W f(\alpha_t)(x).$$

Moreover, along smooth solutions,

$$\frac{\mathrm{d}}{\mathrm{d}t}f(\alpha_t) = -\mathcal{N}_{\gamma,\alpha_t}(v_t)^2.$$

Proof. Write $\alpha = (\mathrm{Id} + \tau v)_\sharp \alpha_t$. The spectral JKO objective equals, up to terms independent of v and $o(\tau)$,

$$\tau \left[\frac{1}{2} \mathcal{N}_{\gamma,\alpha_t}(v)^2 + \int \langle \nabla_W f(\alpha_t)(x), v(x) \rangle \mathrm{d}\alpha_t(x) \right].$$

Minimizing the bracket gives $v = \mathcal{J}_\gamma^{\alpha_t}(\nabla_W f(\alpha_t))$. Substituting this velocity into the transport expansion

$$(\mathrm{Id} + \tau v)_\sharp \alpha_t = \alpha_t - \tau \operatorname{div}(\alpha_t v) + o(\tau)$$

gives (3.46). Finally, the variation formula gives $\frac{\mathrm{d}}{\mathrm{d}t} f(\alpha_t) = \int \langle \nabla_W f(\alpha_t), v_t \rangle \mathrm{d}\alpha_t$. Since v_t minimizes a 2-homogeneous penalized objective, the one-dimensional rescaling $s \mapsto s v_t$ has derivative zero at $s = 1$, hence

$$\int \langle \nabla_W f(\alpha_t), v_t \rangle \mathrm{d}\alpha_t + \mathcal{N}_{\gamma,\alpha_t}(v_t)^2 = 0,$$

which proves the dissipation identity. □

Empirical and Muon limits. For empirical measures $\alpha_X = n^{-1} \sum_i \delta_{x_i}$, stack the velocities $v(x_i)$ into a matrix $V \in \mathbb{R}^{n \times d}$ and the gradients $g(x_i)$ into a matrix $G \in \mathbb{R}^{n \times d}$. By 1-homogeneity of γ,

$$\mathcal{N}_{\gamma,\alpha_X}(v)^2 = \gamma\left(\frac{1}{n} V^\top V\right) = \frac{1}{n} \gamma(V^\top V), \qquad \int \langle g, v \rangle \mathrm{d}\alpha_X = \frac{1}{n} \operatorname{tr}(G^\top V).$$

Thus (3.43) is, up to the harmless factor $1/n$, the finite-dimensional matrix LMO

$$J_\gamma(G) \in \operatorname*{argmin}_{V \in \mathbb{R}^{n \times d}} \left\{ \operatorname{tr}(G^\top V) + \frac{1}{2} \gamma(V^\top V) \right\}. \tag{3.47}$$

The empirical flow is therefore

$$\dot{X}(t) = J_\gamma(G_X(t)), \qquad (G_X(t))_i = \nabla_W f(\alpha_{X(t)})(x_i(t)).$$

Here G_X is the gradient for the empirical Wasserstein metric $\|\dot{X}\|_n^2 = n^{-1} \sum_i \|\dot{x}_i\|^2$, not the ordinary Euclidean gradient of $f(\alpha_X)$. If one rewrites the same rule with the unweighted matrix convention, it corresponds to the lift $X \mapsto n f(\alpha_X)$; using $X \mapsto f(\alpha_X)$ only rescales time by the factor n. If $G = U \operatorname{diag}(\sigma_i) W^\top$ and $\gamma(M) = \lambda_{\max}(M)$, then

$$J_\gamma(G) = -\|G\|_{S_1} U W^\top = -\operatorname{tr}\left((G^\top G)^{1/2} \right) G (G^\top G)^{\dagger/2}. \tag{3.48}$$

The ray of this direction is the polar factor $-UW^\top$, and the scalar factor $\|G\|_{S_1}$ can be absorbed into a time or step-size normalization. This is the exact-polar, full-batch, continuous-time idealization of Muon [49, 74]. Practical Muon implementations replace the polar factor by a few Newton–Schulz iterations for GPU efficiency [57]; this polynomial approximation fits the same spectral-normalization philosophy, although adaptive rescaling and finite momentum make the practical algorithm more than a position-only metric gradient flow.

The effect of this spectral normalization is visible already in the homogeneous two-layer ReLU model discussed in Section 3.4. Figure 3.15 reproduces, in the style of the book figures, the numerical comparison from [74]: the same teacher network, same initialization and same empirical first variation are evolved either by the W_2 particle flow or by the operator-gauge Muon direction.

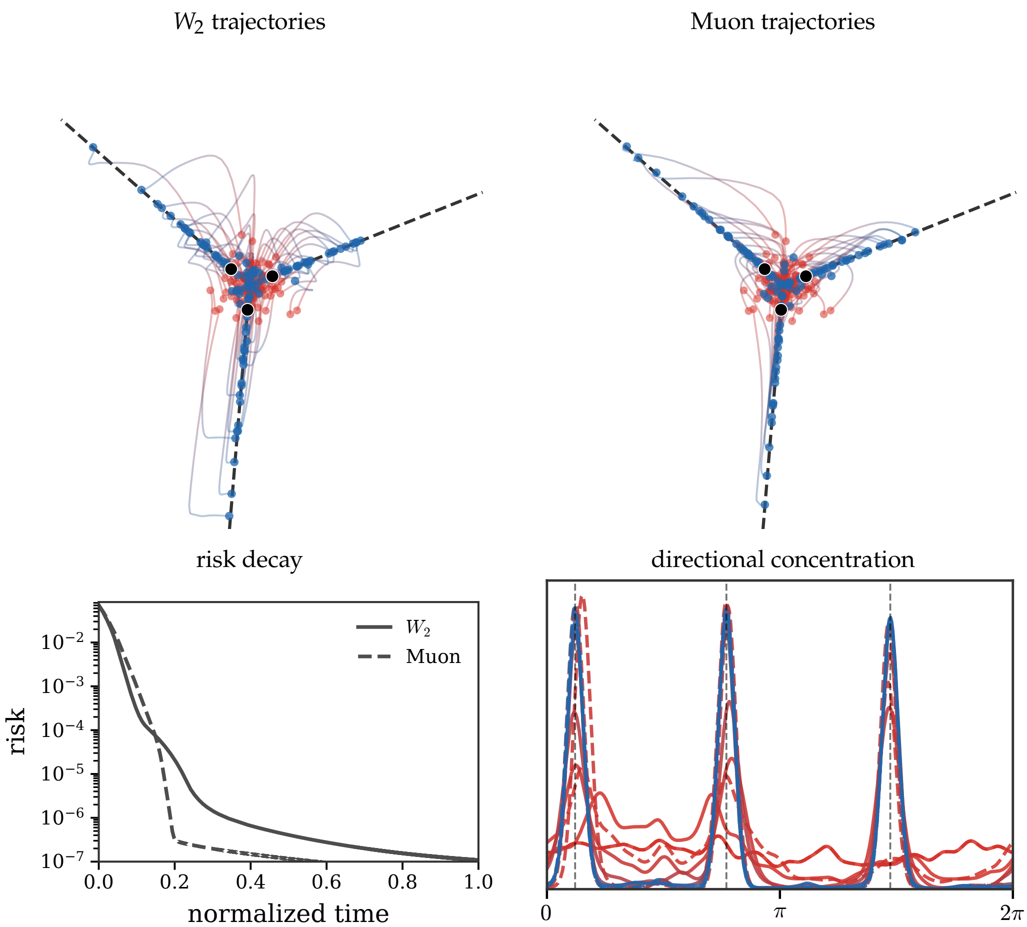


Figure 3.15: Wasserstein versus Muon mean-field training of a homogeneous ReLU model. The top row displays trajectories in the reduced homogeneous coordinates $(|u|v_1, |u|v_2)$; black dashed rays are the teacher directions. The bottom row compares the empirical square-loss decay and the weighted angular density of neurons. Solid curves correspond to the W_2 flow, dashed curves to the operator-gauge Muon flow. Both methods use the same particles and first variation; the spectral normalization produces a more coherent collective alignment and reaches the low-risk regime earlier in normalized time.

Remark 3.31 (Relation with momentum). The section above introduced momentum by adding a velocity variable. Practical Muon instead often averages gradients before applying the spectral normalization. In continuous time this leads to a phase-space system for pairs (x, a),

$$\tau \dot{a}_t = \nabla_W f(\alpha_t)(x_t) - a_t, \qquad \dot{x}_t = -(A_t^\star)^{-1} a_t,$$

where η_t is the law of (x_t, a_t), $\alpha_t = (\pi_x)_\sharp \eta_t$, and $A_t^\star$ is selected from the covariance of the averaged force,

$$A_t^\star \in \operatorname*{argmin}_{A \in \mathcal{B}_\gamma \cap \mathbb{S}^d_{++}} \operatorname{tr}\left(A^{-1} \int a a^\top \mathrm{d}\eta_t(x, a) \right).$$

When $\tau = 0$, the auxiliary force relaxes instantly to $a_t = \nabla_W f(\alpha_t)(x_t)$ and one recovers (3.46). For $\tau > 0$, the dynamics is kinetic and is not, in general, a gradient flow of a distance on the spatial marginal alone.

SECTION 4

Generative Models via Transportation

The preceding gradient-flow calculus is variational. Modern machine-learning models often use the same transportation language more broadly: one may prescribe an interpolation and regress its velocity, fit a one-step generator to a descent field, or view network depth as a continuous transport of token measures. The examples below separate what is genuinely a Wasserstein gradient flow from what is a transportation dynamics with a useful geometric interpretation.

4.1 Generative Models via Flow Matching

Flow matching constructs a generative map by learning the velocity field of an interpolation. The key computational insight is that a constrained continuity-equation problem can be trained by an unconstrained regression.

Generative models aim to build a transportation map T between a reference distribution α (typically an isotropic Gaussian) and the target data distribution β. Since such reference measures are non-atomic, a measurable map with $T_\sharp\alpha = \beta$ exists on standard Borel spaces, for instance by identifying both probability spaces with the unit interval and using a quantile-type rearrangement. This abstract existence statement is much weaker than having an explicit and numerically stable construction of T. Optimal transport is one approach to achieving this, but it is computationally expensive and raises questions about how to estimate it from samples. A different route is to prescribe an interpolation between noise and data, learn its velocity, and obtain T by integrating a time-dependent vector field v_t. This point of view sits at the meeting point of two literatures, surveyed from a transport perspective in [73]. The diffusion branch builds on score matching [48], denoising score matching [100], nonequilibrium noising chains [87], denoising diffusion probabilistic models [47], score-based generative modeling [88], and the continuous-time score-SDE/probability-flow formulation [89]. The deterministic regression branch was introduced, essentially in parallel, under three closely related names: flow matching [56], rectified flow [58], and stochastic interpolants [1]. In all three cases, the computational object is a velocity field whose regression loss avoids simulating the learned ODE during training. This vector field v_t is obtained by constructing an interpolation α_t and then finding v_t using the least-squares formula (2.7). As we will explain, for a specific class of interpolation (obtained by a parametric push-forward), this v_t can be obtained by avoiding explicitly inverting a Laplacian and instead computing a simple conditional expectation. This conditional expectation can itself be estimated by solving another least-squares problem, but this time unconstrained, making the estimation feasible from finite samples of α and β.

Stochastic interpolant. The word "stochastic" can hide two different levels of randomness. We first use the simpler one: after drawing a latent variable $U \sim \pi$, the path $t \mapsto P_t(U)$ is deterministic and differentiable. The randomness only comes from the initial draw of U; after taking the push-forward law, $\alpha_t = (P_t)_\sharp\pi$ is a deterministic curve of measures and obeys an ordinary continuity equation. This is the setting behind the stochastic-interpolant construction recalled in Remark 4.3, and behind the flow-matching and rectified-flow regressions below. Genuine temporal noise, where the path itself has Brownian fluctuations, is different and is discussed in Remark 4.4.

We assume first that α_t is obtained by pushing a latent distribution $\pi \in \mathcal{P}(\mathbb{R}^{d'})$ through a time-dependent map $P_t : \mathbb{R}^{d'} \to \mathbb{R}^d$; the latent dimension d' may be larger than the data dimension d, i.e.

$$\forall t \in [0,1], \quad \alpha_t := (P_t)_\sharp\pi. \tag{4.1}$$

The basic two-endpoint construction already covers most flow-matching paths used in practice.

Example 4.1 (Linear two-endpoint deterministic interpolants). Set $d' = 2d$, write $(x, y) \in \mathbb{R}^d \times \mathbb{R}^d$, and choose $P_0(x, y) = x$ and $P_1(x, y) = y$. If π has marginals (α_0, α_1), then $\alpha_t = (P_t)_\sharp\pi$ interpolates between the two endpoint laws. The simplest choices are the independent coupling $\pi = \alpha_0 \otimes \alpha_1$ and the straight path

$$P_t(x, y) = (1 - t)x + ty.$$

With this linear path and an arbitrary coupling π, the regression below is the common core of flow matching and rectified flow: Lipman et al. emphasize conditional probability paths and simulation-free training of continuous normalizing flows, while rectified flow emphasizes straight couplings, reflow, and the possibility of reducing transport costs and discretization error [56, 58]. More complex constructions are possible when sampling from π remains simple. Static auxiliary randomness is still handled by enlarging the latent variable, while Brownian noise leads to the diffusion correction described below; this is the broader stochastic-interpolant viewpoint connecting deterministic flows, probability-flow ODEs and diffusion SDEs [1].

If $\pi = \alpha \otimes \beta$ and $\alpha = \frac{1}{n}\sum_i \delta_{x_i}$, $\beta = \frac{1}{m}\sum_j \delta_{y_j}$, then α_t consists of $n \times m$ Dirac masses

$$\alpha_t = \frac{1}{nm}\sum_{i,j} \delta_{P_t(x_i,y_j)}.$$

If $\pi = (\mathrm{Id}, T)_\sharp \alpha$ is a Brenier-type coupling, then $\alpha_t = ((1-t)\mathrm{Id} + tT)_\sharp \alpha$ is the so-called McCann OT interpolation.

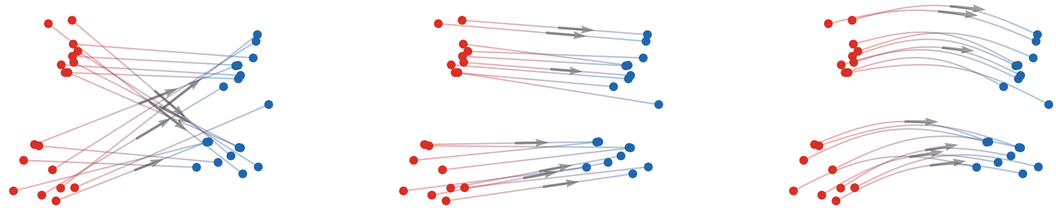


Figure 4.1: Flow matching interpolants between the same empirical source and target measures. A product-style random pairing produces crossing paths, an OT pairing gives direct displacement rays, and a curved bridge changes the path geometry while keeping the same endpoints. Gray arrows mark representative midpoint velocities $\partial_t P_t$.

Flow matching formula. This interpolation is not directly useful for sampling from β, but it can be used to define a flow field v_t so that the Eulerian advection equation (2.2) holds. This flow field is computed by solving an unconstrained least-squares problem, or equivalently, it is a conditional expectation.

Proposition 4.2 (Flow matching vector field)**.** *For each fixed t, assume $\partial_t P_t \in L^2(\pi; \mathbb{R}^d)$. Consider the flow-matching problem over measurable fields $v_t : \mathbb{R}^d \to \mathbb{R}^d$*

$$\min_{v_t} \int_{\mathbb{R}^{d'}} \|v_t(P_t(u)) - [\partial_t P_t](u)\|^2 \,\mathrm{d}\pi(u). \tag{4.2}$$

Its minimizer is characterized α_t-almost everywhere by the conditional expectation

$$v_t(z) = \mathbb{E}_{u\sim\pi}\big([\partial_t P_t](u) \,\big|\, z = P_t(u)\big). \tag{4.3}$$

Then the pair (α_t, v_t) satisfies the continuity equation (2.2).

Proof. We first recall the two equivalent ways of writing the interpolated measure. Formally, one may write

$$\alpha_t(z) = \int_{\mathbb{R}^{d'}} \delta(z - P_t(u))\,\mathrm{d}\pi(u),$$

while the rigorous meaning is that, for every smooth test function φ,

$$\int_{\mathbb{R}^d} \varphi(z)\,\mathrm{d}\alpha_t(z) = \int_{\mathbb{R}^{d'}} \varphi(P_t(u))\,\mathrm{d}\pi(u). \tag{4.4}$$

The minimizer in (4.2) is the orthogonal projection in $L^2(\pi; \mathbb{R}^d)$ of the latent velocity $\partial_t P_t(u)$ onto the closed subspace of functions that depend on u only through $P_t(u)$. This projection is the conditional expectation (4.3). Formally, this can be read as

$$v_t(z) = \frac{1}{\alpha_t(z)} \int_{\mathbb{R}^{d'}} \delta(z - P_t(u))[\partial_t P_t](u)\,\mathrm{d}\pi(u),$$

and rigorously it means that, for every smooth test vector field m,

$$\int \langle m(z),\, v_t(z)\rangle \,\mathrm{d}\alpha_t(z) = \int \langle m(P_t(u)),\, [\partial_t P_t](u)\rangle \,\mathrm{d}\pi(u). \tag{4.5}$$

We now prove that this field transports the curve $(\alpha_t)_t$. The weak form of $\partial_t \alpha_t + \mathrm{div}(\alpha_t v_t) = 0$ is that, for every smooth scalar test function φ,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(z)\,\mathrm{d}\alpha_t(z) - \int \langle v_t(z),\, \nabla\varphi(z)\rangle \,\mathrm{d}\alpha_t(z) = 0. \tag{4.6}$$

Using (4.4) and differentiating under the integral sign gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(z)\mathrm{d}\alpha_t(z) = \int \langle \nabla\varphi(P_t(u)),\, [\partial_t P_t](u)\rangle \mathrm{d}\pi(u). \tag{4.7}$$

On the other hand, applying (4.5) with $m = \nabla\varphi$ gives

$$\int \langle v_t(z),\, \nabla\varphi(z)\rangle \,\mathrm{d}\alpha_t(z) = \int \langle \nabla\varphi(P_t(u)),\, [\partial_t P_t](u)\rangle \mathrm{d}\pi(u). \tag{4.8}$$

Comparing (4.7) and (4.8) yields (4.6), which is the desired continuity equation. □

The conditional expectation in (4.3) has a simple measure-theoretic meaning. Let $\alpha_t = (P_t)_\sharp \pi$ and define the vector-valued measure m_t on $\mathbb{R}^d$ by

$$\int_{\mathbb{R}^d} \langle \psi(z),\, \mathrm{d}m_t(z)\rangle := \int_{\mathbb{R}^{d'}} \langle \psi(P_t(u)),\, [\partial_t P_t](u)\rangle \mathrm{d}\pi(u)$$

for every bounded continuous vector field ψ. Since $\alpha_t(A) = 0$ implies $\pi(P_t^{-1}(A)) = 0$, one has $m_t \ll \alpha_t$. The Radon–Nikodym decomposition of m_t with respect to α_t is therefore

$$\mathrm{d}m_t(z) = v_t(z)\mathrm{d}\alpha_t(z), \qquad v_t = \frac{\mathrm{d}m_t}{\mathrm{d}\alpha_t}.$$

In the language of Lebesgue decomposition, the flux measure m_t has only an absolutely continuous part with respect to α_t and no singular part; the conditional expectation is precisely this density. Equivalently, disintegrating π with respect to the map P_t gives $\pi(\mathrm{d}u) = \pi_{t,z}(\mathrm{d}u)\alpha_t(\mathrm{d}z)$, where $\pi_{t,z}$ is supported on the fiber $\{u \,:\, P_t(u) = z\}$, and

$$v_t(z) = \int_{\{P_t(u)=z\}} [\partial_t P_t](u)\mathrm{d}\pi_{t,z}(u).$$

Thus the solution of (4.2) is the conditional expectation of the velocities $\partial_t P_t$: intuitively, $v_t(z)$ is the average velocity of all trajectories passing through z. Numerically, $(x,t) \to v_t(x)$ can be parameterized by a neural network (e.g., a U-Net for vision tasks) and estimated using stochastic gradient descent on the objective in (4.2).

Algorithm 4.1 Flow matching regression and sampling

Input: Interpolant $P_t(u)$, training source $u \sim \pi$, parametrized field $v_\theta(t,z)$, training steps N.
Output: Learned sampler $X_0 \mapsto X_1$.
Training:
For $q = 1, \ldots, N$ **do**:
  Draw $t_q \sim \mathrm{Unif}(0,1)$ and $u_q \sim \pi$.
  Set $z_q = P_{t_q}(u_q)$ and $w_q = \partial_t P_t(u_q)|_{t=t_q}$.
  Update θ by one stochastic-gradient step on $\|v_\theta(t_q, z_q) - w_q\|^2$.

Sampling:
Draw $X_0 \sim \alpha_0$.
Integrate $\dot{X}_t = v_\theta(t, X_t), \qquad t \in [0,1]$. **Return** X_1.

For the exact field v_t, integrating the ODE $\dot{x} = v_t(x)$ defines a transport map T_t. If v_t is regular enough, or more generally if the continuity equation has a unique solution for this velocity, then $(T_t)_\sharp \alpha_0 = \alpha_t$. Thus the same interpolation as (4.1) is represented by a deterministic flow rather than by the original coupling. The sampling procedure consists in first drawing $X_0 \sim \alpha$, and then integrating the ODE $\dot{X}_t = v_t(X_t)$ starting with $X_{t=0} = X_0$. In the ideal exact-field limit, the resulting $X_{t=1}$ is distributed according to $\alpha_1 = \beta$.

Remark 4.3 (Static-noise stochastic interpolants). In the terminology of Albergo–Boffi–Vanden-Eijnden [1], a stochastic interpolant is not first defined as an SDE. It is an explicit random bridge

$$X_t = I_t(X_0, X_1, Z), \qquad X_0 \sim \alpha_0, \quad X_1 \sim \alpha_1,$$

where Z is an auxiliary random variable, usually Gaussian and independent of the endpoints, and where

$$I_0(x_0, x_1, z) = x_0, \qquad I_1(x_0, x_1, z) = x_1.$$

A typical spatially linear example is

$$X_t = a(t)X_0 + b(t)X_1 + \gamma(t)Z, \qquad \gamma(0) = \gamma(1) = 0.$$

The noise Z is static: conditionally on (X_0, X_1, Z), the path $t \mapsto X_t$ is differentiable. Thus this construction is exactly the previous push-forward framework with $u = (X_0, X_1, Z)$, $\pi = \mathrm{Law}(X_0, X_1, Z)$, and $P_t = I_t$. Its Eulerian velocity is therefore

$$v_t(x) = \mathbb{E}\big[\partial_t I_t(X_0, X_1, Z) \mid X_t = x\big],$$

and the interpolant density satisfies the continuity equation. The associated SDEs in the stochastic-interpolant framework are alternative sampling dynamics having the same one-time marginals; they are not the definition of the interpolant itself.

Remark 4.4 (Brownian realizations of interpolant marginals). One can also represent an interpolating marginal curve by Brownian-in-time dynamics. This is a different construction from the static-noise bridge of Remark 4.3. Let Z_t solve the Itô equation

$$\mathrm{d}Z_t = r_t(U, Z_t)\mathrm{d}t + \Sigma_t(U, Z_t)\mathrm{d}B_t, \qquad \alpha_t = \mathrm{Law}(Z_t),$$

where $U \sim \pi$ is static and B_t is Brownian motion. Define the Eulerian drift and diffusion matrix by conditioning on the observed state,

$$v_t(z) = \mathbb{E}\big[r_t(U, Z_t) \mid Z_t = z\big], \qquad D_t(z) = \mathbb{E}\big[\Sigma_t(U, Z_t)\Sigma_t(U, Z_t)^\top \mid Z_t = z\big].$$

Then, for smooth test functions φ,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi \mathrm{d}\alpha_t = \int \langle \nabla\varphi, v_t\rangle \mathrm{d}\alpha_t + \frac{1}{2}\int \mathrm{Tr}\big(D_t \nabla^2 \varphi\big)\mathrm{d}\alpha_t,$$

or, in distributional form,

$$\partial_t \alpha_t + \mathrm{div}(\alpha_t v_t) = \frac{1}{2}\sum_{i,j} \partial^2_{ij}\big((D_t)_{ij}\alpha_t\big). \tag{4.9}$$

Thus the natural noisy analogue of (4.2) regresses the instantaneous drift,

$$\min_w \mathbb{E}\big[\|w_t(Z_t) - r_t(U, Z_t)\|^2\big],$$

and learns the drift term of a Fokker–Planck equation, not a pure continuity equation unless the diffusion tensor vanishes. When $\alpha_t = \rho_t \mathrm{d}x$ has a smooth positive density, the same marginal curve can be represented, at least formally, by a probability-flow ODE

$$\partial_t \rho_t + \mathrm{div}(\rho_t \bar{v}_t) = 0, \qquad \bar{v}_t = v_t - \frac{1}{2\rho_t}\mathrm{div}(\rho_t D_t),$$

where the divergence of the matrix field $\rho_t D_t$ is taken row-wise. In the scalar spatially homogeneous case $D_t = \sigma_t^2 \mathrm{Id}$, this reduces to

$$\bar{v}_t = v_t - \frac{\sigma_t^2}{2} \nabla \log \rho_t,$$

which is the familiar score correction relating diffusion SDEs to probability-flow ODEs.

Connection with diffusion models. In the special case where $P_t(x, y) = (1-t)x + ty$ is a linear interpolation and $\pi = \alpha \otimes \beta$, the curve α_t is a convolution of rescaled versions of α_0 and α_1. The flow-matching problem (4.2) becomes

$$\min_{(v_t)_t} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|v_t((1-t)x + ty) - (y - x)\|^2 \, \mathrm{d}\alpha_0(x) \mathrm{d}\alpha_1(y).$$

When one endpoint is an isotropic Gaussian, this construction is closely related to the probability-flow formulation of diffusion models, up to the usual change of time parametrization [89]. This is why flow matching can be viewed both as a deterministic alternative to diffusion training and as a common language for diffusion paths, OT-inspired paths, and rectified paths [56, 58, 1]. The next two propositions are written in the noising direction, from a data law α to a Gaussian; reversing time gives the corresponding sampling flow. They also give an explicit closed form for v_t and show that it is a gradient field. In this setting, v_t is also the solution of the constrained least-squares problem (2.7). The regression (4.2) is computationally simpler because the continuity equation has already been enforced by the chosen interpolant. To prove this, we rely on Tweedie's formula, which expresses the optimal Gaussian denoiser through the score, i.e. the gradient of the log-density.

Proposition 4.5 (Tweedie identity)**.** *Let W be a random vector in $\mathbb{R}^d$ with density β. For $\sigma > 0$, observe*

$$Z \;=\; W + \sigma\, \varepsilon, \quad \text{where } \varepsilon \sim \mathcal{N}(0, I_d) \text{ is independent of } W.$$

Denote by

$$\beta_\sigma \;=\; \beta * \mathcal{N}\big(0, \sigma^2 I_d\big)$$

the density of Z. Then

$$\mathbb{E}\big[\, W \mid Z = z \,\big] \;=\; z \;+\; \sigma^2 \nabla \log \beta_\sigma(z) \qquad \text{for all } z \in \mathbb{R}^d.$$

Proof. Bayes' rule gives the conditional density $p_{W|Z}(w \mid z) = \dfrac{\beta(w)\, \varphi_\sigma(z - w)}{\beta_\sigma(z)}$ with φ_σ the $\mathcal{N}(0, \sigma^2 I_d)$ density. Hence

$$\mathbb{E}[W \mid Z = z] = \frac{1}{\beta_\sigma(z)} \int_{\mathbb{R}^d} w\, \beta(w)\, \varphi_\sigma(z - w)\, \mathrm{d}w.$$

Differentiating the Gaussian convolution under the integral sign and using $\nabla_z \varphi_\sigma(z - w) = -\sigma^{-2}(z - w)\, \varphi_\sigma(z - w)$ yields

$$\nabla_z \beta_\sigma(z) = \int \beta(w)\, \nabla_z \varphi_\sigma(z - w)\, \mathrm{d}w = -\sigma^{-2}\Big(z - \mathbb{E}[W \mid Z = z]\Big)\, \beta_\sigma(z).$$

Rearranging finishes the proof. □

Proposition 4.6 (Gaussian-endpoint flow-matching field)**.** *Let $X \sim \alpha$ and $Y \sim \mathcal{N}(0, I_d)$ be independent. For $t \in (0, 1)$ set*

$$Z_t \;=\; (1 - t)\, X + t\, Y, \qquad \alpha_t = \mathit{Law}(Z_t).$$

The regression minimizer $v^\star : \mathbb{R}^d \times (0, 1) \longrightarrow \mathbb{R}^d$ of

$$\min_v \int_0^1 \iint_{\mathbb{R}^d \times \mathbb{R}^d} \big| y - x - v\big((1 - t)x + ty, t\big)\big|^2 \, \mathrm{d}\alpha(x)\, \mathrm{d}\mathcal{N}(y)\, \mathrm{d}t$$

is

$$v^\star(x, t) = -\frac{1}{1 - t}\, x \;-\; \frac{t}{1 - t}\, \nabla \log \alpha_t(x) \qquad (x \in \mathbb{R}^d,\; t \in (0, 1)).$$

In particular, for each $t \in (0, 1)$ this field is a gradient field,

$$v^\star(\cdot, t) = -\nabla \left(\frac{\|\cdot\|^2}{2(1 - t)} + \frac{t}{1 - t} \log \alpha_t \right).$$

Proof. Fix $t \in (0,1)$ and write $W = (1-t)X$, $\sigma = t$, so that $Z_t = W + \sigma Y$ matches the setting of Proposition 4.5. Conditional expectations satisfy $v^\star(z,t) = \mathbb{E}[Y - X \mid Z_t = z] = \frac{1}{t}\mathbb{E}[Z_t - W \mid Z_t = z] - \frac{1}{1-t}\mathbb{E}[W \mid Z_t = z]$. Applying Proposition 4.5 to $\mathbb{E}[W \mid Z_t = z]$ and noting $\mathbb{E}[Y \mid Z_t = z] = -t\nabla \log \alpha_t(z)$ gives the claimed formula. □

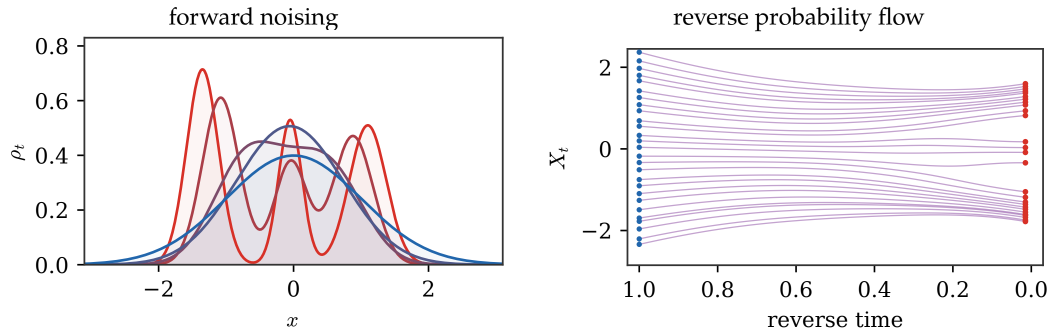


Figure 4.2: One-dimensional diffusion bridge for a Gaussian-mixture data law. The forward path $Z_t = (1-t)X + tY$ smooths the red data density toward a blue Gaussian endpoint. Reversing the probability-flow ODE transports a denser set of blue noise samples back toward the data modes, making the splitting of trajectories across mixture components visible.

The same probability-flow intuition is visible in two dimensions. For a discrete data law, or more generally for a Gaussian mixture, the noising density is a Gaussian mixture whose score can be evaluated explicitly. This makes it possible to draw backward trajectories without training a neural network. In the plots below, the Gaussian endpoint has covariance $\sigma^2\mathrm{Id}$ to keep the geometry visible at the scale of the three atoms. For a scalar noising schedule $Z_t = a_t X + b_t Y$, the intermediate law has component centers $a_t c_j$ and covariance $(b_t\sigma)^2\mathrm{Id}$. For the linear bridge, $p_t(z) = \sum_j w_j \mathcal{N}((1-t)c_j, (t\sigma)^2\mathrm{Id})$, with $s_t = \nabla \log p_t$, and the scaled version of Proposition 4.6 gives $v_t(z) = -(z + t\sigma^2 s_t(z))/(1-t)$.

Algorithm 4.2 Exact probability-flow sampling for a Gaussian mixture

Input: Gaussian-mixture data law, schedule (a_t, b_t), noise level σ, number of samples R.
Output: Backward samples $(Z_0^{(r)})_r$.
Define the noising variable: $Z_t = a_t X + b_t Y, \qquad Y \sim \mathcal{N}(0, \sigma^2\mathrm{Id})$.
Compute closed-form mixture density p_t and score $s_t = \nabla \log p_t$.
Set probability-flow velocity: $v_t(z) = \frac{a_t'}{a_t} z + \left(\frac{a_t' b_t^2}{a_t} - b_t' b_t\right)\sigma^2 s_t(z)$.
For $r = 1, \ldots, R$ **do**:
  Draw $Z_1^{(r)}$ from the Gaussian endpoint.
  Integrate $\dot{Z}_t^{(r)} = v_t(Z_t^{(r)})$ backward from $t = 1$ to $t = 0$.
Return $(Z_0^{(r)})_r$.

When is the induced map optimal? Integrating the learned velocity gives a deterministic map from α_0 to α_1, but this map is not automatically the Brenier optimal map. It is optimal only in special cases where the accumulated flow remains the gradient of a convex potential. The Gaussian product-coupling case already shows the precise obstruction: the interpolated covariances are simple, the velocity is affine, but the terminal map can contain a hidden rotational part. This phenomenon, and its extensions to rectified flows and mixtures, is analyzed in depth in [46].

Proposition 4.7 (Gaussian flow matching and optimality)**.** *Let $\Sigma_0, \Sigma_1 \succ 0$ and let $X_0 \sim \mathcal{N}(0,\Sigma_0)$ and $X_1 \sim \mathcal{N}(0,\Sigma_1)$ be independent. Consider the linear flow-matching interpolation*

$$Z_t = (1-t)X_0 + tX_1, \qquad \alpha_t = \mathrm{Law}(Z_t) = \mathcal{N}(0,\Sigma_t),$$

where

$$\Sigma_t = (1-t)^2\Sigma_0 + t^2\Sigma_1. \tag{4.10}$$

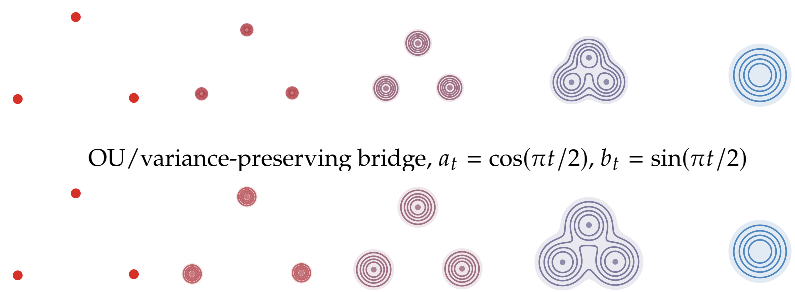


Figure 4.3: Two-dimensional noising paths from three Dirac masses to a single Gaussian. The top row shows the linear interpolation $Z_t = (1-t)X + tY$, whose component centers move linearly toward the origin and whose covariance grows like $(t\sigma)^2\mathrm{Id}$. The bottom row uses the variance-preserving Ornstein–Uhlenbeck coefficients $a_\tau = e^{-\tau}$ and $b_\tau = \sqrt{1-e^{-2\tau}}$, reparametrized by $\tau = -\log\cos(\pi t/2)$ so that $a_t = \cos(\pi t/2)$ and $b_t = \sin(\pi t/2)$. It has the same endpoints but a different speed of contraction and noising.

Then the exact flow-matching velocity is affine, $v_t(z) = A_t z$, with

$$A_t = \big(t\Sigma_1 - (1-t)\Sigma_0\big)\Sigma_t^{-1}. \tag{4.11}$$

The induced flow map T_t^{FM} from α_0 to α_t is

$$T_t^{\mathrm{FM}} = \Sigma_0^{1/2}\Big((1-t)^2\mathrm{Id} + t^2\Sigma_0^{-1/2}\Sigma_1\Sigma_0^{-1/2}\Big)^{1/2}\Sigma_0^{-1/2}. \tag{4.12}$$

In particular,

$$T_1^{\mathrm{FM}} = \Sigma_0^{1/2}\big(\Sigma_0^{-1/2}\Sigma_1\Sigma_0^{-1/2}\big)^{1/2}\Sigma_0^{-1/2}. \tag{4.13}$$

This terminal map coincides with the quadratic optimal transport map

$$T^{\mathrm{OT}} = \Sigma_0^{-1/2}\big(\Sigma_0^{1/2}\Sigma_1\Sigma_0^{1/2}\big)^{1/2}\Sigma_0^{-1/2} \tag{4.14}$$

if and only if $\Sigma_0\Sigma_1 = \Sigma_1\Sigma_0$.

Proof. The conditional-expectation formula gives

$$v_t(z) = \mathbb{E}[X_1 - X_0 \mid Z_t = z].$$

Since all variables are jointly Gaussian, this conditional expectation is linear and

$$v_t(z) = \mathrm{Cov}(X_1 - X_0, Z_t)\,\mathrm{Cov}(Z_t)^{-1}z = \big(t\Sigma_1 - (1-t)\Sigma_0\big)\Sigma_t^{-1}z,$$

which proves (4.11). To solve the characteristic equation, whiten the source by setting

$$C = \Sigma_0^{-1/2}\Sigma_1\Sigma_0^{-1/2}, \qquad \widetilde{Z}_t = \Sigma_0^{-1/2}Z_t.$$

In these coordinates the source covariance is Id and

$$\widetilde{\Sigma}_t = (1-t)^2\mathrm{Id} + t^2 C.$$

Because Id and C commute, the affine flow map in whitened coordinates is simply $\widetilde{T}_t = \widetilde{\Sigma}_t^{1/2}$. Indeed,

$$\frac{\mathrm{d}}{\mathrm{d}t}\widetilde{\Sigma}_t^{1/2} = \big(tC - (1-t)\mathrm{Id}\big)\widetilde{\Sigma}_t^{-1/2},$$

which is exactly the equation $\dot{\widetilde{T}}_t = \widetilde{A}_t\widetilde{T}_t$ with $\widetilde{T}_0 = \mathrm{Id}$. Returning to the original coordinates gives (4.12), and $t = 1$ gives (4.13).

Both T_1^{FM} and T^{OT} push $\mathcal{N}(0,\Sigma_0)$ to $\mathcal{N}(0,\Sigma_1)$. The Brenier map between nondegenerate Gaussians is the unique symmetric positive definite linear map with this property. Hence $T_1^{\mathrm{FM}} = T^{\mathrm{OT}}$ if and only if T_1^{FM} is symmetric positive definite. The map T_1^{FM} is similar to $C^{1/2}$, so if it is symmetric then it is automatically positive definite. It remains to characterize symmetry. Since $C^{1/2}$ is symmetric positive definite,

$$(T_1^{\mathrm{FM}})^\top = \Sigma_0^{-1/2} C^{1/2} \Sigma_0^{1/2}.$$

Thus symmetry of T_1^{FM} is equivalent to $\Sigma_0 C^{1/2} = C^{1/2}\Sigma_0$, hence to $\Sigma_0 C = C\Sigma_0$ by functional calculus. Multiplying this identity on the left and right by $\Sigma_0^{1/2}$ gives $\Sigma_0\Sigma_1 = \Sigma_1\Sigma_0$. Conversely, if Σ_0 and Σ_1 commute, they are orthogonally co-diagonalizable, and both (4.13) and (4.14) reduce in that basis to the diagonal map with entries $\sqrt{\lambda_{1,k}/\lambda_{0,k}}$. This proves the equivalence. □

The proposition gives a compact warning about a common overinterpretation of flow matching.

Remark 4.8 (Changing the bridge speed does not restore optimality). The same terminal map (4.13) is obtained for any scalar schedule $Z_t = a_t X_0 + b_t X_1$ with the same endpoints, because after whitening the covariance path remains $a_t^2 \mathrm{Id} + b_t^2 C$. Thus changing the speed of a scalar Gaussian bridge, for instance by using an OU schedule, cannot repair the non-optimality created by non-commuting covariances.

Commuting covariances reduce the terminal map to independent one-dimensional scalings, whereas non-commuting covariances create a non-symmetric affine map, hence a transport with a rotational or shearing component. More generally, mixture-like paths can create the same obstruction even when every instantaneous velocity looks natural. This distinction is closely related to counterexamples showing that flow maps associated with Fokker–Planck or diffusion-type evolutions do not in general provide optimal transport maps [53]. In particular, starting from an isotropic Gaussian does not by itself guarantee optimality once the target distribution is non-Gaussian; additional structural assumptions on the path or on the coupling are needed.

Variations on the interpolant. The geometry of the generated trajectories depends on the chosen interpolant, not only on the two endpoint laws. There is first a harmless ambiguity: a monotone reparametrization $Z_t = (1-\lambda(t))X + \lambda(t)Y$ of the linear bridge only changes the speed of the flow,

$$v_t(z) = \lambda'(t)\, v^{\mathrm{lin}}_{\lambda(t)}(z), \qquad v^{\mathrm{lin}}_r(z) = \mathbb{E}[Y - X \mid (1-r)X + rY = z].$$

It therefore leaves the spatial integral curves unchanged. Diffusion models use a genuinely different family of noising paths. If

$$Z_t = a_t X + b_t Y, \qquad Y \sim \mathcal{N}(0, \sigma^2 \mathrm{Id}),$$

then both the mixture centers and the component variances are changed. Writing p_t for the density of Z_t and $s_t = \nabla \log p_t$, Tweedie's formula gives, away from times where $a_t = 0$,

$$v_t(z) = a_t'\, \mathbb{E}[X \mid Z_t = z] + b_t'\, \mathbb{E}[Y \mid Z_t = z] = \frac{a_t'}{a_t} z + \left(\frac{a_t' b_t^2}{a_t} - b_t' b_t \right) \sigma^2 s_t(z).$$

For the linear bridge, $a_t = 1-t$ and $b_t = t$, this recovers the formula above. For the variance-preserving Ornstein–Uhlenbeck noising used in diffusion models,

$$a_\tau = e^{-\tau}, \qquad b_\tau = \sqrt{1 - e^{-2\tau}},$$

one obtains the forward probability-flow velocity $v_\tau(z) = -z - \sigma^2 \nabla \log p_\tau(z)$. Sampling follows the reverse field $z + \sigma^2 \nabla \log p_\tau(z)$ as τ decreases. This is the noising law used in the left panel of Figure 4.4; the trajectories are more curved than for the linear bridge because the centers and variances evolve according to the OU/Fokker–Planck scaling rather than by affine interpolation. Numerically, the integration is stopped at a small positive time before the Dirac endpoint, where the score becomes singular.

The finite-time coefficients $a_t = \cos(\pi t/2)$ and $b_t = \sin(\pi t/2)$ are not a new spatial interpolant: they are exactly the OU coefficients after the time change $\tau = -\log\cos(\pi t/2)$. Figure 4.5 therefore compares OU with a genuinely different scalar bridge,

$$a_t = (1-t)(1-2t), \qquad b_t = t,$$

whose data coefficient changes sign before vanishing. This overshooting bridge is mainly a diagnostic example: it keeps the same endpoints, but its intermediate mixture reflects through the origin and produces visibly different reverse trajectories.

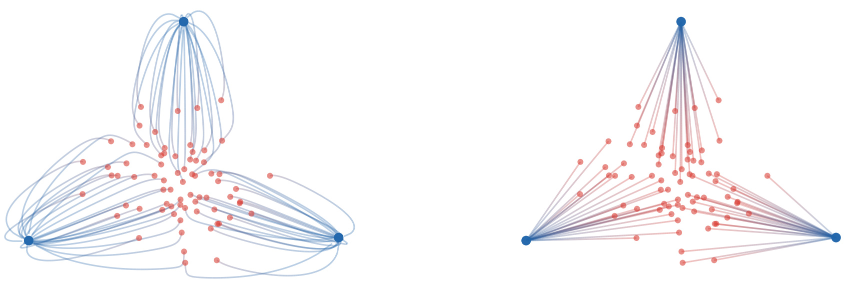


Figure 4.4: Diffusion-style sampling trajectories compared with OT rays in the three-Dirac setting of Figure 4.3. Red particles are sampled from the centered Gaussian endpoint and transported toward the three blue atoms. The left panel integrates the reverse probability-flow ODE for the variance-preserving OU noising $a_\tau = e^{-\tau}$, $b_\tau = \sqrt{1 - e^{-2\tau}}$, using the closed-form Gaussian-mixture score and stopping just before the singular Dirac endpoint. The right panel uses the straight displacement rays selected by a quadratic OT matching to the same atoms.

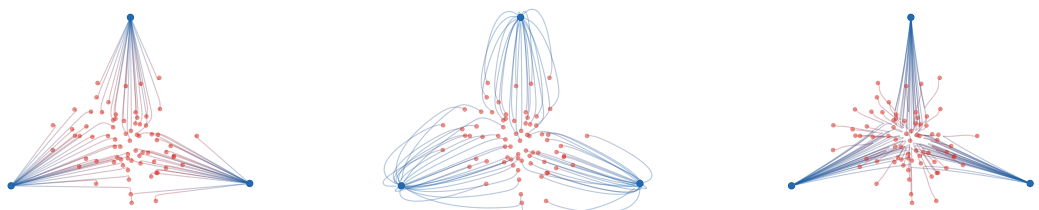


Figure 4.5: Effect of the interpolant on the exact reverse flow for the same three-Dirac target and the same Gaussian endpoint. The linear bridge $a_t = 1 - t$, $b_t = t$ produces almost radial curves. The variance-preserving OU bridge $a_\tau = e^{-\tau}$, $b_\tau = \sqrt{1 - e^{-2\tau}}$ changes the relative speed of contraction and noising. The overshooting bridge $a_t = (1 - t)(1 - 2t)$, $b_t = t$ is not a time reparameterization of either one and produces a more pronounced bending of the reverse trajectories.

4.2 One-Step Generative Models

One-step generative models try to keep the geometric training principle of flows while removing the expensive multi-step integration at sampling time. The idea is to evolve the model distribution during training, but to store the final evolution in a single generator evaluation.

Training a one-step flow. Let ζ be a simple latent distribution and let $\alpha_\theta = (G_\theta)_\sharp \zeta$ be the model distribution. Assume that the target data distribution is β. A Wasserstein-flow construction chooses a discrepancy

$$\mathcal{E}_\beta(\alpha),$$

for instance a smoothed $\mathrm{KL}(\alpha|\beta)$, an MMD/IPM loss, or a debiased Sinkhorn divergence $\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta)$. The associated formal descent is

$$\partial_t \mu_t + \operatorname{div}(\mu_t w_t) = 0, \qquad w_t(x) = -\nabla \delta_\alpha \mathcal{E}_\beta(\mu_t)(x). \tag{4.15}$$

Instead of integrating (4.15) at inference time, one fits a parametric residual field U_η along the current model distribution:

$$\min_\eta \int_0^1 \int \|U_\eta(t, x) - w_t(x)\|^2 \, \mathrm{d}\mu_t(x) \, \mathrm{d}t. \tag{4.16}$$

In a particle or generator implementation, the learned residual is then used to update the current generator by

$$\alpha_\theta^+ = (\mathrm{Id} + \tau U_\eta)_\sharp \alpha_\theta, \qquad \text{or equivalently} \qquad G_\theta^+(z) = G_\theta(z) + \tau U_\eta(G_\theta(z)).$$

Algorithm 4.3 One-step Wasserstein-flow generator update

Input: Generator G_{θ_k}, latent law ζ, data law β, numerical descent-field oracle W_β, step size τ, batch size B.
Output: Updated generator $G_{\theta_{k+1}}$.
Draw $z_b \sim \zeta$ for $b = 1, \dots, B$.
Set $x_b = G_{\theta_k}(z_b)$.
Set $w_k(x) = W_\beta[\alpha_{\theta_k}](x)$, where $W_\beta[\alpha] = -\nabla\delta_\alpha\mathcal{E}_\beta(\alpha)$.
Set η_k by minimizing the empirical least-squares loss: $\frac{1}{B}\sum_{b=1}^B \|U_\eta(x_b) - w_k(x_b)\|^2$.
Update by composition: $G_{\theta_{k+1}}(z) = G_{\theta_k}(z) + \tau U_{\eta_k}(G_{\theta_k}(z))$. **Return** $G_{\theta_{k+1}}$.

After many training updates, the accumulated generator is evaluated once at test time. This is the organizing principle behind recent one-step methods based on Wasserstein gradient flows: W-Flow uses such a construction with the Sinkhorn divergence as a tractable global discrepancy [43], while drifting methods evolve the generated distribution during training through a fitted vector field and also admit one-step inference [28]. The gradient-flow interpretation of drifting models, and its relation to KL, MMD, sliced-Wasserstein and Sinkhorn-type discrepancies, is analyzed in [41, 45]. These ideas are also connected to the Sinkhorn-type normalization dynamics used to model attention in Sinkformers [82].

Self-corrected drifting fields. Drifting methods need not start from an exact Wasserstein gradient. They often prescribe an attraction-minus-repulsion field and then regress this field in $L^2(\mu_t)$. A simple continuous version uses a positive kernel $K_\varepsilon(x,y)$ and defines, for any measure ν,

$$B_\varepsilon[\nu](x) := \frac{\int (y-x)K_\varepsilon(x,y)\,\mathrm{d}\nu(y)}{\int K_\varepsilon(x,y)\,\mathrm{d}\nu(y)}. \tag{4.17}$$

For the Gaussian kernel $K_\varepsilon(x,y) = \exp(-\|x-y\|^2/(2\varepsilon))$, this normalized field is a score of a smoothed density:

$$B_\varepsilon[\nu](x) = \varepsilon\nabla\log\left(\int K_\varepsilon(x,y)\,\mathrm{d}\nu(y)\right). \tag{4.18}$$

The drifting velocity is then

$$u_t(x) = B_\varepsilon[\beta](x) - B_\varepsilon[\mu_t](x) = \varepsilon\nabla\log\frac{\int K_\varepsilon(x,y)\,\mathrm{d}\beta(y)}{\int K_\varepsilon(x,y)\,\mathrm{d}\mu_t(y)}. \tag{4.19}$$

The first term pulls samples toward data, while the second term corrects self-attraction and prevents all particles from collapsing onto the same high-density region. For a fixed reference measure, $B_\varepsilon[\nu]$ is precisely the Gaussian mean-shift displacement in (4.26): it moves x toward the local kernel barycenter of ν. Hence self-corrected drifting can be read as the difference between a target mean-shift field and the current model's own mean-shift field. Sinkhorn drifting replaces these one-sided kernel normalizations by two-sided entropic OT couplings, so that the cross and self terms are normalized by Sinkhorn scaling rather than by a single denominator [45].

Proposition 4.9 (Drifting as a time-dependent Wasserstein gradient). *Let μ_t be a smooth curve of positive densities and let $u_t = \nabla\varphi_t$ be a smooth time-dependent gradient field. Define the semi-relaxed functional*

$$\mathcal{R}_t(\alpha|\mu_t) := -\int \varphi_t(x)\,\mathrm{d}\alpha(x) + \int \varphi_t(x)\,\mathrm{d}\mu_t(x). \tag{4.20}$$

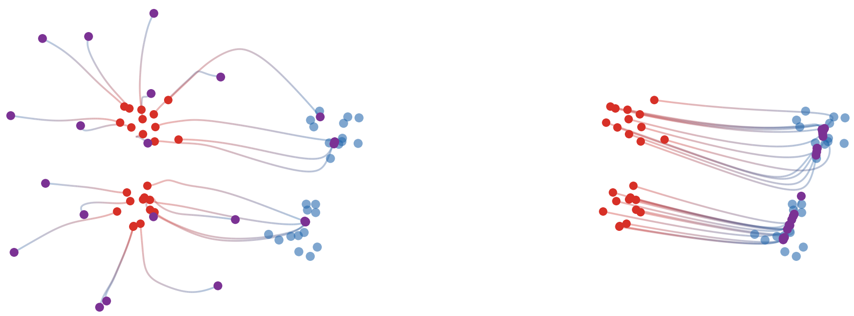
Here μ_t and φ_t are frozen when taking the first variation with respect to the first argument α. Then the continuity equation

$$\partial_t\mu_t + \mathrm{div}(\mu_t u_t) = 0$$

is the formal Wasserstein gradient descent of the time-dependent functional $\alpha \mapsto \mathcal{R}_t(\alpha|\mu_t)$.

Algorithm 4.4 Self-corrected drifting particle update

Input: Particles x_i^k for μ_k, data samples $(y_b)_{b=1}^B$ from β, kernel scale ε, step h.
Output: Updated particles x_i^{k+1}.
For each particle i **do**:
  Set $Z_{\beta,i} = \sum_{b=1}^B K_\varepsilon(x_i^k, y_b)$ and $b_i^k = Z_{\beta,i}^{-1} \sum_{b=1}^B (y_b - x_i^k) K_\varepsilon(x_i^k, y_b)$.
  Set $Z_{\mu,i} = \sum_{j=1}^n K_\varepsilon(x_i^k, x_j^k)$ and $m_i^k = Z_{\mu,i}^{-1} \sum_{j=1}^n (x_j^k - x_i^k) K_\varepsilon(x_i^k, x_j^k)$.
  Set $u_i^k = b_i^k - m_i^k$.
  Update $x_i^{k+1} = x_i^k + h\, u_i^k$.
Return $(x_i^{k+1})_i$.



raw kernel drift self-corrected drift

Figure 4.6: Drifting trajectories for a small particle generator. The raw Laplacian-kernel drift has weak long-range attraction and can leave particles away from the data modes. The self-corrected field uses the difference $B_\varepsilon[\beta] - B_\varepsilon[\mu_t]$, so a longer integration brings particles to the blue modes while repelling them from their own current concentration.

Proof. Since μ_t and φ_t are fixed in the variation with respect to α, the first variation is

$$\delta_\alpha \mathcal{R}_t(\alpha|\mu_t)(x) = -\varphi_t(x).$$

By Proposition 3.2,

$$\nabla_{\mathcal{W}} \mathcal{R}_t(\alpha|\mu_t) = \nabla \delta_\alpha \mathcal{R}_t(\alpha|\mu_t) = -\nabla \varphi_t = -u_t.$$

The Wasserstein gradient-descent velocity is the negative of this gradient, namely u_t. Substituting this velocity in the continuity equation gives the claimed flow. □

Example 4.10 (Kernel drifting as a semi-relaxed divergence). For the Gaussian-kernel drift (4.19), set

$$\varphi_t(x) = \varepsilon \log \frac{\int K_\varepsilon(x,y)\, \mathrm{d}\beta(y)}{\int K_\varepsilon(x,y)\, \mathrm{d}\mu_t(y)}.$$

Then $u_t = \nabla \varphi_t$, so Proposition 4.9 shows that kernel drifting is the Wasserstein gradient descent of

$$\mathcal{R}_t^{\mathrm{drift}}(\alpha|\mu_t) = \varepsilon \int \log \frac{\int K_\varepsilon(x,y)\, \mathrm{d}\mu_t(y)}{\int K_\varepsilon(x,y)\, \mathrm{d}\beta(y)}\, \mathrm{d}\alpha(x) + \text{constant}.$$

It is "semi-relaxed" because the current model μ_t is used to build the potential, but it is not varied inside the denominator when computing the first variation in α.

Remark 4.11 (General fields and projection onto gradients). A general regressed field b_t is not necessarily a Wasserstein gradient, since Wasserstein tangent vectors are represented by gradient fields modulo $L^2(\mu_t)$-null directions. The gradient component is obtained by the weighted projection

$$\nabla\varphi_t = \operatorname*{argmin}_{\nabla\varphi} \int \|\nabla\varphi(x) - b_t(x)\|^2 \, d\mu_t(x).$$

One may first normalize b_t pointwise, for instance by $b_t/(\|b_t\| + \eta)$, or globally by $\|b_t\|_{L^2(\mu_t)}$, before this projection. Proposition 4.9 then applies to the projected field. Non-gradient components can still be useful in a parametric model, but they are not descent directions of a scalar functional for the $\mathcal{W}_2$ Riemannian metric.

4.3 Moment Measures

Moment measures give another way to make a whole distribution from one convex potential. Instead of first fixing a simple source law and then learning a transport map, one asks for a convex function whose own log-concave density is pushed forward by its gradient. This couples sampling and mapping in a rigid way: the same potential defines both the source density and the Brenier map. The reward is a hidden convex structure: after a suitable optimal-transport reformulation, a nonlinear equation on convex functions becomes a convex minimization problem for probability densities. This is one of the cleanest places where optimal transport, Prékopa-type inequalities and convex geometry meet.

Definition 4.12 (Moment measure of a convex function)**.** Let $u : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ be a proper lower-semicontinuous convex function such that

$$Z_u := \int_{\mathbb{R}^d} e^{-u(x)} \, dx < +\infty.$$

The normalized log-concave measure associated with u is

$$\rho_u := Z_u^{-1} e^{-u(x)} \, dx.$$

Whenever ∇u is defined ρ_u-almost everywhere, the moment measure of u is

$$\mathfrak{M}(u) := (\nabla u)_\sharp \rho_u.$$

The normalization removes additive constants in u. Translations of the argument are another invariance: if $u_a(x) = u(x-a)$, then ρ_{u_a} is the translate of ρ_u, while $\nabla u_a(x) = \nabla u(x-a)$, hence $\mathfrak{M}(u_a) = \mathfrak{M}(u)$. A first obstruction is immediate. Formally, if u is smooth and e^{-u} decays fast enough for the boundary term to vanish, then

$$\int y \, d\mathfrak{M}(u)(y) = Z_u^{-1} \int \nabla u(x) e^{-u(x)} \, dx = -Z_u^{-1} \int \nabla(e^{-u(x)}) \, dx = 0.$$

Thus moment measures are necessarily centered. The non-smooth theory uses essentially continuous convex functions, meaning convex functions whose restriction to the closure of their effective domain is continuous except possibly on a boundary set of zero $\mathcal{H}^{d-1}$ measure.

Figure 4.7 shows the forward construction in one dimension. Even before solving the inverse problem, one sees the two coupled effects of the same convex potential: e^{-u} chooses the source law, while u' transports it to the moment measure.

Theorem 4.13 (Cordero–Erausquin–Klartag)**.** *Let $\mu \in \mathcal{P}_1(\mathbb{R}^d)$ be a probability measure. There exists an essentially continuous convex function u such that $\mu = \mathfrak{M}(u)$ if and only if*

$$\int y \, d\mu(y) = 0 \qquad \textit{and} \qquad \operatorname{supp}(\mu) \textit{ is not contained in a hyperplane.}$$

Under these conditions, u is unique up to translations of the argument and additive constants.

This theorem is due to Cordero–Erausquin and Klartag [24]. It is a functional analogue of a Minkowski-type problem: the target measure prescribes how the gradient image of a log-concave density should be distributed. The hyperplane condition is the natural non-degeneracy assumption; otherwise the prescribed gradient image lives in a lower-dimensional affine direction and no coercive full-dimensional convex potential can be recovered.

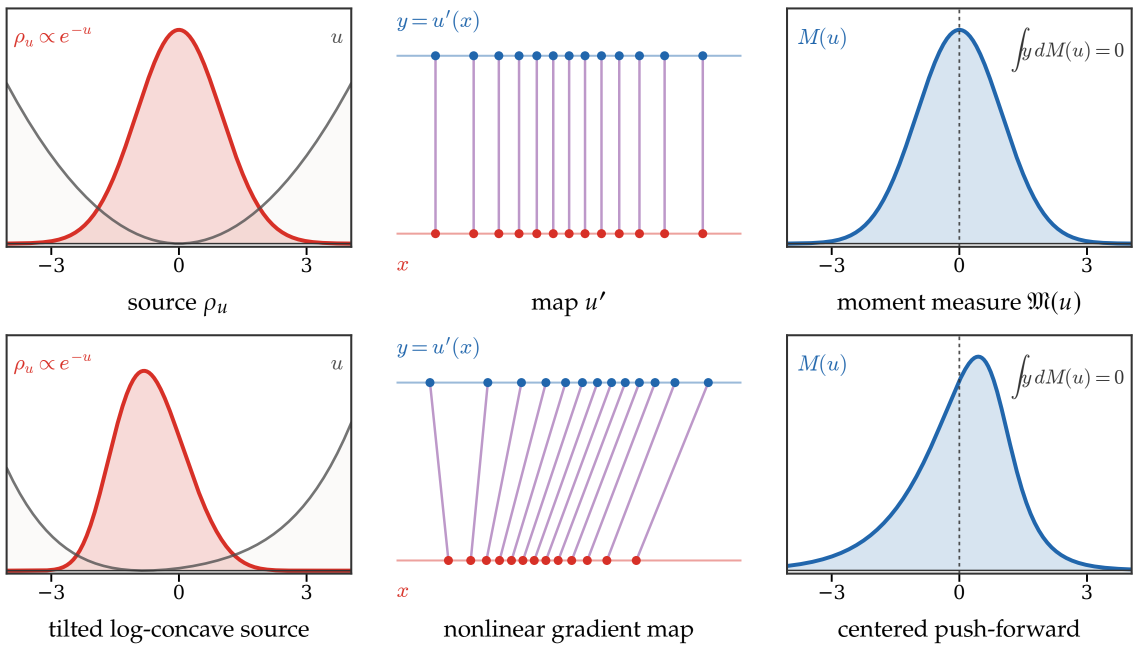


Figure 4.7: Forward moment-measure construction in one dimension. The top row uses $u(x) = x^2/2$, so ρ_u and $\mathfrak{M}(u)$ are both standard Gaussian and $u'(x) = x$. The bottom row uses $u(x) = x^2/2 + 0.18\,x^4/4 + 0.92\,x$, which tilts the log-concave source and produces a nonlinear monotone map. In both cases the blue push-forward has barycenter zero, illustrating the identity $\int y \, \mathrm{d}\mathfrak{M}(u)(y) = 0$.

Example 4.14 (Quadratic potentials). Let $A \in \mathbb{R}^{d\times d}$ be symmetric positive definite and

$$u(x) = \frac{1}{2} x^\top A x + c.$$

Then $\rho_u = \mathcal{N}(0, A^{-1})$ and $\nabla u(x) = Ax$, so

$$\mathfrak{M}(u) = \mathcal{N}(0, A).$$

Thus centered non-degenerate Gaussians are moment measures. This example also shows the self-dual flavor of the construction: the covariance of the log-concave source is inverted by the linear gradient map.

Optimal-transport variational formulation. Santambrogio [83] reformulates the moment-measure problem as a minimization over probability densities. For a centered $\mu \in \mathcal{P}_1(\mathbb{R}^d)$, define the maximal-correlation transport functional, with values in $\mathbb{R} \cup \{+\infty\}$,

$$C_\mu(\rho) := \sup_{\pi \in \mathcal{U}(\rho,\mu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} x \cdot y \, \mathrm{d}\pi(x, y). \tag{4.21}$$

By Kantorovich duality for the scalar-product cost,

$$C_\mu(\rho) = \inf_{v \text{ convex}} \left\{ \int v(x) \, \mathrm{d}\rho(x) + \int v^*(y) \, \mathrm{d}\mu(y) \right\}, \tag{4.22}$$

where v^* is the Legendre transform. If $\rho, \mu \in \mathcal{P}_2(\mathbb{R}^d)$, then

$$C_\mu(\rho) = \frac{1}{2} \int \|x\|^2 \, \mathrm{d}\rho(x) + \frac{1}{2} \int \|y\|^2 \, \mathrm{d}\mu(y) - \frac{1}{2} \mathcal{W}_2^2(\rho, \mu). \tag{4.23}$$

The variational problem attached to a centered target μ is

$$\min_{\rho \in \mathcal{P}_1(\mathbb{R}^d)} \left\{ \mathcal{H}(\rho) + C_\mu(\rho) \right\}, \qquad \mathcal{H}(r \, \mathrm{d}x) := \int r(x) \log r(x) \, \mathrm{d}x, \tag{4.24}$$

with $\mathcal{H}(\rho) = +\infty$ when ρ is not absolutely continuous. The centering of μ makes this functional invariant under translations of ρ, since translating ρ by a vector a changes $\int x \cdot y \, d\pi$ by $a \cdot \int y \, d\mu(y) = 0$.

Proposition 4.15 (Variational characterization of moment measures). *Let $\mu \in \mathcal{P}_1(\mathbb{R}^d)$ be centered and not supported on a hyperplane. Then the minimization problem (4.24) admits a solution, unique up to translations. Every minimizer is a log-concave density of the form*

$$\rho = Z_u^{-1} e^{-u} \, dx,$$

where u is convex, essentially continuous, and satisfies the moment-measure equation

$$\mu = (\nabla u)_\sharp \rho.$$

Conversely, any essentially continuous convex u satisfying this equation yields a global minimizer. In the finite-second-moment case $\mu \in \mathcal{P}_2(\mathbb{R}^d)$, the objective $\rho \mapsto \mathcal{H}(\rho) + C_\mu(\rho)$ is displacement convex along $\mathcal{W}_2$ geodesics on $\mathcal{P}_2(\mathbb{R}^d)$ whenever the terms are finite. For general $\mu \in \mathcal{P}_1(\mathbb{R}^d)$, the same statement is obtained by approximation and lower semicontinuity.

Proof. The existence proof has two ingredients. First, since μ is centered, translating ρ does not change either $\mathcal{H}(\rho)$ or $C_\mu(\rho)$, so one can center a minimizing sequence. Second, the assumption that μ is not supported on a hyperplane gives a coercive lower bound of the form $C_\mu(\rho) \geqslant c_\mu \int \|x\| \, d\rho(x)$ for centered absolutely continuous ρ, with $c_\mu > 0$. Together with the lower semicontinuity estimates for entropy and maximal correlation, this yields a minimizer [83].

Let $\rho = r \, dx$ be a minimizer and let u be a convex optimizer in the dual formula (4.22). Keeping u fixed and varying ρ in (4.24) gives the Euler equation

$$\log r(x) + 1 + u(x) = \text{constant} \qquad \text{on } \{r > 0\},$$

so $\rho = Z_u^{-1} e^{-u} dx$. The optimality condition for the scalar-product transport problem says that an optimal coupling is supported on $\{(x, y) : y \in \partial u(x)\}$. Since ρ is absolutely continuous and u is convex, $\partial u(x) = \{\nabla u(x)\}$ for ρ-almost every x, hence $\mu = (\nabla u)_\sharp \rho$.

Conversely, assume $\rho = Z_u^{-1} e^{-u} dx$ and $\mu = (\nabla u)_\sharp \rho$. Let ν be a smooth compactly supported competitor, and let T be the Brenier map from ρ to ν. Along the geodesic $\rho_t = ((1-t)\mathrm{Id} + tT)_\sharp \rho$, the right derivative of the entropy at $t = 0$ is

$$\frac{d}{dt} \mathcal{H}(\rho_t) \Big|_{t=0^+} = -\int \langle T(x) - x, \nabla u(x) \rangle \, d\rho(x),$$

where the identity follows by differentiating the Jacobian formula and integrating by parts. Santambrogio's derivative estimate for the maximal-correlation functional gives

$$\frac{d}{dt} C_\mu(\rho_t) \Big|_{t=0^+} \geqslant \int \langle T(x) - x, \nabla u(x) \rangle \, d\rho(x),$$

because $(\mathrm{Id}, \nabla u)_\sharp \rho$ is optimal for the scalar-product problem. The first-order terms cancel, hence the one-sided derivative of $\mathcal{H} + C_\mu$ at ρ in every such direction is nonnegative. Displacement convexity then implies global minimality, and approximation removes the smooth compact-support restriction. Strict displacement convexity of entropy gives uniqueness, except for translations; translations do not change C_μ because μ is centered.

It remains to justify the convexity assertion. The entropy term is displacement convex by McCann's theorem, recalled in Proposition 3.11. If μ has finite second moment, identity (4.23) writes C_μ as the sum of the 1-convex moment term $\rho \mapsto \frac{1}{2} \int \|x\|^2 \, d\rho$ and the (-1)-convex term $\rho \mapsto -\frac{1}{2} \mathcal{W}_2^2(\rho, \mu)$, hence C_μ is displacement convex. For merely finite first moment μ, the same convexity and the variational characterization follow by the approximation and semicontinuity arguments of Santambrogio. □

Remark 4.16 (Where the convexity is hidden). If one eliminates ρ first, the problem becomes the convex-potential functional

$$u \mapsto \int u^*(y) \, d\mu(y) - \log\left(\int e^{-u(x)} \, dx \right).$$

> This is not visibly convex as a function of u: the first term is convex, while the logarithmic partition term is concave in this parametrization. Cordero–Erausquin and Klartag reveal convexity after passing to the dual potential u^* and using a Prékopa-type argument. Santambrogio's formulation reveals the same mechanism in transport language: the difficult convexity becomes the displacement convexity of an entropy-plus-maximal-correlation functional in the density variable ρ.

Conjugate moment measures for generation. The moment-measure factorization suggests a generative recipe: sample X from the log-concave law $Z_u^{-1}e^{-u}$ and output $\nabla u(X)$. This ties sampling and mapping through the same convex potential. Vesseron, Béthune and Cuturi [96] argue that this direct factorization can be poorly adapted to practical generative modeling, and propose instead the conjugate factorization

$$\beta = (\nabla w^*)_\sharp \left(Z_w^{-1} e^{-w(z)} \, \mathrm{d}z \right). \tag{4.25}$$

Here ∇w^* is the Brenier map from the learned log-concave source to the target distribution β. This keeps the single-convex-potential philosophy, but places the transport map on the conjugate side; it can be parameterized by input-convex neural networks and trained using OT solvers. From the viewpoint of this section, moment measures are therefore a rigorous convex-analytic prototype for one-step generators based on gradients of convex potentials.

4.4 Evolution in Depth of Transformers

Deep residual architectures can be read as time discretizations of ODEs or PDEs. For transformers, the transported objects are token measures and the velocity is induced by attention.

Transformers were introduced as sequence-to-sequence architectures driven by self-attention [95] and have since become a central architecture for language and vision models [9, 30]. Their distinctive feature is that each token is updated by a data-dependent average of all other tokens. This makes an attention layer permutation-equivariant before positional encoding, context dependent after conditioning on the input sequence, and naturally compatible with a measure viewpoint in which a prompt is regarded as an empirical distribution of tokens.

The mathematical limit used below concerns depth rather than model scale: one lets the number of residual attention layers grow while each layer makes a small update, as in continuous-depth neural networks [15]. For attention, the resulting velocity is nonlinear in the current token law because it is normalized by the whole context. This measure-theoretic view appears in the analysis of attention as a Lipschitz or interacting-particle operator [102, 39], in the Sinkhorn-normalized dynamics of Sinkformers [82], and in recent well-posedness and mean-field-limit results for several attention mechanisms [13]. It also separates the infinite-depth limit studied here from the token-limit question, where one controls how a finite empirical context approximates its limiting attention operator [6].

We now consider very deep transformers, focusing on a single-head attention mechanism for simplicity while ignoring MLP layers, layer normalization, causality, and masking. This stripped-down framework is best suited to modeling encoders and vision transformers; the references above indicate which parts of this simplified picture extend to richer attention mechanisms.

Attention as a context-dependent velocity. After tokenization, embedding, and positional encoding, each input (from a set of tokens) is represented as a point cloud $(x_i)_{i=1}^n$ of n points in the space of vectorized tokens. An attention layer with skip connection and rescaling by $1/T$ (where T is the depth) defines a transformation of the tokens:

$$x_i \mapsto x_i + \frac{1}{T} \sum_j \frac{e^{\langle Qx_i, Kx_j \rangle} V x_j}{\sum_\ell e^{\langle Qx_i, Kx_\ell \rangle}},$$

where $\theta = (K, Q, V)$ are the parameters of the attention layer, represented by three matrices.

Token measure evolution. To handle an arbitrary number of tokens, we define $\alpha = \frac{1}{n} \sum_i \delta_{x_i}$ as the empirical measure of tokens and rewrite the transformer mapping as:

$$x_i \mapsto x_i + \frac{1}{T} \Gamma_\theta[\alpha](x_i),$$

where

$$\Gamma_\theta[\alpha](x) := \frac{\int e^{\langle Qx, Ky \rangle} V y \, \mathrm{d}\alpha(y)}{\int e^{\langle Qx, Kz \rangle} \, \mathrm{d}\alpha(z)}.$$

At the level of the token distribution, the layer pushes α forward by the "in-context" mapping $\Gamma_{\theta_t}[\alpha]$, which depends on the context α, the tokens, and the depth-dependent parameters θ_t. Denoting $t \in [0,1]$ as the depth and $\tau = 1/T$ as the step size, this gives:

$$\alpha_{t+\tau} = (\mathrm{Id} + \tau\Gamma_{\theta_t}[\alpha_t])_\sharp \alpha_t.$$

As $\tau \to 0$, this converges formally to the conservation equation

$$\partial_t \alpha_t + \mathrm{div}(\alpha_t \Gamma_{\theta_t}[\alpha_t]) = 0.$$

Algorithm 4.5 Residual attention depth evolution

Input: Tokens $(x_i^0)_{i=1}^n$, depth T, layer parameters (Q_k, K_k, V_k).
Output: Final token measure α_T.
Initialize: $\alpha_0 = \frac{1}{n}\sum_{i=1}^n \delta_{x_i^0}, \qquad \tau = 1/T.$
For $k = 0, \ldots, T-1$ **do**:
  For $i = 1, \ldots, n$ **do**
    $\Gamma_{\theta_k}[\alpha_k](x_i^k) = \frac{\sum_j \exp(\langle Q_k x_i^k, K_k x_j^k\rangle) V_k x_j^k}{\sum_j \exp(\langle Q_k x_i^k, K_k x_j^k\rangle)}.$
    $x_i^{k+1} = x_i^k + \tau\, \Gamma_{\theta_k}[\alpha_k](x_i^k).$
  Set $\alpha_{k+1} = (\mathrm{Id} + \tau\Gamma_{\theta_k}[\alpha_k])_\sharp \alpha_k.$
Return α_T.

L^2 attention and mean shift. A particularly geometric variant replaces the dot-product score $\langle Qx, Ky\rangle$ by a negative squared Euclidean score $s_\varepsilon(x,y) = -\|x-y\|^2/(2\varepsilon)$. Take, for simplicity, the same token space for queries, keys and values, and set

$$K_\varepsilon(x,y) = \exp(-\|x-y\|^2/(2\varepsilon)), \qquad \rho_\varepsilon[\alpha](x) = \int K_\varepsilon(x,y)\mathrm{d}\alpha(y), \qquad m_\varepsilon[\alpha](x) = \frac{\int y K_\varepsilon(x,y)\mathrm{d}\alpha(y)}{\rho_\varepsilon[\alpha](x)}.$$

The map $x \mapsto m_\varepsilon[\alpha](x)$ is exactly Gaussian-kernel attention, i.e. normalized kernel regression over tokens; such L^2 or Gaussian-kernel scores are used explicitly in transformer variants motivated by Lipschitz control and projection-free attention [51, 52]. Classical mean shift, however, uses the displacement from the current point to this local barycenter. This gives

$$M_\varepsilon[\alpha](x) := m_\varepsilon[\alpha](x) - x = \frac{\int (y-x)K_\varepsilon(x,y)\mathrm{d}\alpha(y)}{\rho_\varepsilon[\alpha](x)} = \varepsilon\nabla \log \rho_\varepsilon[\alpha](x) \tag{4.26}$$

and, when α is empirical, $\rho_\varepsilon[\alpha]$ is a Gaussian kernel density estimate up to normalization. Thus $M_\varepsilon[\alpha]$ is the classical Gaussian mean-shift vector [36, 16, 23]. Consequently, the barycentric residual update

$$x_i^{k+1} = (1-\tau)x_i^k + \tau m_\varepsilon[\alpha_k](x_i^k) = x_i^k + \tau M_\varepsilon[\alpha_k](x_i^k)$$

is an explicit Euler step of the continuous-time mean-shift equation

$$\partial_t \alpha_t + \mathrm{div}\big(\alpha_t M_\varepsilon[\alpha_t]\big) = 0.$$

With time step one this recovers the usual mean-shift iteration $x \leftarrow m_\varepsilon[\alpha](x)$; with small residual steps it becomes a transport PDE that moves each token uphill along the log of the smoothed token density. This distinction between the raw barycentric attention output m_ε and the velocity $M_\varepsilon = m_\varepsilon - \mathrm{Id}$ is important: adding m_ε directly as a residual would produce a different drift. The mean-shift form isolates a purely metric attention mechanism from the learned bilinear geometry of $\langle Qx, Ky\rangle$.

Gradient structure and limitations. When the token space has dimension d and the query/key space has dimension r, take $Q, K \in \mathbb{R}^{r\times d}$ and $V \in \mathbb{R}^{d\times d}$. If $V = Q^\top K$, the field $\Gamma_\theta[\alpha]$ is a gradient vector field in the token variable. Indeed, define the log-partition potential

$$\Phi_\alpha(x) = \int \exp(\langle Qx, Ky\rangle)\mathrm{d}\alpha(y), \qquad U_\alpha(x) = \log \Phi_\alpha(x).$$

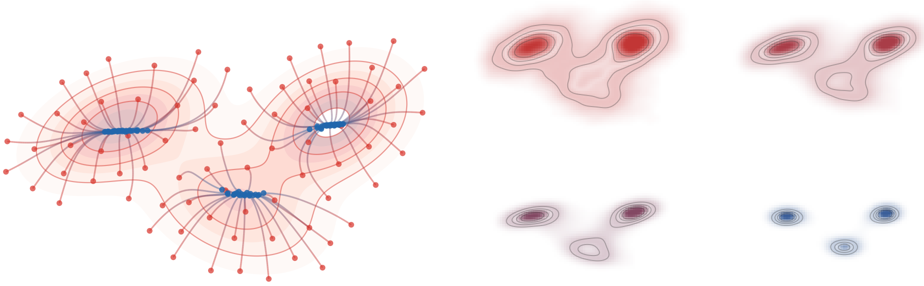


Figure 4.8: Continuous-time mean shift for a densely sampled three-Gaussian mixture. Left: initial density level sets, in red, and representative particle paths of $\dot{x} = M_\varepsilon[\alpha_t](x)$, colored from red to blue. Right: four later kernel-density renderings of α_t at increasing times, with the same red-to-blue time palette; the initial density is omitted because it is shown on the left. The snapshots are chosen before complete mode collapse, so that the flow visibly advects mass uphill along $\log \rho_\varepsilon[\alpha_t]$ and sharpens the overlapping modes.

Then

$$\nabla_x U_\alpha(x) = \frac{\int Q^\top K y \exp(\langle Qx, Ky\rangle) \mathrm{d}\alpha(y)}{\int \exp(\langle Qx, Kz\rangle) \mathrm{d}\alpha(z)} = \Gamma_\theta[\alpha](x).$$

This is an instantaneous gradient in x. It is not, however, the gradient of the first variation of a fixed functional of α, because the potential U_α itself depends on the current measure through the same attention normalization. Thus the PDE is generally a transportation dynamics, not a Wasserstein gradient flow. Special variants recover additional structure: Sinkhorn attention can be interpreted through doubly stochastic normalization and Wasserstein-type gradient flows [82, 13], while layer normalization leads naturally to dynamics on the sphere and to modified metrics. The key open difficulty for the present viewpoint is training: after the architecture has been rewritten as a controlled transport equation, learning corresponds to optimizing the time-dependent parameters $(\theta_t)_t$ rather than merely analyzing the forward PDE for fixed parameters.

4.5 Flows over the Gaussian Manifold

Gaussian measures provide a useful testing ground for the preceding dynamics. They are not invariant under a general Wasserstein gradient flow: a nonlinear velocity usually creates non-Gaussian densities immediately. The useful substitute is to either identify affine velocities, which exactly preserve Gaussianity, or to project the dynamics onto the Gaussian manifold. In both cases the measure PDE reduces to matrix ODEs for the mean and covariance. This viewpoint is emphasized in the survey [73] and is useful for comparing diffusion paths, Wasserstein gradient flows, drifting fields and transformer-type dynamics.

For constrained gradient flows on this family, the covariance equation is the finite-dimensional Bures–Wasserstein gradient flow on positive definite matrices. Thus Gaussian closure is not just a computational shortcut: it is the restriction of Wasserstein geometry to the Gaussian submanifold, where affine gradient fields encode tangent vectors.

Gaussianity preservation. The first question is invariance: one wants a simple criterion ensuring that the continuity equation does not leave the finite-dimensional Gaussian family.

Proposition 4.17 (Affine velocities preserve Gaussianity). *Let $\alpha_t = \mathcal{N}(\mathbf{m}_t, \mathbf{\Sigma}_t)$, with $\mathbf{\Sigma}_t$ positive definite, solve the continuity equation with an affine velocity*

$$v_t(x) = b_t + A_t(x - \mathbf{m}_t).$$

Then α_t remains Gaussian and its moments solve

$$\dot{\mathbf{m}}_t = b_t, \qquad \dot{\mathbf{\Sigma}}_t = A_t \mathbf{\Sigma}_t + \mathbf{\Sigma}_t A_t^\top.$$

Conversely, any smooth Gaussian curve with positive definite covariance can be generated by such an affine velocity. If one wants the velocity to be a Wasserstein tangent gradient, one chooses the unique symmetric solution of the Lyapunov equation

$$A_t \mathbf{\Sigma}_t + \mathbf{\Sigma}_t A_t = \dot{\mathbf{\Sigma}}_t.$$

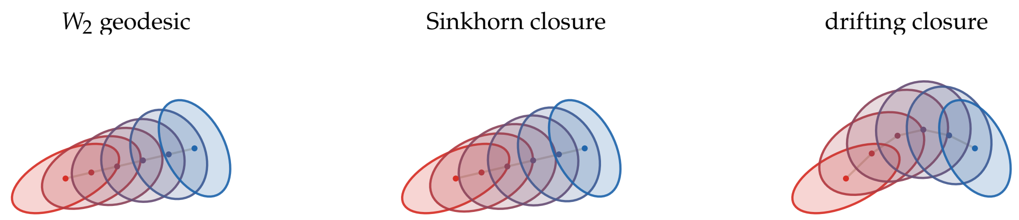


Figure 4.9: Gaussian closures of transport dynamics between two overlapping anisotropic Gaussians. The left panel is the exact W_2 Gaussian geodesic. The middle panel shows a regularized Sinkhorn-style closure, where the same mean path is accompanied by inflated intermediate covariances. The right panel shows a drifting-style closure with a curved mean path and moment-matched covariance ellipses. These finite-dimensional pictures keep only means and covariances, and therefore discard higher-order shape information that would be created by a genuinely nonlinear velocity field.

Proof. Let X_t follow the characteristic ODE $\dot{X}_t = b_t + A_t(X_t - \mathbf{m}_t)$. This linear ODE maps Gaussian random variables to Gaussian random variables. Taking expectation gives $\dot{\mathbf{m}}_t = b_t$. Writing $\tilde{X}_t = X_t - \mathbf{m}_t$, one has $\dot{\tilde{X}}_t = A_t \tilde{X}_t$, hence

$$\dot{\mathbf{\Sigma}}_t = \frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}(\tilde{X}_t \tilde{X}_t^\top) = A_t \mathbf{\Sigma}_t + \mathbf{\Sigma}_t A_t^\top.$$

For the converse, set $b_t = \dot{\mathbf{m}}_t$ and choose any matrix A_t satisfying the covariance equation. Since $\mathbf{\Sigma}_t$ is positive definite, the Lyapunov map $A \mapsto A\mathbf{\Sigma}_t + \mathbf{\Sigma}_t A$ is invertible on symmetric matrices, which gives the unique symmetric choice when a gradient velocity is required. In that case v_t is the gradient of the quadratic potential $x \mapsto \langle b_t, x\rangle + \langle A_t(x - \mathbf{m}_t), x - \mathbf{m}_t\rangle/2$. □

Constrained evolution on the Gaussian manifold. For non-affine velocities, the finite-dimensional substitute is to project the Wasserstein dynamics onto the Gaussian manifold.

Let

$$\mathcal{G} = \{\mathcal{N}(\mathbf{m}, \mathbf{\Sigma}) : \mathbf{m} \in \mathbb{R}^d,\ \mathbf{\Sigma} \succ 0\}$$

be the Gaussian submanifold of $\mathcal{P}_2(\mathbb{R}^d)$. The Wasserstein gradient of a functional constrained to a smooth submanifold $\mathcal{M} \subset \mathcal{P}_2$ is defined as the Riesz representative of the differential restricted to tangent velocities of $\mathcal{M}$. Equivalently, it is the small-step limit of the constrained JKO scheme

$$\alpha^{k+1} \in \operatorname*{argmin}_{\alpha \in \mathcal{M}} \frac{1}{2\tau} \mathcal{W}_2^2(\alpha, \alpha^k) + f(\alpha).$$

For $\mathcal{M} = \mathcal{G}$, tangent velocities are affine gradient fields $v(x) = b + A(x - \mathbf{m})$ with $A = A^\top$. The constrained gradient is therefore the $L^2(\mathcal{N}(\mathbf{m}, \mathbf{\Sigma}))$ projection of the ambient Wasserstein gradient onto this finite-dimensional affine space, whenever the ambient gradient exists.

Proposition 4.18 (Gaussian-constrained Wasserstein gradients)**.** *Let f be a smooth functional and assume that its restriction to nondegenerate Gaussian measures can be written as*

$$f(\mathcal{N}(\mathbf{m}, \mathbf{\Sigma})) = F(\mathbf{m}, \mathbf{\Sigma}).$$

Then the Wasserstein gradient constrained to the Gaussian family is the affine vector field

$$v_F(x) = \nabla_{\mathbf{m}} F(\mathbf{m}, \mathbf{\Sigma}) + 2\nabla_{\mathbf{\Sigma}} F(\mathbf{m}, \mathbf{\Sigma})(x - \mathbf{m}),$$

where $\nabla_{\mathbf{\Sigma}} F$ denotes the symmetric matrix derivative. Equivalently, v_F is the $L^2(\mathcal{N}(\mathbf{m}, \mathbf{\Sigma}))$ projection of the ambient Wasserstein gradient onto affine gradient fields, whenever the ambient gradient exists. Hence the gradient descent flow constrained to Gaussian measures satisfies

$$\dot{\mathbf{m}}_t = -\nabla_{\mathbf{m}} F(\mathbf{m}_t, \mathbf{\Sigma}_t), \qquad \dot{\mathbf{\Sigma}}_t = -2\big(\mathbf{\Sigma}_t \nabla_{\mathbf{\Sigma}} F(\mathbf{m}_t, \mathbf{\Sigma}_t) + \nabla_{\mathbf{\Sigma}} F(\mathbf{m}_t, \mathbf{\Sigma}_t)\mathbf{\Sigma}_t\big), \tag{4.27}$$

and the descent velocity is affine.

Proof. Test the functional along a Gaussian tangent vector, represented by an affine gradient field

$$v(x) = b + A(x - \mathbf{m})$$

with A symmetric. The induced first-order variations are $\dot{\mathbf{m}} = b$ and $\dot{\mathbf{\Sigma}} = A\mathbf{\Sigma} + \mathbf{\Sigma}A$. Therefore

$$\mathrm{d}F(\mathbf{m}, \mathbf{\Sigma})[b, A\mathbf{\Sigma} + \mathbf{\Sigma}A] = \langle \nabla_{\mathbf{m}} F,\, b\rangle + \mathrm{tr}(\nabla_{\mathbf{\Sigma}} F(A\mathbf{\Sigma} + \mathbf{\Sigma}A))\,.$$

Since A, $\mathbf{\Sigma}$ and $\nabla_{\mathbf{\Sigma}} F$ are symmetric, the second term equals

$$2\,\mathrm{tr}(\nabla_{\mathbf{\Sigma}} F\, A\mathbf{\Sigma}) = \int \langle 2\nabla_{\mathbf{\Sigma}} F(x - \mathbf{m}),\, A(x - \mathbf{m})\rangle \mathrm{d}\mathcal{N}(\mathbf{m}, \mathbf{\Sigma})(x).$$

Together with the mean term, this gives

$$\mathrm{d}F(\mathbf{m}, \mathbf{\Sigma})[\dot{\mathbf{m}}, \dot{\mathbf{\Sigma}}] = \int \langle v_F(x),\, v(x)\rangle \mathrm{d}\mathcal{N}(\mathbf{m}, \mathbf{\Sigma})(x)$$

for all affine gradient fields v. This identifies the constrained Wasserstein gradient in the induced $L^2(\alpha)$ metric, or equivalently the projection of the ambient gradient when it exists. Substituting the descent velocity $-v_F$ in Proposition 4.17 gives (4.27). □

This proposition should be read as the organizing rule for Gaussian closures: once the scalar energy has been reduced to a function of $(\mathbf{m}, \mathbf{\Sigma})$, its constrained Wasserstein gradient is automatically affine and the covariance follows the Bures-type ODE (4.27). When the first variation of f is quadratic, this constrained gradient coincides with the full Wasserstein gradient.

Gaussian-preserving gradient flows. The next examples show that many familiar energies already have affine Wasserstein gradients on Gaussian inputs, so their full flow remains inside the Gaussian family.

Example 4.19 (Gaussian energies and affine gradients). Proposition 4.18 turns many standard energies into explicit affine fields:

- *Quadratic potential energy.* If

$$f(\alpha) = \int \left(\frac{1}{2}x^\top Hx + \langle \ell,\, x\rangle\right)\mathrm{d}\alpha(x), \qquad H = H^\top,$$

  then

$$\nabla_W f(\alpha)(x) = Hx + \ell = (H\mathbf{m} + \ell) + H(x - \mathbf{m}).$$

  This is the Gaussian form of transport under a quadratic confinement.

- *Quadratic interaction energy.* If

$$f(\alpha) = \frac{1}{4}\iint (x - y)^\top G(x - y)\mathrm{d}\alpha(x)\mathrm{d}\alpha(y), \qquad G = G^\top,$$

  then $F(\mathbf{m}, \mathbf{\Sigma}) = \frac{1}{2}\mathrm{tr}(G\mathbf{\Sigma})$ and

$$\nabla_W f(\alpha)(x) = G(x - \mathbf{m}).$$

  The mean is unchanged and the covariance contracts or expands according to the signs of G.

- *Relative entropy to a Gaussian.* For $\bar{\alpha} = \mathcal{N}(\bar{\mathbf{m}}, \bar{\mathbf{\Sigma}})$,

$$f(\alpha) = \mathrm{KL}(\alpha|\bar{\alpha})$$

  has

$$\nabla_W f(\alpha)(x) = \bar{\mathbf{\Sigma}}^{-1}(\mathbf{m} - \bar{\mathbf{m}}) + (\bar{\mathbf{\Sigma}}^{-1} - \mathbf{\Sigma}^{-1})(x - \mathbf{m}).$$

  The descent equations are the Ornstein–Uhlenbeck moment equations

$$\dot{\mathbf{m}}_t = -\bar{\mathbf{\Sigma}}^{-1}(\mathbf{m}_t - \bar{\mathbf{m}}), \qquad \dot{\mathbf{\Sigma}}_t = 2\mathrm{Id} - \bar{\mathbf{\Sigma}}^{-1}\mathbf{\Sigma}_t - \mathbf{\Sigma}_t\bar{\mathbf{\Sigma}}^{-1}.$$

- *Squared Wasserstein distance to a Gaussian.* For

$$f(\alpha) = \frac{1}{2}\mathcal{W}_2^2(\alpha,\bar{\alpha}), \qquad \bar{\alpha} = \mathcal{N}(\bar{\mathbf{m}},\bar{\mathbf{\Sigma}}),$$

  the Gaussian Brenier map $T_{\alpha\to\bar{\alpha}}$ is affine,

$$T_{\alpha\to\bar{\alpha}}(x) = \bar{\mathbf{m}} + M(x-\mathbf{m}), \qquad M = \mathbf{\Sigma}^{-1/2}(\mathbf{\Sigma}^{1/2}\bar{\mathbf{\Sigma}}\mathbf{\Sigma}^{1/2})^{1/2}\mathbf{\Sigma}^{-1/2}.$$

  Hence

$$\nabla_{\mathcal{W}} f(\alpha)(x) = x - T_{\alpha\to\bar{\alpha}}(x) = (\mathbf{m}-\bar{\mathbf{m}}) + (\mathrm{Id}-M)(x-\mathbf{m}),$$

  and descent moves each Gaussian infinitesimally along the Bures–Wasserstein geodesic toward $\bar{\alpha}$.

- *Gaussian-only losses.* Sliced SW_2^2 losses to a Gaussian, Gaussian Sinkhorn divergences, and any smooth closed formula depending only on $(\mathbf{m},\mathbf{\Sigma})$ fit the same constrained-gradient template:

$$v_F(x) = \nabla_{\mathbf{m}}F + 2\nabla_{\mathbf{\Sigma}}F(x-\mathbf{m}).$$

  For Gaussian Sinkhorn divergences this finite-dimensional flow is studied in [44].

Not every PDE preserves Gaussianity exactly. For instance, Wasserstein flows of relative Fisher information, related to quantum-drift or higher-order diffusion equations, typically require a Gaussian projection to close on $(\mathbf{m},\mathbf{\Sigma})$. Such projected closures are still useful: they expose the finite-dimensional dynamics predicted by a variational model and make it easy to compare variational flows with non-variational affine dynamics such as drifting fields or the Gaussian transformer closure below.

Proposition 4.20 (Centered Gaussian covariance catalogue)**.** *Let $\gamma = \mathcal{N}(0,\mathrm{Id})$ and let $\mu_t = \mathcal{N}(0,C_t)$ with $C_t \succ 0$. For the normalizations displayed below, the Wasserstein descent constrained to the centered Gaussian manifold satisfies $\dot{C}_t = h(C_t)$, with*

$$\begin{array}{rcl}
\mathrm{KL}(\mu|\gamma) & : & h(C) = 2(\mathrm{Id}-C),\\
\frac{1}{2}\mathcal{I}(\mu|\gamma) & : & h(C) = 2(C^{-1}-C),\\
\mathcal{W}_2^2(\mu,\gamma) & : & h(C) = 4(C^{1/2}-C),\\
\mathrm{MMD}_k^2(\mu,\gamma), \quad k(x,y)=\langle x,\,y\rangle^2 & : & h(C) = 8(C-C^2),\\
S_\varepsilon(\mu,\gamma) & : & h(C) = 4\left(C+\frac{\varepsilon^2}{16}\mathrm{Id}\right)^{1/2} - 2\left(C^2+\frac{\varepsilon^2}{16}\mathrm{Id}\right)^{1/2} - 2C - \frac{\varepsilon}{2}\mathrm{Id},\\
\mathrm{SW}_2^2(\mu,\gamma) & : & h(C) = V(C)C + CV(C),
\end{array}$$

where S_ε is the debiased Sinkhorn divergence for the quadratic cost $\|x-y\|^2$ and KL regularization strength ε, and

$$V(C) = 2\int_{\mathbb{S}^{d-1}}\left(\frac{1}{\sqrt{\theta^\top C\theta}}-1\right)\theta\theta^\top\,\mathrm{d}\sigma(\theta)$$

for the normalized spherical measure σ. Here

$$\mathcal{I}(\mu|\gamma) = \int \left|\nabla\log\rho(x)+x\right|^2\rho(x)\,\mathrm{d}x \qquad (\mu=\rho\,\mathrm{d}x).$$

Thus the unhalved Fisher divergence has right-hand side $4(C^{-1}-C)$. Multiplying any of these energies by a constant simply rescales the corresponding right-hand side.

Proof. Each row is obtained by identifying the affine descent velocity $v(x) = M_C x$ generated by the corresponding Gaussian-constrained calculation and then applying Proposition 4.17, which gives $\dot{C} = M_C C + C M_C^\top$. For $\mathrm{KL}(\cdot|\gamma)$, the Fokker–Planck velocity is $(C^{-1}-\mathrm{Id})x$, hence $\dot{C} = 2(\mathrm{Id}-C)$. For the Fisher row, the restriction of $\frac{1}{2}\mathcal{I}$ to centered Gaussians is

$$\frac{1}{2}\left(\mathrm{tr}(C)+\mathrm{tr}(C^{-1})-2d\right).$$

Using Proposition 4.18 gives the descent velocity $(C^{-2} - \mathrm{Id})x$, hence $\dot{C} = 2(C^{-1} - C)$. This row should be read as a Gaussian projected closure of the fourth-order Fisher flow.

For $\mathcal{W}_2^2(\cdot, \gamma)$, the Brenier map from $\mathcal{N}(0, C)$ to γ is $C^{-1/2}x$, so the descent velocity for the unhalved squared distance is $2(C^{-1/2} - \mathrm{Id})x$, giving $4(C^{1/2} - C)$. For the polynomial MMD row, centered Gaussians satisfy $\mathrm{MMD}_k^2(\mu, \gamma) = \|C - \mathrm{Id}\|_F^2$; the first variation is quadratic and its descent velocity is $4(\mathrm{Id} - C)x$, giving $8(C - C^2)$.

Gaussian Sinkhorn dual potentials are quadratic, so the velocity is again linear; differentiating the closed Gaussian formula yields the displayed spectral expression. The square roots are spectral functions of C, hence commute with C, which is why the covariance ODE closes as a matrix function of C alone. For sliced Wasserstein, each one-dimensional projection is a Gaussian transport with velocity $2((\theta^\top C\theta)^{-1/2} - 1)\langle\theta, x\rangle\theta$; averaging these velocities over $\mathbb{S}^{d-1}$ gives $v(x) = V(C)x$ and thus $\dot{C} = V(C)C + CV(C)$. □

Example 4.21 (Linear mean-field networks as cross-moment flows). In the two-layer model above, take the linear activation $\sigma(s) = s$, so that

$$\psi((u, v), z) = v\,\langle u, z\rangle.$$

The predictor is the linear map

$$G_{\alpha_t}(z) = Q_t z, \qquad Q_t = \int vu^\top \mathrm{d}\alpha_t(u, v) \in \mathbb{R}^{d' \times d}.$$

Thus the energy depends on the neuron law only through the cross moment Q_t, a subblock of the raw second moment of $x = (u, v)$. For square loss, set

$$S = \int zz^\top \mathrm{d}\rho(z, y), \qquad R = \int yz^\top \mathrm{d}\rho(z, y), \qquad H_t = Q_t S - R.$$

The first variation is

$$\delta f(\alpha_t)(u, v) = \langle H_t, vu^\top\rangle = v^\top H_t u.$$

Hence the particle velocity in parameter space is linear:

$$-\nabla_{(u,v)}\delta f(\alpha_t)(u, v) = -\begin{pmatrix} 0 & H_t^\top \\ H_t & 0 \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix}.$$

Therefore a Gaussian law of neurons remains Gaussian. Its mean and covariance follow Proposition 4.17, with a matrix depending only on the current cross moment Q_t, a raw second-moment subblock that becomes a covariance subblock when the neuron law is centered. This exact closure is special to the linear activation; for nonlinear activations, Gaussian closures are usually projections rather than invariant families.

Non-variational Gaussian-preserving flows. The last examples are not ordinary gradient flows of a fixed scalar energy on the full Wasserstein space. They preserve Gaussianity because the prescribed velocity field is affine when evaluated on Gaussian measures.

Example 4.22 (Flow matching and diffusion paths between Gaussians). Consider a prescribed Gaussian interpolation $\alpha_t = \mathcal{N}(\mathbf{m}_t, \mathbf{\Sigma}_t)$. Proposition 4.17 shows that an exact flow-matching velocity can be taken affine:

$$v_t(x) = \dot{\mathbf{m}}_t + A_t(x - \mathbf{m}_t), \qquad A_t\mathbf{\Sigma}_t + \mathbf{\Sigma}_t A_t = \dot{\mathbf{\Sigma}}_t.$$

In the isotropic case $\mathbf{\Sigma}_t = s_t^2\mathrm{Id}$, this reduces to the transparent formula

$$v_t(x) = \dot{\mathbf{m}}_t + \frac{\dot{s}_t}{s_t}(x - \mathbf{m}_t).$$

For instance, the diffusion noising path

$$X_t = a_t X_0 + \sigma_t Z, \qquad Z \sim \mathcal{N}(0, \mathrm{Id}),$$

has $\mathbf{m}_t = a_t\mathbf{m}_0$ and $\mathbf{\Sigma}_t = a_t^2\mathbf{\Sigma}_0 + \sigma_t^2\mathrm{Id}$. Thus, in the Gaussian case, diffusion paths and flow-matching paths reduce to the same mean-covariance bookkeeping, although the corresponding training objectives are different.

Example 4.23 (Gaussian kernel drifting). Let the target be $\beta = \mathcal{N}(\bar{\mathbf{m}}, \bar{\boldsymbol{\Sigma}})$ and assume $\mu_t = \mathcal{N}(\mathbf{m}_t, \boldsymbol{\Sigma}_t)$. For the Gaussian kernel

$$K_\varepsilon(x, y) = \exp(-\|x - y\|^2/(2\varepsilon)),$$

the normalized field (4.17) satisfies

$$B_\varepsilon[\mu_t](x) = -\varepsilon(\boldsymbol{\Sigma}_t + \varepsilon \mathrm{Id})^{-1}(x - \mathbf{m}_t).$$

Indeed the smoothed density $x \mapsto \int K_\varepsilon(x, y) \mathrm{d}\mu_t(y)$ is proportional to the Gaussian density with mean $\mathbf{m}_t$ and covariance $\boldsymbol{\Sigma}_t + \varepsilon \mathrm{Id}$. Thus $B_\varepsilon[\mu_t]$ is the mean-shift vector of a Gaussian density: it points linearly toward the Gaussian mode, with strength set by the bandwidth. The drifting velocity (4.19) is therefore the difference of two affine mean-shift fields; it is affine and preserves Gaussianity. With

$$A_t = (\boldsymbol{\Sigma}_t + \varepsilon \mathrm{Id})^{-1}, \qquad \bar{A} = (\bar{\boldsymbol{\Sigma}} + \varepsilon \mathrm{Id})^{-1},$$

the ODE is

$$\dot{\mathbf{m}}_t = \varepsilon \bar{A}(\bar{\mathbf{m}} - \mathbf{m}_t), \qquad \dot{\boldsymbol{\Sigma}}_t = \varepsilon\big((A_t - \bar{A})\boldsymbol{\Sigma}_t + \boldsymbol{\Sigma}_t(A_t - \bar{A})\big).$$

This finite-dimensional model explains the stabilizing role of the self-normalized repulsion term in drifting: without it, the covariance equation loses the $A_t\boldsymbol{\Sigma}_t + \boldsymbol{\Sigma}_t A_t$ contribution.

Example 4.24 (Gaussian closure of attention dynamics). For the transformer PDE, assume $\alpha = \mathcal{N}(\mathbf{m}, \boldsymbol{\Sigma})$. Since exponential tilting preserves Gaussianity,

$$\frac{\int e^{\langle Qx, Ky\rangle}\, y \,\mathrm{d}\alpha(y)}{\int e^{\langle Qx, Kz\rangle}\, \mathrm{d}\alpha(z)} = \mathbf{m} + \boldsymbol{\Sigma} K^\top Q x.$$

Therefore

$$\Gamma_\theta[\alpha](x) = V\mathbf{m} + V\boldsymbol{\Sigma} K^\top Q x$$

is affine. The Gaussian token law is preserved and satisfies

$$\dot{\mathbf{m}}_t = (V_t + V_t\boldsymbol{\Sigma}_t K_t^\top Q_t)\mathbf{m}_t, \qquad \dot{\boldsymbol{\Sigma}}_t = B_t\boldsymbol{\Sigma}_t + \boldsymbol{\Sigma}_t B_t^\top, \qquad B_t = V_t\boldsymbol{\Sigma}_t K_t^\top Q_t.$$

When $V_t = Q_t^\top K_t$, the matrix $B_t = Q_t^\top K_t \boldsymbol{\Sigma}_t K_t^\top Q_t$ is symmetric positive semidefinite, matching the gradient-field case mentioned above. This closure is not a convergence theorem for trained transformers. It is instead a tractable model of how attention can shear, amplify or contract a cloud of tokens through its covariance.

Contractive Gaussian projection. The preceding examples show when Gaussianity is preserved or imposed by projection. Gelbrich's inequality [38] gives a useful variational explanation: replacing a measure by the Gaussian with the same first two moments cannot increase its Wasserstein distance to another similarly projected measure.

Theorem 4.25 (Gelbrich theorem). *For $\mu \in \mathcal{P}_2(\mathbb{R}^d)$, let*

$$\mathcal{R}\mu := \mathcal{N}(\mathbf{m}_\mu, \boldsymbol{\Sigma}_\mu)$$

be the Gaussian with the same mean and covariance as μ. Then

$$\mathcal{W}_2^2(\mathcal{R}\mu, \mathcal{R}\nu) = \|\mathbf{m}_\mu - \mathbf{m}_\nu\|^2 + \mathcal{B}^2(\boldsymbol{\Sigma}_\mu, \boldsymbol{\Sigma}_\nu) \leqslant \mathcal{W}_2^2(\mu, \nu).$$

Proof. Take any coupling (X, Y) of μ and ν, center the variables, and write $C = \mathbb{E}[(X - \mathbf{m}_\mu)(Y - \mathbf{m}_\nu)^\top]$. In the positive definite case, positivity of the block covariance matrix implies the factorization $C = \boldsymbol{\Sigma}_\mu^{1/2} K \boldsymbol{\Sigma}_\nu^{1/2}$ with $\|K\|_{\mathrm{op}} \leqslant 1$, and therefore, by operator/nuclear norm duality,

$$\operatorname{tr} C \leqslant \operatorname{tr}\left((\boldsymbol{\Sigma}_\mu^{1/2}\boldsymbol{\Sigma}_\nu\boldsymbol{\Sigma}_\mu^{1/2})^{1/2}\right).$$

The semidefinite case follows by adding $\eta \mathrm{Id}$ to both covariance matrices and letting $\eta \downarrow 0$. Expanding $\mathbb{E}\|X - Y\|^2$ gives the lower bound

$$\mathbb{E}\|X - Y\|^2 \geqslant \|\mathbf{m}_\mu - \mathbf{m}_\nu\|^2 + \mathcal{B}^2(\boldsymbol{\Sigma}_\mu, \boldsymbol{\Sigma}_\nu).$$

Taking the infimum over couplings proves the inequality, while equality for Gaussian laws is Proposition 1.12. □

The following preservation criterion is a direct consequence of Gelbrich's theorem and was explained to us by Hugo Lavenant. It says that a functional which does not increase under moment-matched Gaussian projection admits Gaussian minimizing movements from Gaussian initial data.

Theorem 4.26 (Hugo Lavenant Gaussian-preservation criterion). *Let $F : \mathcal{P}_2(\mathbb{R}^d) \to (-\infty, +\infty]$ satisfy*

$$F(\mathcal{R}\mu) \leqslant F(\mu) \qquad \forall \mu \in \mathcal{P}_2(\mathbb{R}^d),$$

with $\mathcal{R}$ defined in Theorem 4.25. If γ is Gaussian and ν minimizes the JKO step

$$\eta \mapsto F(\eta) + \frac{1}{2\tau} \mathcal{W}_2^2(\gamma, \eta),$$

then $\mathcal{R}\nu$ is also a minimizer. If the JKO minimizer is unique, it is Gaussian. Thus any unique Wasserstein gradient flow obtained as the limit of this JKO scheme preserves Gaussian initial data.

Proof. For the JKO claim, $\mathcal{R}\gamma = \gamma$ because γ is Gaussian. Hence, for any competitor η,

$$F(\mathcal{R}\eta) + \frac{1}{2\tau} \mathcal{W}_2^2(\gamma, \mathcal{R}\eta) \leqslant F(\eta) + \frac{1}{2\tau} \mathcal{W}_2^2(\gamma, \eta).$$

Applying this to a minimizer $\eta = \nu$ shows that $\mathcal{R}\nu$ is again a minimizer. Uniqueness forces $\nu = \mathcal{R}\nu$. □

Remark 4.27 (Gaussian barycenters from contraction). The same projection argument also explains why quadratic Wasserstein barycenters of Gaussian measures are Gaussian. If β_s are Gaussian and

$$F(\mu) = \sum_s \lambda_s \mathcal{W}_2^2(\mu, \beta_s),$$

then $\mathcal{R}\beta_s = \beta_s$, and Theorem 4.25 gives $F(\mathcal{R}\mu) \leqslant F(\mu)$. Thus the moment-matched Gaussian projection of any barycenter is again a barycenter; when the barycenter is unique, it must itself be Gaussian. This is the contraction viewpoint behind Gaussian barycenter preservation.

Conclusion

The central message of this review is that many machine-learning methods can be read as PDEs on probability measures. Optimal transport supplies the geometry behind this statement: it turns a loss into a metric gradient flow, a coupling into a geodesic, a score into a Fokker–Planck drift, and a generative model into a transported law. This language is valuable because it separates modelling choices from representation choices. A finite particle algorithm, a neural parametrization and a continuum PDE may then be understood as different approximations of the same measure-valued evolution.

The same viewpoint also clarifies the limits of current theory. Empirical measures are singular, neural parametrizations are non-convex, high-dimensional transport is statistically expensive, and generative flows often use interpolants that are transportation dynamics without being Wasserstein geodesics or gradient flows. These tensions are not defects of the PDE viewpoint; they identify where mathematical analysis is still needed. For machine learning, PDEs are therefore both an explanatory tool and a source of new problems.

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